

ULTIMATE TEACH YOURSELF BOOKLET

Computer Science XI

Learn every chapter on your own — explanations, worked examples, memory tricks, practice questions and full answer keys.

Bilingual support: English + اردو

کمپیوٹر سائنس — گیارہویں جماعت: خود سیکھیں، خود پرکھیں

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Final Revision: one-page summaries, mock paper and answers

How to use this booklet

This booklet is written so that you can learn every chapter **on your own**, without a teacher sitting next to you. Each topic explains the idea in simple English, shows a worked example, gives you a memory trick, warns you about common mistakes, explains it again in Urdu, and ends with quick questions so you can check that you really understood.

یہ کتابچہ اس طرح لکھا گیا ہے کہ آپ استاد کے بغیر خود ہر باب سیکھ سکیں۔ ہر موضوع پہلے آسان انگریزی میں سمجھایا گیا ہے، پھر حل شدہ مثال، یاد رکھنے کا طریقہ، عام غلطیاں، اردو میں وضاحت اور آخر میں خود کو پرکھنے کے سوالات دیے گئے ہیں۔

What the coloured boxes mean

LEARN IT

The main explanation. Read it slowly and make sure every sentence makes sense.

WORKED EXAMPLE

A solved example, step by step. Cover the solution and try it yourself first.

REMEMBER IT

A short trick, analogy or mnemonic that helps you recall the idea in the exam.

COMMON MISTAKE

Mistakes students often make. Read these before every test.

اردو میں سمجھیں

اسی موضوع کی اردو میں آسان وضاحت۔

EXAM TIP

How Golden topics are asked in board exams and what examiners look for.

CHECK YOURSELF

Two or three quick questions. Answer them before looking at the answers underneath.

GOLDEN TOPIC

★ **Golden** marks a high-yield topic that appears often in examinations. Give these topics extra time.

The 5-step self-study method

1. **Preview (5 minutes).** Read the chapter opener and the "What you will learn" list. Turn each objective into a question you want answered.
2. **Learn one topic at a time.** Read the explanation, then work through the example with a pen and paper. Do not just read the solution — reproduce it.
3. **Check yourself straight away.** Answer the "Check yourself" questions from memory. If you get one wrong, re-read that topic before moving on.
4. **Practise at the end of the chapter.** Do the multiple-choice, short and long questions in exam conditions (no notes), then mark your work using the answer key.
5. **Review and repeat.** Tick the "I can..." checklist. Revisit anything you could not tick after 1 day, 3 days and 7 days. Spaced repetition makes knowledge stick.

Tips for board exam answers

- **Define, explain, example.** A strong short answer gives a clear definition, one or two lines of explanation and a real-life example.

- **Use tables for "differentiate" questions.** Compare at least four points side by side.
- **Draw and label.** Logic gates, diagrams, charts and data structures earn marks when drawn neatly and labelled.
- **Show every step.** In K-maps, sorting traces, trace tables, statistics and code output questions, examiners give marks for the working, not only the final answer.
- **Write correct code.** Indent Python carefully, use meaningful variable names, and write the expected output under your program.

Computer Systems

This chapter shows you how a computer thinks in 0s and 1s, how logic gates turn those bits into decisions, how software is planned and built step by step, and how data travels across a network. These three ideas — digital logic, software development and communication models — are the foundation of everything else in Computer Science.

اس باب میں آپ سیکھیں گے کہ کمپیوٹر صفر اور ایک کی زبان میں کیسے کام کرتا ہے، لاجک گیٹس فیصلے کیسے کرتے ہیں، سافٹ ویئر مرحلہ وار کیسے بنایا جاتا ہے، اور نیٹ ورک پر ڈیٹا ایک جگہ سے دوسری جگہ کیسے پہنچتا ہے۔

What you will learn

- Tell the difference between discrete and continuous quantities, and between analog and digital signals, with examples.
- Use the Boolean operations AND, OR and NOT, and apply the order of precedence.
- Draw the symbol, write the expression and give the truth table of AND, OR, NOT, NAND, NOR, XOR and XNOR gates.
- Simplify Boolean expressions using 2-variable and 3-variable Karnaugh maps.
- Describe the six phases of the SDLC and compare the Waterfall and Agile models using case studies.
- State the advantages of digital systems.
- Build a complete truth table for any expression with 2 or 3 inputs.
- Write minterms (SOP) and maxterms (POS) from a truth table, and draw a logic diagram from an expression.
- Build and test a simple circuit in LogiSim Evolution.
- Explain the seven layers of the OSI model and map them to the four layers of the TCP/IP model.

1.1.1 Discrete and Continuous Quantities

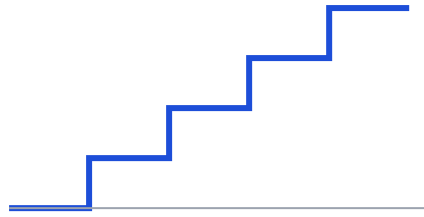
LEARN IT

Everything around you that can be measured or counted is a **quantity**. Quantities come in two kinds.

Discrete quantities are separate, countable items with specific values. You can count them one by one: 1, 2, 3 ... There is nothing "in between". You cannot have 2.5 students in a classroom.

Continuous quantities can take *any* value within a range. They flow smoothly without breaks. Water can be 30 °C, 30.5 °C, 30.51 °C, and so on — there is always another value in between.

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Discrete (Stairs)



Continuous (Ramp)

Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

REMEMBER IT

Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask yourself: "Do I *count* it or *measure* it?" Count → discrete. Measure → continuous.

COMMON MISTAKE

Students sometimes say "money is continuous because amounts are large". No — rupees and paisa are counted in fixed units, so money is discrete. Size does not matter; what matters is whether values in between are possible.

اروین سمجھیں

منفصل (Discrete) مقادیر وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔ مسلسل (Continuous) مقادیر کسی حد کے اندر کوئی بھی قیمت لے سکتی ہیں، جیسے پانی کا درجہ حرارت۔ سیڑھیاں منفصل ڈیٹا کی مثال ہیں اور ڈھلوان (ریمپ) مسلسل ڈیٹا کی۔

CHECK YOURSELF

1. Is the number of pages in a book discrete or continuous?
2. Is your height discrete or continuous?
3. In the stairs/ramp analogy, which one shows continuous data?

Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.

1.1.2 Introduction to Digital Systems

LEARN IT

A **digital system** is an electronic system that uses only two values: **0** and **1**. These two values are called **binary digits**, or **bits** for short.

- **0** represents OFF, LOW, or FALSE.
- **1** represents ON, HIGH, or TRUE.

Why only two values? Because it is easy for an electronic circuit to tell "voltage present" from "no voltage". Every number, letter, photo and song in a computer is finally stored as a long pattern of these bits.

Advantages of digital systems

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Examples of digital systems

- **Computers:** process data as 0s and 1s.
- **Smartphones:** store photos and music as binary data.
- **Digital cameras:** capture images as binary code.

WORKED EXAMPLE

Why is copying digital data "accurate"?

Step 1: A song is stored as bits, for example 10110010...

Step 2: When you copy it to a USB, the computer copies each bit exactly: 1 stays 1, 0 stays 0.

Step 3: Even after 100 copies the bits are identical, so the song sounds the same. An old cassette tape (analog) loses a little quality every time it is copied.

REMEMBER IT

Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

اردو میں سمجھیں

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔ 0 کا مطلب بند، کم یا غلط ہے اور 1 کا مطلب چالو، زیادہ یا درست ہے۔ ڈیجیٹل نظام قابل بھروسا، درست، بنانے میں آسان اور سافٹ ویئر سے قابو میں رہنے والے ہوتے ہیں۔

CHECK YOURSELF

1. What does a 1 represent in a digital system?
2. Give two advantages of digital systems.

Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).

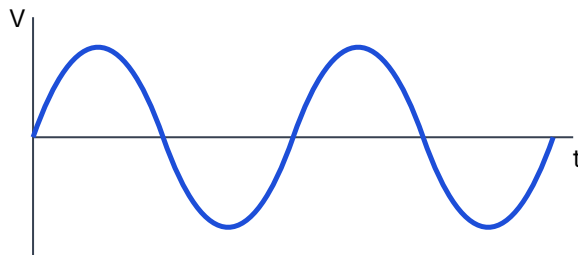
1.1.3 Analog and Digital Signals ★ Golden

LEARN IT

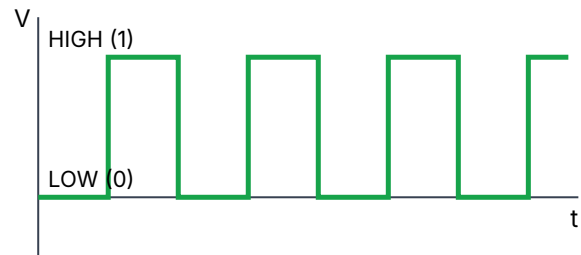
A **signal** is a changing voltage (or other quantity) that carries information.

Analog signals are continuous signals that change smoothly over time, like a **sine wave**. Examples: the human voice, a traditional mercury thermometer.

Digital signals have only two levels: **HIGH (1)** and **LOW (0)**. They jump instantly between these values, which creates a **square wave**. Example: computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.

WORKED EXAMPLE

Why is digital more resistant to noise?

Step 1: Suppose a wire carries 5 V for HIGH and 0 V for LOW.

Step 2: Noise adds a small error, so 5 V arrives as 4.6 V and 0 V arrives as 0.3 V.

Step 3: The receiver only asks "Is it closer to HIGH or LOW?" 4.6 V is still read as 1 and 0.3 V is still read as 0. The data is perfect.

Step 4: In an analog signal, every exact value carries meaning, so 4.6 V instead of 5 V really changes the sound or reading. The noise cannot be removed.

REMEMBER IT

Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

COMMON MISTAKE

Do not draw a digital signal with sloping sides. A digital signal jumps *instantly*, so its edges are straight vertical lines. Also do not confuse "digital" with "electronic": an analog signal can also be electronic.

اینا لاگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔ ڈیجیٹل سگنل کی صرف دو سطحیں ہوتی ہیں، (1) HIGH اور (0) LOW، اور یہ اچانک ایک سے دوسری پر جاتا ہے، اس لیے مربع لہر (Square Wave) بنتی ہے۔ ڈیجیٹل سگنل شور سے کم متاثر ہوتا ہے، اس لیے زیادہ قابلِ بھروسہ ہے۔

EXAM TIP

Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave shape, reliability, example). Always label your diagrams: axes V and t, and HIGH(1)/LOW(0) on the square wave.

CHECK YOURSELF

1. What shape of wave does a digital signal make?
2. Which type of signal is more affected by noise?
3. Give one example of an analog signal.

Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).

1.1.4 Boolean Algebra

LEARN IT

Boolean algebra was developed by the English mathematician **George Boole**. It is a kind of algebra that deals with only two values: **TRUE (1)** and **FALSE (0)**. It is the foundation of digital electronics and computer programming — every **if** condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, $Y = A + B$

Some simple rules (laws) follow directly from the meaning of AND, OR and NOT. They are useful when you check or simplify answers:

OR rules	AND rules	NOT rules
$A + 0 = A$	$A \cdot 1 = A$	$(A')' = A$
$A + 1 = 1$	$A \cdot 0 = 0$	$A + A' = 1$
$A + A = A$	$A \cdot A = A$	$A \cdot A' = 0$

WORKED EXAMPLE

Using a law: Simplify $Y = A'B + AB$.

Step 1: Both terms contain B, so take B out: $Y = B(A' + A)$.

Step 2: $A' + A = 1$ (a variable OR its opposite is always 1).

Step 3: $Y = B \cdot 1 = B$.

COMMON MISTAKE

In Boolean algebra "+" means OR, not addition. So $1 + 1 = 1$ (TRUE OR TRUE is TRUE), not 2. There is no value bigger than 1.

اردو میں سمجھیں

بولین الجبرا جارج بول (George Boole) نے بنایا۔ اس میں صرف دو قیمتیں ہوتی ہیں: درست (1) اور غلط (0)۔ متغیرات (Variables) جیسے A اور B صرف 0 یا 1 رکھتے ہیں، آپریشنز AND، OR اور NOT ہیں، اور ان کے ملاپ سے ایکسپریشن (Expression) بنتا ہے۔

CHECK YOURSELF

1. Who developed Boolean algebra?
2. In Boolean algebra, what is $1 + 1$?

Answers: 1. George Boole. 2. 1 (because + means OR).

1.1.5 Boolean Operations

LEARN IT

There are three basic Boolean operations. Every other operation can be made from them.

1. AND operation ($\cdot$) — gives output 1 **only when ALL inputs are 1**. Expression: $Y = A \cdot B$ (often written just AB).

2. OR operation ($+$) — gives output 1 **when ANY input is 1**. Expression: $Y = A + B$.

3. NOT operation (' or $\bar{}$) — **inverts** (flips) the input. If the input is 1 the output is 0, and if the input is 0 the output is 1. It is a **unary** operation, meaning it works on only one operand. Expression: $Y = A'$ (also written as A with a bar on top, $\bar{A}$).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

WORKED EXAMPLE

Evaluate each expression for $A = 1$, $B = 0$.

- (a) $A \cdot B = 1 \cdot 0 = 0$ (not all inputs are 1).
- (b) $A + B = 1 + 0 = 1$ (at least one input is 1).
- (c) $B' = 0' = 1$ (NOT flips 0 to 1).
- (d) $A \cdot B' = 1 \cdot 1 = 1$.

REMEMBER IT

AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

اردو میں سمجھیں

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ). OR آپریشن تب 1 دیتا ہے جب کوئی ایک ان پٹ بھی 1 ہو (دو دروازوں والا کمرہ). NOT آپریشن ان پٹ کو الٹ دیتا ہے اور صرف ایک ان پٹ پر کام کرتا ہے، اس لیے اسے یونری (Unary) کہتے ہیں۔

CHECK YOURSELF

1. What symbol is used for the OR operation?
2. Why is NOT called a unary operation?
3. Find $A + B'$ when $A = 0$, $B = 1$.

Answers: 1. "+". 2. Because it works on only one operand (one input). 3. $B' = 0$, so $0 + 0 = 0$.

1.1.6 Truth Tables

LEARN IT

A **truth table** shows **all possible input combinations** and the output for each one, for a logic gate or a circuit. It is the "complete record" of how the circuit behaves.

How to create a truth table:

1. Count the number of inputs (n).
2. Calculate total rows = 2^n .
3. List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
4. Calculate the output for each combination.

2-input example ($n = 2$, rows = 4)

A	B	Combination
0	0	Row 1
0	1	Row 2
1	0	Row 3

3-input example ($n = 3$, rows = 8)

A	B	C	Decimal
0	0	0	0
0	0	1	1
0	1	0	2

1	1	Row 4
---	---	-------

0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

REMEMBER IT

Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write 0,0,0,0,1,1,1,1 (change every 4 rows). For n inputs the leftmost column changes every 2^{n-1} rows.

WORKED EXAMPLE

Make the truth table of a 3-input AND: $Y = A \cdot B \cdot C$.

Step 1: $n = 3$, so rows = $2^3 = 8$.

Step 2: List the 8 combinations in binary order.

Step 3: AND gives 1 only when all inputs are 1, so only row 111 gives 1.

A	B	C	$Y = A \cdot B \cdot C$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

COMMON MISTAKE

Do not list rows in a random order — you may miss or repeat a combination. Always count in binary from 000 to 111. Also remember rows = 2^n , not $2 \times n$: 3 inputs give 8 rows, not 6.

اردو میں سمجھیں

ٹرو تھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔ اگر ان پٹ n ہوں تو قطاریں (Rows) 2 کی طاقت n ہوں گی؛ دو ان پٹ کے لیے 4 اور تین ان پٹ کے لیے 8 قطاریں۔ ان پٹ ہمیشہ بائینری ترتیب میں لکھیں۔

CHECK YOURSELF

1. How many rows does a truth table with 4 inputs have?
2. In a 3-input table, what is the 6th row (counting from row 1 = 000)?

Answers: 1. $2^4 = 16$. 2. 101.

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

LEARN IT

Logic gates are electronic circuits that perform Boolean operations. They take one or more input bits and produce one output bit. They are the **building blocks of all digital systems** — a processor contains billions of them.

The three basic gates match the three basic operations: the **AND gate**, the **OR gate** and the **NOT gate** (also called an **inverter**).



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

AND gate — $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR gate — $Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate — $Y = A'$

A	Y
0	1
1	0

WORKED EXAMPLE

A school bell circuit. The bell (Y) should ring when the timer says "period over" ($A = 1$) **and** the school is open ($B = 1$).

Step 1: The bell needs both conditions, so use an AND gate: $Y = A \cdot B$.

Step 2: Sunday, timer fires: $A = 1, B = 0 \rightarrow Y = 1 \cdot 0 = 0$. No bell.

Step 3: Monday, timer fires: $A = 1, B = 1 \rightarrow Y = 1$. The bell rings.

If instead the bell should ring when *either* the timer fires *or* the principal presses a button, you would use an OR gate.

REMEMBER IT

Shortcut: AND output is 0 if *any* input is 0. OR output is 1 if *any* input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

COMMON MISTAKE

Do not mix up the AND and OR symbols. AND has a **flat** input side (like the letter D). OR has a **curved** input side and a **pointed** output, like a shield or arrowhead.

اروین سمجھیں

لاجک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔ AND گیٹ صرف تب 1 دیتا ہے جب دونوں ان پٹ 1 ہوں، OR گیٹ صرف تب 0 دیتا ہے جب دونوں ان پٹ 0 ہوں، اور NOT گیٹ ان پٹ کو الٹ دیتا ہے۔

EXAM TIP

You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y. Write the full truth table in binary order. A frequent MCQ: "Which gate gives 1 only when all inputs are 1?" (AND).

CHECK YOURSELF

1. What is the output of an OR gate when $A = 0$ and $B = 0$?
2. What is another name for the NOT gate?
3. For which input does a 2-input AND gate output 1?

Answers: 1. 0. 2. Inverter. 3. Only $A = 1, B = 1$.

1.1.7 Universal Logic Gates (NAND and NOR)

LEARN IT

NAND means "NOT AND": it is an AND gate followed by a NOT. **NOR** means "NOT OR": an OR gate followed by a NOT. On the symbol, the NOT is shown as a small bubble at the output.

NAND and NOR are called **universal gates** because **any logic circuit in the world can be built using only NAND gates (or only NOR gates)**. Factories like this: they can mass-produce one type of chip and build everything from it.



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

NAND (NOT AND) — $Y = (A \cdot B)'$

A	B	$A \cdot B$	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOR (NOT OR) — $Y = (A + B)'$

A	B	$A + B$	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

WORKED EXAMPLE

Show that a NAND gate can work as a NOT gate.

Step 1: Connect the same input A to both inputs of the NAND gate. Then $Y = (A \cdot A)'$.

Step 2: Since $A \cdot A = A$, we get $Y = A'$.

Step 3: Check: $A = 0 \rightarrow (0 \cdot 0)' = 0' = 1$. $A = 1 \rightarrow (1 \cdot 1)' = 1' = 0$. This is exactly the NOT truth table.

In the same way, a NOR gate with both inputs tied together gives $(A + A)' = A'$. And a NAND followed by a NAND-as-NOT gives $((A \cdot B)')' = A \cdot B$, an AND gate. This is why they are "universal".

REMEMBER IT

NAND is the exact **opposite** of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

COMMON MISTAKE

Do not forget the bubble. Without the small circle, a NAND symbol becomes an AND symbol and your answer is wrong. Also, $(A \cdot B)'$ is **not** the same as $A' \cdot B'$: for $A = 1$, $B = 0$, $(1 \cdot 0)' = 1$ but $0 \cdot 1 = 0$.

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR۔ انہیں یونیورسل (عالمگیر) گیٹ کہتے ہیں کیونکہ صرف NAND یا صرف NOR گیٹس سے دنیا کا کوئی بھی لاجک سرکٹ بنایا جا سکتا ہے۔ NAND صرف تب 0 دیتا ہے جب دونوں ان پٹ 1 ہوں، اور NOR صرف تب 1 دیتا ہے جب دونوں ان پٹ 0 ہوں۔

CHECK YOURSELF

1. Why are NAND and NOR called universal gates?
2. What is the NOR output for A = 0, B = 1?
3. How do you make a NOT gate from a NAND gate?

Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.

1.1.7 Advanced Logic Gates (XOR and XNOR)

LEARN IT

XOR (Exclusive OR) gives output 1 when the inputs are **different**. Expression: $Y = A \oplus B$. In terms of basic operations, $A \oplus B = A'B + AB'$.

XNOR (Exclusive NOR) is XOR followed by NOT. It gives output 1 when the inputs are the **same**. Expression: $Y = (A \oplus B)' = A'B' + AB$.

The XOR symbol is an OR symbol with an extra curved line at the input side. XNOR adds a bubble at the output.



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

XOR — $Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

XNOR — $Y = (A \oplus B)'$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

All seven gates at a glance

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1

0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

(NOT has one input: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.)

WORKED EXAMPLE

A staircase light. A light has one switch at the bottom (A) and one at the top (B) of a staircase. Flipping either switch should change the light.

Step 1: Start with both switches down: $A = 0$, $B = 0$, light off ($Y = 0$).

Step 2: Flip the bottom switch: $A = 1$, $B = 0 \rightarrow$ light on ($Y = 1$).

Step 3: Walk up and flip the top switch: $A = 1$, $B = 1 \rightarrow$ light off ($Y = 0$).

Step 4: The pattern 0, 1, 1, 0 is the XOR truth table. So the light is $Y = A \oplus B$.

REMEMBER IT

Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the **same**. XNOR is sometimes called the "equality gate".

COMMON MISTAKE

XOR is not the same as OR. They differ only in the last row: for $A = 1$, $B = 1$, OR gives 1 but XOR gives 0.

اروین سمجھیں

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔
XNOR دراصل XOR کے بعد NOT ہے۔ OR اور XOR میں فرق صرف آخری قطار میں ہے جہاں دونوں ان پٹ 1 ہوں۔

CHECK YOURSELF

- What is $1 \oplus 1$?
- Which gate outputs 1 when both inputs are the same?
- Write XOR using AND, OR and NOT.

Answers: 1. 0. 2. XNOR. 3. $A \oplus B = A'B + AB'$.

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

LEARN IT

A **logic expression** (Boolean expression) combines variables with operations, for example $Y = A + B' \cdot C$. To evaluate it correctly you must follow the **order of precedence**, just like BODMAS in maths:

1. Parentheses () $\rightarrow$ 2. NOT (') $\rightarrow$ 3. AND ($\cdot$) $\rightarrow$ 4. OR (+)

WORKED EXAMPLE**Worked example 1: truth table of $Y = A + B' \cdot C$**

Step 1: 3 inputs, so 8 rows.

Step 2: Precedence: first find B' (NOT), then $B' \cdot C$ (AND), then $A + (B' \cdot C)$ (OR). Make a column for each step.

A	B	C	B'	$B' \cdot C$	$Y = A + B' \cdot C$
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

Step 3: $Y = 1$ in rows 1, 4, 5, 6, 7 (decimal). Check one row by hand: $A=0, B=0, C=1 \rightarrow B' = 1, B' \cdot C = 1, Y = 0 + 1 = 1$. ✓

WORKED EXAMPLE**Worked example 2: truth table of $Y = (A + B) \cdot C'$**

Step 1: Parentheses first: find $A + B$. Then NOT: C' . Then AND the two.

A	B	C	$A+B$	C'	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Step 2: $Y = 1$ in rows 2, 4 and 6 (decimal).

Minterms (SOP) and Maxterms (POS)

LEARN IT

Sometimes you are given a truth table and must *find* the expression. There are two standard ways.

Minterm (used in SOP — Sum of Products): a **product (AND)** of all the variables, written for a row whose output is **1**. A variable is **complemented (A')** if its value is 0, and **uncomplemented (A)** if its value is 1. OR-ing all the minterms gives the SOP expression.

Maxterm (used in POS — Product of Sums): a **sum (OR)** of all the variables, written for a row whose output is **0**. A variable is **complemented (A')** if its value is 1, and **uncomplemented (A)** if its value is 0. AND-ing all the maxterms gives the POS expression.

A	B	Output (Y)	Minterm (SOP)	Maxterm (POS)
0	0	0	–	$A + B$
0	1	1	$A'B$	–
1	0	0	–	$A' + B$
1	1	1	AB	–

Final SOP expression: $Y = A'B + AB$. **Final POS expression:** $Y = (A + B)(A' + B)$. Both describe the same table. (Both simplify to $Y = B$ — look at the table: Y always equals B. You will confirm this with a K-map later.)

WORKED EXAMPLE

Worked example 3: write the SOP for a 3-input "majority" circuit. $Y = 1$ when at least two of A, B, C are 1.

Row	A	B	C	Y	Minterm
0	0	0	0	0	–
1	0	0	1	0	–
2	0	1	0	0	–
3	0	1	1	1	$A'BC$
4	1	0	0	0	–
5	1	0	1	1	$AB'C$
6	1	1	0	1	ABC'
7	1	1	1	1	ABC

Step 1: Pick the rows with $Y = 1$: rows 3, 5, 6, 7.

Step 2: Row 3 is $A=0, B=1, C=1 \rightarrow A$ is 0 so write A' ; B and C are 1 so write B, C $\rightarrow A'BC$.

Step 3: Do the same for the other rows and OR them: **$Y = A'BC + AB'C + ABC' + ABC$** . In short form: $Y = \Sigma m(3, 5, 6, 7)$. We will simplify this with a K-map in 1.1.13.

REMEMBER IT

SOP = 1s, POS = 0s. Minterm: 0 $\rightarrow$ bar, 1 $\rightarrow$ plain. Maxterm: the reverse (1 $\rightarrow$ bar, 0 $\rightarrow$ plain). Think "Min likes 1s, Max likes 0s".

COMMON MISTAKE

Two classic errors: (1) writing minterms for the 0-rows — minterms come only from rows where $Y = 1$; (2) forgetting precedence — in $A + B \cdot C$ you must do $B \cdot C$ first. $A + B \cdot C$ is **not** $(A + B) \cdot C$: for $A=1, B=0, C=0$ the first gives 1 and the second gives 0.

اروین سمجھیں

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR۔ منٹرم (Minterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 1 ہو؛ اس میں 0 والے متغیر پر بار (A) لگتی ہے۔ میکس ٹرم (Maxterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 0 ہو؛ اس میں 1 والے متغیر پر بار لگتی ہے۔

CHECK YOURSELF

1. Which comes first: AND or OR?
2. Write the minterm for $A = 1, B = 0, C = 1$.
3. Write the maxterm for $A = 0, B = 1$.

Answers: 1. AND. 2. $AB'C$. 3. $A + B'$.

1.1.11 Logic Diagrams ★ Golden

LEARN IT

A **logic diagram** shows how logic gates are connected to create a circuit. It is a "picture" of a Boolean expression. To draw one, use the order of precedence: the operation with the **highest priority is drawn first** (nearest the inputs), and the last operation is the gate that gives the final output.

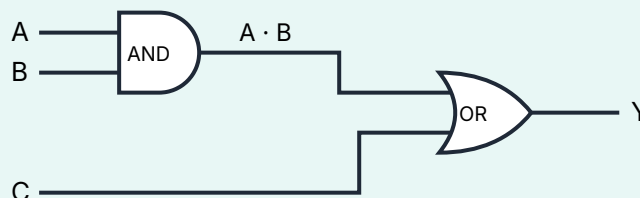
WORKED EXAMPLE

Example 1: Draw the logic diagram for $Y = A \cdot B + C$.

Step 1: Find the operations: AND ($\cdot$) and OR ($+$).

Step 2: AND has higher precedence than OR, so draw the AND gate first, with inputs A and B. Its output is $A \cdot B$.

Step 3: Draw an OR gate. Feed it $A \cdot B$ and C. Its output is Y.



$Y = A \cdot B + C$: AND first, then OR.

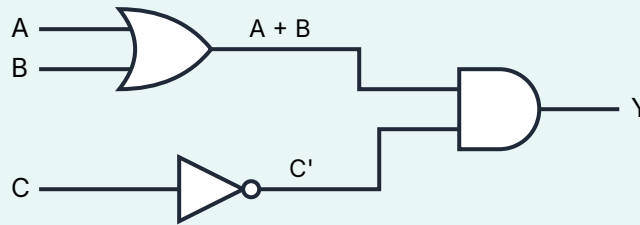
WORKED EXAMPLE

Example 2: Draw the logic diagram for $Y = (A + B) \cdot C'$.

Step 1: Parentheses have the highest priority, so draw the OR gate for $A + B$ first.

Step 2: NOT is next: draw a NOT gate on input C to get C' .

Step 3: The last operation is AND: feed $(A + B)$ and C' into an AND gate to get Y .



$Y = (A + B) \cdot C'$: OR and NOT first, then AND.

You can check this circuit against the truth table in 1.1.8 (worked example 2): $Y = 1$ only for rows 010, 100 and 110.

REMEMBER IT

Read the expression and ask "What is done **last**?" That gate goes on the far right, next to Y . Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2 gates; $Y = (A+B) \cdot C'$ needs 3.

COMMON MISTAKE

Do not draw a separate wire for a variable each time it appears and then leave one unconnected. Every gate input must be joined to an input variable or to another gate's output, and every wire should be clearly labelled in exams.

اردو میں سمجھیں

لاجک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔ ترجیحی ترتیب کے مطابق جو آپریشن پہلے ہوتا ہے اس کا گیٹ پہلے (ان پٹ کے قریب) بنایا جاتا ہے، اور آخری آپریشن کا گیٹ آؤٹ پٹ Y دیتا ہے۔ مثلاً $Y = A \cdot B + C$ میں پہلے AND گیٹ اور پھر OR گیٹ بنتا ہے۔

EXAM TIP

Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols with labelled intermediate outputs (like $A \cdot B$), and finish with the truth table. Marks are lost for wrong gate shapes and missing bubbles.

CHECK YOURSELF

1. In $Y = A + B \cdot C$, which gate is drawn first?
2. How many gates are needed for $Y = A' \cdot B$?

Answers: 1. The AND gate (for $B \cdot C$). 2. Two: one NOT gate (for A') and one AND gate.

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

LEARN IT

A **Karnaugh map (K-map)** is a graphical tool used to **simplify Boolean expressions**. It is a grid of cells — one cell for each row of the truth table. The cells are arranged so that **neighbouring (adjacent) cells differ by only ONE bit**. This ordering is called **Gray code**.

You write the output (1 or 0) of each row into its cell, then **group the 1s** in rectangles of size 1, 2, 4 or 8 (powers of 2). Each group becomes one simple term. This simplifies the equation without long algebra.

Two-variable K-map (A, B)

2 variables need $2^2 = 4$ cells.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Three-variable K-map (A, B, C)

3 variables need $2^3 = 8$ cells. The columns follow the Gray code sequence **00, 01, 11, 10** (not 00, 01, 10, 11). This makes adjacent cells differ by only one variable.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	m_0 (A'B'C')	m_1 (A'B'C)	m_3 (A'BC)	m_2 (A'BC')
A = 1	m_4 (AB'C')	m_5 (AB'C)	m_7 (ABC)	m_6 (ABC')

LEARN IT

Grouping rules

1. Group only 1s. Groups must be rectangles of size 1, 2, 4 or 8 — never 3, 5 or 6.
2. Make groups **as large as possible**. Bigger groups give simpler terms.
3. Use **as few groups as possible**, but every 1 must be in at least one group.
4. **Overlapping is allowed** — a 1 may be used in more than one group.
5. **Wrapping around the edges is allowed**: a cell in the far-left column is adjacent to the cell in the far-right column (BC = 00 is next to BC = 10).
6. To read a group: variables that **change** (0 to 1) inside the group are **eliminated**. Variables that stay the same are kept — plain if they stay 1, complemented if they stay 0.

Group size tells you how many variables disappear: in a 3-variable map, a group of 2 leaves 2 variables, a group of 4 leaves 1 variable, and a group of 8 means $Y = 1$.

WORKED EXAMPLE

K-map example 1 (2 variables): simplify $Y = A'B + AB$.

Step 1: The minterms are $A'B$ (cell 1) and AB (cell 3). Put 1 in those cells, 0 elsewhere.

	B = 0	B = 1
A = 0	0	1
A = 1	0	1

Step 2: The two 1s are in the same column, one above the other. Make one group of 2.

Step 3: Inside the group, A changes (0 then 1) → eliminate A. B stays 1 → keep B.

Result: $Y = B$. This agrees with the algebra in 1.1.4 and the minterm table in 1.1.8.

WORKED EXAMPLE

K-map example 2 (2 variables, overlapping): simplify $Y = A'B' + A'B + AB$.

Step 1: Put 1s in cells 0, 1 and 3.

	B = 0	B = 1
A = 0	1	1
A = 1	0	1

Step 2: Group 1 = the top row (cells 0 and 1). A stays 0, B changes → term A' .

Step 3: Group 2 = the right column (cells 1 and 3). A changes, B stays 1 → term B . Cell 1 is used twice — overlapping is allowed.

Result: $Y = A' + B$. Check: $A' + B$ is 0 only when $A = 1$ and $B = 0$, which is exactly the one 0 cell (cell 2). ✓

WORKED EXAMPLE

K-map example 3 (3 variables): simplify the majority function $Y = \Sigma m(3, 5, 6, 7) = A'BC + AB'C + ABC' + ABC$.

Step 1: Fill the map. Careful — the columns are in Gray order 00, 01, 11, 10, so m_3 is in the third column and m_2 in the fourth.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	0	1 (m_3)	0
A = 1	0	1 (m_5)	1 (m_7)	1 (m_6)

Step 2: No group of 4 is possible (there is no 2×2 block or full row of 1s). So make groups of 2, each using m_7 :

- Group 1: $m_3 + m_7$ (column BC = 11). A changes; B = 1, C = 1 stay $\rightarrow$ **BC**.
- Group 2: $m_5 + m_7$ (row A = 1, columns 01 and 11). B changes; A = 1, C = 1 stay $\rightarrow$ **AC**.
- Group 3: $m_7 + m_6$ (row A = 1, columns 11 and 10). C changes; A = 1, B = 1 stay $\rightarrow$ **AB**.

Result: $Y = AB + BC + AC$. Four 3-variable terms (needing 4 three-input AND gates) became three 2-variable terms.

Check one row: A=1, B=0, C=1 $\rightarrow$ AB = 0, BC = 0, AC = 1 $\rightarrow$ Y = 1. In the truth table, row 5 is 1. ✓

WORKED EXAMPLE

K-map example 4 (3 variables, wrap-around): simplify $Y = \Sigma m(0, 2, 4, 6)$.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	1	0	0	1
A = 1	1	0	0	1

Step 1: The 1s are in the far-left column (BC = 00) and the far-right column (BC = 10).

Step 2: Because the map wraps around, these two columns are adjacent. Make **one group of 4**: m_0, m_4, m_2, m_6 .

Step 3: In this group A changes (0, 1) and B changes (0, 1). Only C stays the same: C = 0 $\rightarrow$ term **C'**.

Result: $Y = C'$. Without wrapping you would get two groups ($B'C' + BC'$), which is correct but not the simplest.

WORKED EXAMPLE

K-map example 5 (3 variables): simplify $Y = \Sigma m(1, 4, 5, 6, 7)$ — this is the output of $Y = A + B'C$ from 1.1.8.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	0	0
A = 1	1	1	1	1

Step 1: Group of 4: the whole row $A = 1$ (m_4, m_5, m_7, m_6). B and C both change; A stays 1 → **A**.

Step 2: m_1 is still alone. Pair it with m_5 below it (column BC = 01). A changes; B = 0, C = 1 stay → **B'C**.

Result: $Y = A + B'C$ — the K-map confirms the original expression was already in its simplest form.

WORKED EXAMPLE

Going further — a 4-variable K-map: simplify $Y = \Sigma m(0, 2, 5, 7, 8, 10, 13, 15)$. (Not required by the slides, but a good test of the same rules.) 4 variables need 16 cells; both rows (AB) and columns (CD) use Gray order 00, 01, 11, 10.

	CD = 00	CD = 01	CD = 11	CD = 10
AB = 00	1 (m_0)	0	0	1 (m_2)
AB = 01	0	1 (m_5)	1 (m_7)	0
AB = 11	0	1 (m_{13})	1 (m_{15})	0
AB = 10	1 (m_8)	0	0	1 (m_{10})

Step 1: The centre 2×2 block (m_5, m_7, m_{13}, m_{15}) is a group of 4. A and C change; B = 1 and D = 1 stay → **BD**.

Step 2: The four corners (m_0, m_2, m_8, m_{10}) are adjacent through wrap-around (left-right *and* top-bottom). A and C change; B = 0 and D = 0 stay → **B'D'**.

Result: $Y = BD + B'D'$ (this is B XNOR D). Eight 4-variable minterms became two 2-variable terms.

REMEMBER IT

Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges touch. For reading groups: "**If it changes, it goes.**"

COMMON MISTAKE

Most common K-map mistakes: (1) writing the columns as 00, 01, 10, 11 — this puts m_2 and m_3 in the wrong places; (2) making groups of 3 or 6 — only 1, 2, 4, 8 are allowed; (3) grouping diagonally — groups must be rectangles of side-by-side cells; (4) stopping at small groups when a bigger group (or a wrap-around group) is possible.

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔ اس کے ساتھ والے خانوں میں صرف ایک ہٹ کا فرق ہوتا ہے (گرے کوڈ: 00، 01، 11، 10)۔ 1 والے خانوں کو 1، 2، 4 یا 8 کے گروپوں میں جتنا بڑا ممکن ہو جوڑیں؛ گروپ ایک دوسرے پر چڑھ سکتے ہیں اور کناروں سے گھوم کر بھی بن سکتے ہیں۔ گروپ کے اندر جو متغیر بدلتا ہے وہ ختم ہو جاتا ہے۔

EXAM TIP

K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s too), circle the groups clearly, write the term for each group beside it, and give the final simplified expression. Examiners often add "draw the logic diagram of the result" — so practise drawing it too.

CHECK YOURSELF

1. In a 3-variable K-map, which column is next to $BC = 01$ on the right?
2. Simplify $Y = \sum m(2, 3)$ for two variables A, B.
3. Simplify $Y = \sum m(4, 5, 6, 7)$ for three variables.

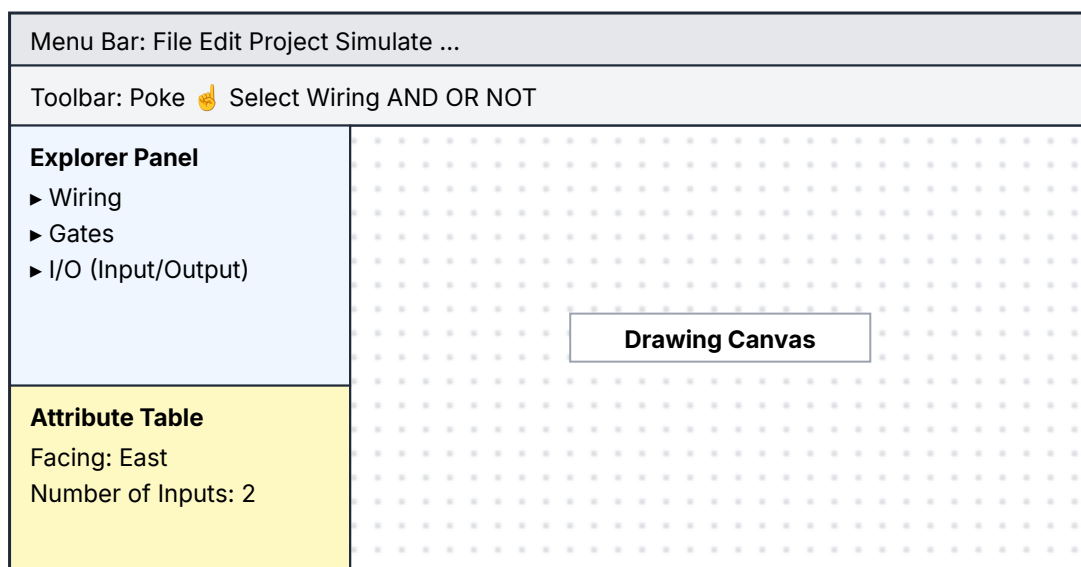
Answers: 1. $BC = 11$. 2. Cells 2 and 3 are the row $A = 1$; B changes $\rightarrow Y = A$. 3. The whole row $A = 1$ is a group of 4 $\rightarrow Y = A$.

LogiSim Evolution: Simulation Tool

LEARN IT

LogiSim Evolution is a free graphical tool for **designing and simulating logic circuits**. "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change. It lets you see the truth tables and Boolean expressions you learn in theory.

1. The interface areas



The main areas of the LogiSim Evolution window.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.

Wiring colours during simulation

Colour	Meaning
Bright green	Logic 1 (HIGH)
Dark green	Logic 0 (LOW)
Blue	Unknown / unconnected pin
Red	Error / short circuit

REMEMBER IT

Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is wrong".

COMMON MISTAKE

If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin. Zoom in and make sure every wire end sits exactly on a connection point.

ارو میں سمجھیں

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔ اس میں مینو بار، ٹول بار، ایکسپلورر پینل، ایٹری بیوٹ ٹیبل اور ڈرائنگ کینوس ہوتے ہیں۔ چمکدار سبز تار 1، گہری سبز 0، نیلی تار غیر منسلک اور سرخ تار غلطی کو ظاہر کرتی ہے۔

CHECK YOURSELF

1. Where do you change the number of inputs of a gate?
2. What does a red wire mean?

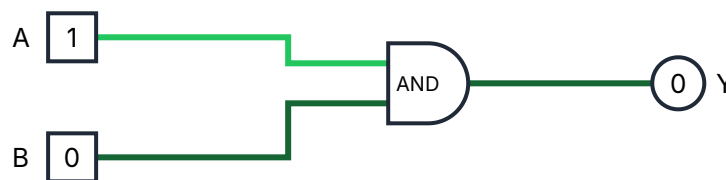
Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.

Building a Circuit in LogiSim

WORKED EXAMPLE

Step by step: building $Y = A \cdot B$

1. Select the **AND gate** from the Toolbar (or from the Gates library) and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate. Label them A and B.
3. Select the **Output Pin** (circle icon with a 0) and place it to the right. Label it Y.
4. Use the **Wiring Tool** to click and drag wires from the input pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the output pin Y.
6. Select the **Poke Tool** (pointing finger) and click inputs A and B to toggle them between 0 and 1. Test the circuit!



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

WORKED EXAMPLE

Testing the circuit = making its truth table. Poke the inputs through all four combinations and write what Y shows:

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green

This matches the AND truth table, so the circuit is correct. Try the same with $Y = A \cdot B + C$ from 1.1.11 (add a second gate and a third input pin).

REMEMBER IT

Input pin = **square** (you push it). Output pin = **circle** (it shows a light). Poke = finger — you "poke" inputs to change them.

COMMON MISTAKE

Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it. Switch to the Poke Tool first.

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائرنگ ٹول سے تاریں جوڑیں۔ آخر میں پوک ٹول (انگلی) سے ان پٹ کو 0 اور 1 کر کے سرکٹ کو جانچیں۔

CHECK YOURSELF

1. Which icon is the input pin: square or circle?
2. Which tool do you use to change an input between 0 and 1?

Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

LEARN IT

The **Software Development Life Cycle (SDLC)** is a structured process that guides software development from beginning to end. It ensures **clear goals, better planning, risk reduction and high quality**. Without it, teams forget features, go over budget and deliver buggy software.

SDLC has **6 phases**:



Phases 1–3 in detail

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Phases 4–6 in detail

Phase	Activities and details	Deliverable
4. Testing	Find and fix bugs (errors), using Black Box and White Box testing.	Test Reports, bug-free software
5. Deployment	Release the software to end-users: Direct, Phased, or Pilot release.	Live Software

6. Maintenance	Provide support and updates, and fix post-release issues.	Patches, Updated Versions
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Deployment methods: In a **direct** release the old system is switched off and the new one switched on at once. In a **phased** release the new system is introduced part by part. In a **pilot** release one small group (for example one branch or one class) uses it first before everyone else.

Testing methods

Black Box Testing

- The tester does **not** see the code.
- Tests functionality: give inputs, check outputs.
- Done by dedicated testers.

White Box Testing

- The tester **can** see the code.
- Tests the internal logic and structure.
- Done by developers.

WORKED EXAMPLE

Building a house = SDLC.

Phase 1 (Requirements): you ask the family what rooms they need — 3 bedrooms, a kitchen, a garage.

Phase 2 (Design): the architect draws the blueprints.

Phase 3 (Implementation): the builders lay the bricks.

Phase 4 (Testing): an inspector checks the wiring and the plumbing.

Phase 5 (Deployment): the family moves in.

Phase 6 (Maintenance): repairs, painting and a new room added later.

WORKED EXAMPLE

Black box vs white box on a login screen. A black-box tester types a correct password and a wrong password and checks that the first is accepted and the second rejected — without looking at the code. A white-box tester (the developer) reads the code and checks that every `if` branch, for example the "account locked after 3 tries" branch, actually runs correctly.

REMEMBER IT

Remember the order with "**R**ead **D**evelopers **I**mplement, **T**est, **D**eploy, **M**aintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass box (you can see inside).

COMMON MISTAKE

Do not say the SDLC ends at deployment. Maintenance is a full phase and often lasts the longest. Also, the SRS is the deliverable of Requirement Analysis, not of Design.

اردو میں سمجھیں

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ٹیسٹنگ، تنصیب (Deployment) اور دیکھ بھال (Maintenance)۔ پہلا مرحلہ سب سے اہم ہے اور اس سے SRS دستاویز بنتی ہے۔ بلیک باکس ٹیسٹنگ میں کوڈ نظر نہیں آتا جبکہ وائٹ باکس ٹیسٹنگ میں کوڈ دیکھ کر اندرونی منطق جانچی جاتی ہے۔

EXAM TIP

"Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of each (SRS, Design Document, Source Code, Test Reports, Live Software, Patches). A short question often asks you to differentiate black box and white box testing — answer in a 3-row table (sees code?, what is tested?, who does it?).

CHECK YOURSELF

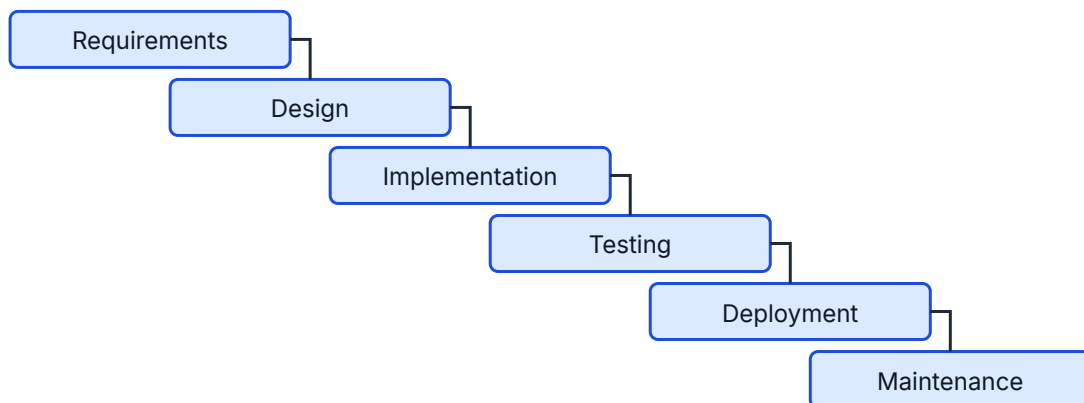
1. Which phase produces the SRS document?
2. Who usually performs white box testing?
3. Name the three ways to deploy software.

Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.

1.2.4 The Waterfall Model ★ Golden

LEARN IT

The **Waterfall model** is a **linear, sequential** approach to the SDLC. Each phase must be completed before the next one begins, like water falling down steps. There is no easy way to go back to a previous phase.



The Waterfall model: each phase "falls" into the next, in one direction only.

When to use it

- Requirements are clear, fixed and well understood.
- The project is small to medium in size.
- The technology is stable (not new or experimental).

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.

WORKED EXAMPLE

Case study: Al-Noor Library Management System. The textbook uses this system as a Waterfall case study because the requirements were **100% fixed before coding began**.

Step 1: The librarian listed exactly what was needed: add books, issue books, return books, calculate fines. Nothing would change.

Step 2: Because the list was fixed, the team could finish the full design, then all the coding, then all the testing, in order.

Step 3: Every phase produced a document, so the library has a complete record for future maintenance.

REMEMBER IT

Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

COMMON MISTAKE

Do not choose Waterfall for a project where the customer is still unsure what they want. A late change means going back through design, coding and testing, which is slow and costly.

اردو میں سمجھیں

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔ یہ تب بہتر ہے جب ضروریات واضح اور طے شدہ ہوں، جیسے النور لائبریری مینجمنٹ سسٹم میں۔ یہ سمجھنے میں آسان ہے مگر تبدیلیوں کے لیے لچکدار نہیں اور ٹیسٹنگ دیر سے ہوتی ہے۔

EXAM TIP

Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to use" conditions, and give at least two pros and two cons. Mention the Al-Noor Library case study as your example.

CHECK YOURSELF

1. Why is the Waterfall model called "sequential"?
2. Why did the Al-Noor Library system use Waterfall?

Answers: 1. Because each phase must be completed before the next begins, in a fixed order. 2. Because its requirements were 100% fixed before coding began.

1.2.4 The Agile Model ★ Golden

LEARN IT

The **Agile model** is an **iterative** approach: the software is built in small, repeated cycles called **sprints** (usually 2–4 weeks long). Each sprint plans, builds and tests a small part of the software, and then shows it to the customer. **Customer feedback is gathered continuously** and used in the next sprint.



When to use it

- Requirements are unclear or may change.
- The customer wants frequent progress updates (early delivery).
- The project is complex or innovative.

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

WORKED EXAMPLE

Case study: Al-Noor Student Mobile Application. This app used Agile because teachers, parents and students **constantly requested new features** during development.

Sprint 1 (2 weeks): build login and the timetable screen. Show it to users.

Feedback: parents ask, "Can we see attendance too?"

Sprint 2: add attendance. Teachers now ask for homework notifications.

Sprint 3: add notifications. Each sprint delivers a working app, and new ideas are welcomed instead of causing a restart.

REMEMBER IT

Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

COMMON MISTAKE

"Less documentation" does not mean "no documentation" or "no planning". Agile teams still plan every sprint; they simply value working software over long documents.

اروین سمجھیں

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔ ہر اسپرینٹ کے بعد گاہک کی رائے لی جاتی ہے۔ یہ تبدیلیوں کے لیے بہت لچکدار ہے اور جلد کام کرنے والا سافٹ ویئر دیتا ہے، مگر اس میں دستاویزات کم ہوتی ہیں اور وقت و لاگت کا اندازہ مشکل ہوتا ہے۔ انور اسٹوڈنٹ موبائل ایپ اس کی مثال ہے۔

EXAM TIP

Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous customer feedback". Use the Al-Noor Student Mobile App as your case study.

CHECK YOURSELF

1. What is a sprint and how long does it usually last?
2. Give one disadvantage of Agile.

Answers: 1. A short development cycle that delivers a working part of the software; usually 2–4 weeks. 2. Less documentation, or unpredictable end-time and cost.

1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

LEARN IT

Both models follow the same SDLC phases. The difference is **how** they move through them: Waterfall goes through each phase **once**, in a straight line; Agile goes through all phases **many times**, in small loops.

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

WORKED EXAMPLE

Choosing a model — think like the textbook.

Case A: Al-Noor Library Management System. Requirements fixed 100% before coding → no changes expected → **Waterfall**.

Case B: Al-Noor Student Mobile Application. Teachers, parents and students keep asking for features → requirements change → **Agile**.

Practice case: A bank wants a salary-calculation program for a fixed government formula that will not change. Requirements are clear and fixed, the project is medium-sized, and the technology is stable → **Waterfall**.

Practice case: A start-up wants a new food-delivery app and is not sure which features customers will like. Requirements are unclear and the project is innovative → **Agile**.

REMEMBER IT

One question decides it: **"Will the requirements change?"** No → Waterfall. Yes / not sure → Agile.

COMMON MISTAKE

Do not say one model is always "better". Each suits a different situation. In exams, justify your choice using the project's features (fixed vs changing requirements, size, customer involvement).

اردو میں سمجھیں

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔ ایجائل ماڈل لچکدار ہے، گاہک ہر اسپرنٹ کے بعد شامل ہوتا ہے، سافٹ ویئر بار بار حصوں میں ملتا ہے اور ٹیسٹنگ مسلسل ہوتی ہے۔ اگر ضروریات طے شدہ ہوں تو وائرفال اور اگر بدلتی رہیں تو ایجائل منتخب کریں۔

CHECK YOURSELF

1. In which model is the customer involved only at the start and the end?
2. Which model delivers software in frequent, partial deliveries?
3. A school wants an exam-result system with rules that never change. Which model?

Answers: 1. Waterfall. 2. Agile. 3. Waterfall (fixed, clear requirements).

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model

★ Golden

LEARN IT

A **communication model** explains how data travels from one device to another through a network. Because sending data is complicated, the job is split into **layers**; each layer does one part of the job and passes the data to the next layer.

The **OSI (Open Systems Interconnection) model** was developed by **ISO** (International Organization for Standardization) in **1984**. It divides communication into **7 layers**. When you send data, it goes **down** the layers on your device (7 → 1), travels as bits over the cable or air, and goes **up** the layers on the receiving device (1 → 7).



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

REMEMBER IT

Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top to bottom you can use "All People Seem To Need Data Processing".)

OSI layers — top layers (7 to 4)

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)

OSI layers — bottom layers (3 to 1)

Layer	Function	Protocols / hardware
3. Network	Logical addressing (IP addresses) and routing packets across networks.	IP, ICMP (Ping), Routers
2. Data Link	Physical addressing (MAC addresses) and error detection in frames.	Ethernet, MAC, Switches
1. Physical	Transmits raw bits (0, 1) over cables or wireless media.	Cables, Hubs, Wi-Fi radio

TCP vs UDP at the Transport layer

	TCP	UDP
Main feature	Reliable — checks that all data arrives correctly and in order.	Fast — sends data without waiting for checks.
Use when	Accuracy matters.	Speed matters more than perfection.
Examples	Web browsing, emails	Video streaming, online gaming

WORKED EXAMPLE

Tracing an email through the OSI layers (sender side).

Layer 7 (Application): you press Send; your email program uses **SMTP**.

Layer 6 (Presentation): the text is encoded (ASCII) and may be encrypted with TLS.

Layer 5 (Session): a session is opened with the mail server.

Layer 4 (Transport): **TCP** breaks the data into segments and numbers them, so nothing is lost (email needs accuracy).

Layer 3 (Network): each packet gets the **IP address** of the destination; **routers** choose the path.

Layer 2 (Data Link): packets are put in **frames** with **MAC addresses**; a **switch** forwards them on the local network.

Layer 1 (Physical): the frames are sent as raw **bits** over the cable or Wi-Fi radio.

At the receiver, the same steps happen in reverse order (1 → 7).

COMMON MISTAKE

Do not mix up the two addresses: **IP address** = logical address = **Network layer (3)**; **MAC address** = physical address = **Data Link layer (2)**. Also, routers work at layer 3, switches at layer 2 and hubs at layer 1.

اردو میں سمجھیں

کمپیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔ او ایس آئی (OSI) ماڈل 1984 میں آئی ایس او (ISO) نے بنایا اور اس کی سات تہیں (Layers) ہیں: فزیکل، ڈیٹا لنک، نیٹ ورک، ٹرانسپورٹ، سیشن، پریزنٹیشن اور ایپلی کیشن۔ ہر تہ کا الگ کام ہے، جیسے نیٹ ورک تہ IP ایڈریس اور روٹنگ سنبھالتی ہے۔ TCP درستگی کے لیے اور UDP رفتار کے لیے استعمال ہوتا ہے۔

EXAM TIP

A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two protocols/devices for each layer. MCQs often ask: "Which layer handles routing?" (Network), "Which layer does encryption?" (Presentation), "Who developed OSI and when?" (ISO, 1984), "Which protocol is used for video streaming?" (UDP).

CHECK YOURSELF

1. Which OSI layer is responsible for encryption and compression?
2. At which layer do switches and MAC addresses work?
3. Should online gaming use TCP or UDP? Why?

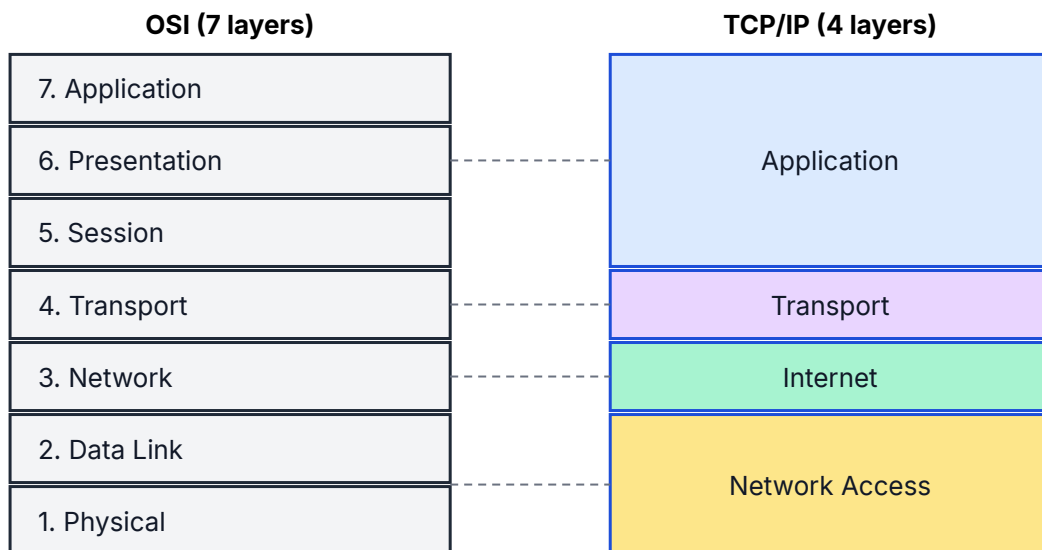
Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

LEARN IT

The OSI model is mainly a **reference** model used for teaching and design. The **TCP/IP model** is the **practical model used on the real Internet**. It condenses the 7 OSI layers into just **4 layers**.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Point	OSI model	TCP/IP model
Number of layers	7	4
Use	Reference / teaching model	Practical model used on the real Internet
Top layers	Separate Application, Presentation and Session layers	All three combined into one Application layer

Bottom layers	Separate Data Link and Physical layers	Combined into one Network Access layer
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WORKED EXAMPLE

Where does each protocol/device go in TCP/IP?

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

REMEMBER IT

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total $3+1+1+2 = 7$.

COMMON MISTAKE

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

اردو میں سمجھیں

ٹی سی پی/آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

CHECK YOURSELF

1. How many layers does the TCP/IP model have?
2. Which TCP/IP layer corresponds to OSI layers 2 and 1?
3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

Chapter 1 Review

Chapter summary

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (') inverts its single input (unary).
- A truth table lists all 2^n input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Key terms

Term	Meaning	اردو
Discrete quantity	A countable quantity with separate, fixed values.	منفصل مقدار
Continuous quantity	A quantity that can take any value within a range.	مسلسل مقدار
Bit	A binary digit, 0 or 1.	بائتری ہندسہ
Digital system	An electronic system that uses only the values 0 and 1.	ڈیجیٹل نظام
Analog signal	A continuous signal that changes smoothly, like a sine wave.	اینالوگ سگنل (مسلسل اشارہ)

Digital signal	A signal with only two levels, HIGH and LOW (square wave).	ڈیجیٹل سگنل
Noise	Unwanted interference that disturbs a signal.	شور (مداخلت)
Boolean algebra	Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole.	بولین الجبرا
Truth table	A table of all input combinations and their outputs.	ٹروٹھ ٹیبل (صداقت جدول)
Logic gate	An electronic circuit that performs a Boolean operation.	منطقی گیٹ
Universal gate	A gate (NAND or NOR) from which any circuit can be built.	عالمگیر گیٹ
Unary operation	An operation with only one operand, e.g. NOT.	یک عددی عمل
Minterm	A product (AND) of all variables for a row with output 1.	منٹرم
Maxterm	A sum (OR) of all variables for a row with output 0.	میکس ٹرم
SOP	Sum of Products: minterms joined with OR.	حاصل ضرب کا مجموعہ
POS	Product of Sums: maxterms joined with AND.	مجموعوں کا حاصل ضرب
Karnaugh map (K-map)	A grid used to simplify Boolean expressions by grouping 1s.	کارنو نقشہ
Gray code	An ordering where neighbouring values differ by only one bit.	گرے کوڈ
Simulation	Running a computer model of a circuit to test it.	نقلی عمل (سمیولیشن)
SDLC	Software Development Life Cycle: the 6-phase process for building software.	سافٹ ویئر کی تیاری کا دورِ حیات
SRS	Software Requirements Specification: document listing what the software must do.	سافٹ ویئر ضروریات کی دستاویز
Black box testing	Testing inputs and outputs without seeing the code.	بلیک باکس جانچ
White box testing	Testing the internal logic while seeing the code.	وائٹ باکس جانچ
Waterfall model	A linear SDLC model; each phase is finished before the next.	آبشار ماڈل

Agile model	An iterative SDLC model using short sprints and constant feedback.	ایجائل (لچکدار) ماڈل
Sprint	A short development cycle in Agile, usually 2–4 weeks.	اسپرینٹ (مختصر ترقیاتی دور)
Protocol	A set of rules for communication, e.g. HTTP, TCP.	پروٹوکول (ضوابطِ ابلاغ)
OSI model	ISO's 7-layer reference model of network communication (1984).	او ایس آئی ماڈل
TCP/IP model	The practical 4-layer model used on the Internet.	ٹی سی پی/آئی پی ماڈل

Practice: multiple choice

- Which of the following is a continuous quantity?
 - Number of students in a class
 - Number of cars in a parking lot
 - Temperature of water
 - Number of books on a shelf
- In a digital system, the value 1 represents:
 - OFF
 - LOW
 - FALSE
 - ON / HIGH / TRUE
- A square wave represents a:
 - Digital signal
 - Analog signal
 - Sine wave
 - Noise signal
- Boolean algebra was developed by:
 - Charles Babbage
 - George Boole
 - Alan Turing
 - Maurice Karnaugh
- How many rows does a truth table with 4 inputs have?
 - 4
 - 8
 - 16
 - 32
- Which gate gives output 1 only when all inputs are 1?
 - OR
 - AND
 - NOR
 - XOR
- Which pair of gates are called universal gates?
 - AND and OR
 - XOR and XNOR
 - NOT and AND
 - NAND and NOR
- What is the output of a NOR gate when $A = 0$ and $B = 0$?
 - 1
 - 0
 - Unknown
 - Depends on the gate size

9. An XOR gate gives output 1 when:
- Both inputs are 1
 - Both inputs are 0
 - The inputs are different
 - The inputs are the same
10. After parentheses, which operation has the highest precedence?
- OR
 - NOT
 - AND
 - XOR
11. The minterm for the row $A = 1, B = 0$ is:
- AB'
 - $A'B$
 - $A + B'$
 - $A' + B$
12. The columns of a 3-variable K-map (BC) are written in the order:
- 00, 01, 10, 11
 - 00, 10, 01, 11
 - 11, 10, 01, 00
 - 00, 01, 11, 10
13. In black box testing, the tester:
- Reads the source code line by line
 - Does not see the code and tests inputs/outputs
 - Is always the developer
 - Tests only the internal structure
14. Which OSI layer handles logical (IP) addressing and routing?
- Data Link
 - Transport
 - Network
 - Session
15. The TCP/IP Network Access layer corresponds to which OSI layers?
- 7, 6 and 5
 - 4
 - 3
 - 2 and 1

Practice: short questions

- Define discrete and continuous quantities. Give two examples of each.
- State four advantages of digital systems.
- Differentiate between analog and digital signals (any three points).
- Why is NOT called a unary operation? Write its truth table.
- How many rows are needed in a truth table for 3 inputs? List all the input combinations.
- Why are NAND and NOR called universal gates? Show how a NAND gate can be used as a NOT gate.
- A 2-input truth table has $Y = 1$ only for rows $A=0, B=1$ and $A=1, B=0$. Write the SOP expression and name the gate.
- Name the deliverable of each SDLC phase.
- Differentiate between black box testing and white box testing.
- When should TCP be used and when should UDP be used? Give one example of each.

Practice: long / problem questions

- Draw the symbol, write the Boolean expression and give the truth table for each of these gates: AND, OR, NOT, NAND, NOR, XOR and XNOR.
- A function is given as $F(A, B, C) = \sum m(1, 3, 5, 6, 7)$. (a) Make its truth table. (b) Simplify it using a 3-variable K-map, explaining each group. (c) Draw the logic diagram of the simplified expression.

- For $Y = A \cdot B' + C$: (a) state the order in which the operations are done; (b) make the complete truth table; (c) draw the logic diagram.
- Explain the six phases of the Software Development Life Cycle with the activities and deliverable of each phase. Use the house-building analogy.
- Compare the Waterfall and Agile models under five headings. Explain why the Al-Noor Library Management System used Waterfall and the Al-Noor Student Mobile Application used Agile.
- Explain the seven layers of the OSI model with the function and one protocol/device for each layer. Then show how they map onto the four layers of the TCP/IP model.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. c | 4. b | 7. d | 10. b | 13. b |
| 2. d | 5. c | 8. a | 11. a | 14. c |
| 3. a | 6. b | 9. c | 12. d | 15. d |

Short questions — model answers

- Discrete quantities are separate, countable items with specific values (number of students in a classroom, number of cars in a parking lot). Continuous quantities can take any value within a range and flow without breaks (temperature of water, speed of a moving car).
- Reliable (less affected by noise and interference); accurate (no loss of quality when copying); easy to design (only two states, 0 and 1); programmable (easily controlled by software).
- Nature: analog is a continuous flow; digital has discrete values (bits, HIGH/LOW). Shape: analog is a sine wave; digital is a square wave. Reliability: analog is low (affected by noise); digital is high (resistant to noise). Examples: human voice vs computer data.
- NOT works on only one operand (one input), so it is unary. Truth table: $A = 0 \rightarrow Y = 1$; $A = 1 \rightarrow Y = 0$.
- $2^3 = 8$ rows: 000, 001, 010, 011, 100, 101, 110, 111.
- Any logic circuit can be built using only NAND gates, or only NOR gates. NAND as NOT: join both inputs and feed A: $Y = (A \cdot A)' = A'$. Check: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.
- Minterms: $A'B$ (row 01) and AB' (row 10). $Y = A'B + AB'$. This is the XOR gate, $Y = A \oplus B$.
- Requirement Analysis $\rightarrow$ SRS document; System Design $\rightarrow$ Design Document; Implementation $\rightarrow$ Working Source Code; Testing $\rightarrow$ Test Reports / bug-free software; Deployment $\rightarrow$ Live Software; Maintenance $\rightarrow$ Patches / Updated Versions.
- Black box: tester does not see the code, tests functionality (inputs/outputs), done by dedicated testers. White box: tester can see the code, tests internal logic/structure, done by developers.
- Use TCP when accuracy matters, e.g. web browsing or email. Use UDP when speed matters more than perfection, e.g. video streaming or online gaming.

Long questions — model answers

- Symbols as drawn in 1.1.7 (AND = D shape; OR = curved back, pointed front; NOT = triangle + bubble; NAND = AND + bubble; NOR = OR + bubble; XOR = OR with extra back curve; XNOR = XOR + bubble). Expressions: AND $Y = A \cdot B$; OR $Y = A + B$; NOT $Y = A'$; NAND $Y = (A \cdot B)'$; NOR $Y = (A + B)'$; XOR $Y = A \oplus B$; XNOR $Y = (A \oplus B)'$. Outputs for inputs 00, 01, 10, 11: AND 0,0,0,1; OR 0,1,1,1; NAND 1,1,1,0; NOR 1,0,0,0; XOR 0,1,1,0; XNOR 1,0,0,1. NOT: $0 \rightarrow 1$, $1 \rightarrow 0$.
- (a) Truth table ($ABC \rightarrow F$): 000 $\rightarrow$ 0, 001 $\rightarrow$ 1, 010 $\rightarrow$ 0, 011 $\rightarrow$ 1, 100 $\rightarrow$ 0, 101 $\rightarrow$ 1, 110 $\rightarrow$ 1, 111 $\rightarrow$ 1.
(b) K-map:

	BC = 00	BC = 01	BC = 11	BC = 10
--	---------	---------	---------	---------

A = 0	0	1	1	0
A = 1	0	1	1	1

Group 1 (size 4): the two middle columns, m_1, m_3, m_5, m_7 . A changes and B changes; $C = 1$ stays $\rightarrow C$.

Group 2 (size 2): m_7 and m_6 in row $A = 1$. C changes; $A = 1, B = 1$ stay $\rightarrow AB$ (m_7 is shared — overlapping is allowed). $F = C + AB$. Check: $C + AB$ is 1 for rows 1, 3, 5, 7 ($C = 1$) and 6, 7 ($AB = 1$) = {1, 3, 5, 6, 7}. ✓

(c) Logic diagram: an AND gate with inputs A and B gives AB; an OR gate takes AB and C and gives F.

3. (a) First NOT (B'), then AND ($A \cdot B'$), then OR ($+ C$).

(b) Truth table:

A	B	C	B'	A·B'	Y
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

(c) Logic diagram: input B goes through a NOT gate to give B' . A and B' go into an AND gate to give $A \cdot B'$. $A \cdot B'$ and C go into an OR gate to give Y. (3 gates.)

- Requirement Analysis: meet clients, understand needs, list features; most important phase; deliverable SRS (house: asking the family what rooms they need). System Design: high-level design (architecture), low-level design (database, UI screens), flowcharts; deliverable Design Document (architect draws blueprints). Implementation: write code in Python, Java etc. from the design; deliverable Working Source Code (builders lay bricks). Testing: find and fix bugs using black box and white box testing; deliverable Test Reports / bug-free software (inspector checks the house). Deployment: release to end-users by direct, phased or pilot release; deliverable Live Software (family moves in). Maintenance: support, updates, fixing post-release issues; deliverable Patches / Updated Versions (repairs and extensions).
- Comparison: Flexibility — Waterfall rigid, Agile very flexible. Customer involvement — Waterfall only at start and end, Agile continuous after every sprint. Delivery — Waterfall single delivery at the end, Agile frequent partial deliveries. Testing — Waterfall late in a dedicated phase, Agile continuous in every sprint. Documentation — Waterfall heavy, Agile light. The Al-Noor Library system used Waterfall because its requirements were 100% fixed before coding began, so a simple, well-documented linear process fitted. The Al-Noor Student App used Agile because teachers, parents and students kept requesting new features, so short sprints with continuous feedback were needed.
- OSI (ISO, 1984), top to bottom: 7 Application — network services to user applications (HTTP, HTTPS, FTP, SMTP, DNS); 6 Presentation — translation, encryption, compression (SSL/TLS, JPEG, MP3, ASCII); 5 Session — opens, maintains, closes sessions (NetBIOS, RPC); 4 Transport — reliable end-to-end delivery, error checking, port numbers (TCP, UDP); 3 Network — logical IP addressing and routing (IP, ICMP, routers); 2 Data Link — physical MAC addressing and error detection in frames (Ethernet, switches); 1 Physical — raw bits over cables/wireless (cables, hubs, Wi-Fi radio). Mnemonic bottom to top: "Please Do Not Throw Sausage Pizza Away". TCP/IP mapping: Application = OSI 7, 6, 5; Transport = 4; Internet = 3; Network Access = 2, 1.

Self-assessment: I can...

- ☐ ...tell discrete from continuous quantities and analog from digital signals, and draw both waves.
- ☐ ...evaluate AND, OR and NOT expressions and follow the order of precedence.
- ☐ ...draw all seven gate symbols and write their truth tables from memory.
- ☐ ...write the SOP (minterms) and POS (maxterms) from a truth table.
- ☐ ...simplify expressions with 2- and 3-variable K-maps, including overlap and wrap-around groups.
- ☐ ...explain the six SDLC phases and choose between Waterfall and Agile for a project.
- ☐ ...list four advantages of digital systems.
- ☐ ...build a complete truth table for 2 or 3 inputs without missing a row.
- ☐ ...explain why NAND and NOR are universal gates.
- ☐ ...draw a logic diagram from an expression.
- ☐ ...build and test a simple circuit in LogiSim and read the wire colours.
- ☐ ...name the 7 OSI layers in order, state their functions, and map them to the 4 TCP/IP layers.

Computational Thinking & Algorithm

This chapter teaches you how to think like a computer scientist: break a problem into parts, spot patterns, ignore what does not matter, and write clear step-by-step algorithms. You will then learn four famous algorithms — Bubble Sort, Selection Sort, Linear Search and Binary Search — and how to judge and choose between them.

اس باب میں آپ سیکھیں گے کہ کسی مسئلے کو کمپیوٹر سائنس دان کی طرح کیسے سوچ کر حل کیا جاتا ہے: مسئلے کو چھوٹے حصوں میں توڑنا، پیٹرن پہچاننا، غیر ضروری باتوں کو نظر انداز کرنا اور مرحلہ وار الگورتھم لکھنا۔ اس کے بعد آپ ترتیب دینے (Sorting) اور تلاش کرنے (Searching) کے چار اہم الگورتھم سیکھیں گے۔

What you will learn

- Define computational thinking and name its four parts: decomposition, pattern recognition, abstraction and algorithms.
- Compare an algorithm with pseudocode on at least five points.
- Dry-run Bubble Sort and Selection Sort pass by pass on a given list, and write them in pseudocode and Python.
- List the advantages and disadvantages of each sorting and searching algorithm.
- Evaluate an algorithm for correctness (using a trace table), efficiency and clarity.
- Define an algorithm and state five reasons why algorithms are important.
- Apply decomposition, pattern recognition and abstraction to real-life problems.
- Trace Linear Search step by step and Binary Search using a low / mid / high table.
- State the best-case and worst-case time complexity of the four algorithms.
- Choose the right algorithm for a situation and justify your choice.

2.1 Computational Thinking

LEARN IT

Computational Thinking (CT) is a systematic approach to solving problems. "Systematic" means you follow a fixed, organised method instead of guessing. CT involves four activities:

- **Decomposition** — breaking a big problem down into smaller parts.
- **Pattern recognition** — identifying things that repeat or follow a rule.
- **Abstraction** — hiding (ignoring) details that do not matter.
- **Algorithms** — developing clear step-by-step instructions to solve the problem.

CT is not "thinking like a computer". It is thinking in a way that lets *you* describe a solution so clearly that a computer (or another person) could carry it out.

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

You already use computational thinking in daily life. When you plan your exam preparation, you split the syllabus into chapters (decomposition), notice that every chapter needs the same routine of reading,

making notes and practising (pattern recognition), ignore things that are not in the syllabus (abstraction), and make a day-by-day timetable (algorithm).

LEARN IT

Did you know? Pseudocode is a human-readable representation of an algorithm using plain English and structured programming keywords. It is the bridge between human logic and machine code. You will study it in detail in section 2.2.2.

REMEMBER IT

Remember the four parts of CT with the phrase "**Don't Pick Any Apple**": **D**ecomposition, **P**attern recognition, **A**bstraction, **A**lgorithm.

COMMON MISTAKE

Students sometimes think computational thinking means "writing code". It does not. CT happens *before* coding. You can do all four steps with pen and paper, without a computer.

ارو میں سمجھیں

کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔ اس کے چار حصے ہیں: مسئلے کو چھوٹے حصوں میں توڑنا (Decomposition)، دہرائی جانے والی باتوں کو پہچاننا (Pattern Recognition)، غیر ضروری تفصیلات کو چھپانا (Abstraction) اور مرحلہ وار حل لکھنا (Algorithm)۔ یہ سوچنے کا طریقہ ہے، کوڈ لکھنا نہیں۔

CHECK YOURSELF

1. Name the four parts of computational thinking.
2. Is computational thinking done before or after writing code?
3. What is pseudocode called a "bridge" between?

Answers: 1. Decomposition, pattern recognition, abstraction, algorithms. 2. Before coding (it is planning the solution). 3. Between human logic and machine code.

2.2 Algorithm ★ Golden

LEARN IT

An **algorithm** is a step-by-step logical procedure used to solve a problem. It is **conceptual** in nature (it is an idea, a plan) and completely **independent of any programming language**. The same algorithm can be written later in Python, C++ or Java.

A good algorithm has:

- **Input** — the data it starts with.
- **Clear steps** — each step has only one meaning.
- **A definite end** — it must finish after a limited number of steps.
- **Output** — the result it produces.

WORKED EXAMPLE

Algorithm to find the larger of two numbers

Step 1: Start

Step 2: Read A and B

Step 3: If $A > B$, print "A is larger"

Step 4: Else print "B is larger or equal"

Step 5: End

Notice: there is no Python or C++ here. Anyone — even a non-programmer — can follow these steps. That is what "language-independent" means.

2.2.1 Importance of Algorithms

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

REMEMBER IT

Remember the five reasons as **S-O-R-A-B**: **S**tructured, **O**ptimised, **R**eusable, **A**ccurate, **B**lueprint.

COMMON MISTAKE

Do not write "an algorithm is a program". A program is the algorithm *written in a programming language* so that a computer can run it. The algorithm itself is only the logic.

اردو میں سمجھیں

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔ یہ ایک تصور ہے اور کسی بھی پروگرامنگ زبان پر منحصر نہیں۔ الگورتھم کام کو منظم بناتے ہیں، وقت اور میموری بچاتے ہیں، دوبارہ استعمال ہو سکتے ہیں، غلطیاں کم کرتے ہیں اور کوڈ لکھنے سے پہلے نقشے کا کام دیتے ہیں۔

EXAM TIP

Very common short question: "Define algorithm and write its importance." Write the definition word for word — *a step-by-step logical procedure used to solve a problem; conceptual and independent of any programming language* — then list at least four of the five importance points. If asked to "write an algorithm", always begin with **Step 1: Start** and finish with **End**, and number every step.

CHECK YOURSELF

1. Why is an algorithm called "language-independent"?
2. Give two reasons why algorithms are important.
3. Write a 5-step algorithm to find the sum of two numbers.

Answers: 1. Because it describes only the logic; it is not tied to Python, C++ or any other language. 2. Any two: structured approach, optimise time/memory, reusable, accurate and consistent, blueprint before coding. 3. Step 1: Start; Step 2: Read A and B; Step 3: SUM = A + B; Step 4: Print SUM; Step 5: End.

2.2.2 Algorithm vs Pseudocode ★ Golden

LEARN IT

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase. The **algorithm** says *what* must be done. The **pseudocode** shows *how* it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on a computer.

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

WORKED EXAMPLE

The same problem written both ways: print whether a number is even or odd.

Algorithm

Step 1: Start
Step 2: Read a number N
Step 3: Divide N by 2 and find the remainder
Step 4: If the remainder is 0, print "Even", otherwise print "Odd"
Step 5: End

Pseudocode

```
START
INPUT N
SET R TO N MOD 2
IF R = 0 THEN
    PRINT "Even"
ELSE
    PRINT "Odd"
END IF
END
```

The equivalent Python program (real, executable code):

```
n = 7
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

OUTPUT
Odd

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

REMEMBER IT

Teacher's analogy: an **algorithm** is like a recipe ("Bake a cake at 350 degrees"). **Pseudocode** is like a structured kitchen manual using standard verbs (SET temperature TO 350; WHILE cake is not baked, WAIT). The program is the oven actually doing it.

COMMON MISTAKE

Pseudocode is **not** executable. If you type pseudocode into Python, it will give errors. Also, do not mix exact Python syntax (colons, brackets) into pseudocode answers unless asked — use keywords like IF ... THEN ... END IF.

اروین سمجھیں

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔ الگورتھم سادہ مرحلوں میں لکھا جاتا ہے اور ہر کوئی اسے سمجھ سکتا ہے۔ سوڈو کوڈ میں IF، FOR، WHILE جیسے الفاظ استعمال ہوتے ہیں اور یہ زیادہ تر پروگرامرز کے لیے ہوتا ہے، لیکن اسے کمپیوٹر پر چلایا نہیں جا سکتا۔

EXAM TIP

"Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least **five** points (definition, nature, purpose, form, audience). Adding the recipe vs kitchen-manual analogy and one small example of each earns full marks.

CHECK YOURSELF

1. Which one — algorithm or pseudocode — uses keywords like FOR and WHILE?
2. Which one is meant for non-programmers too?
3. Can pseudocode be executed by a computer?

Answers: 1. Pseudocode. 2. Algorithm. 3. No; it is semi-formal and not executable.

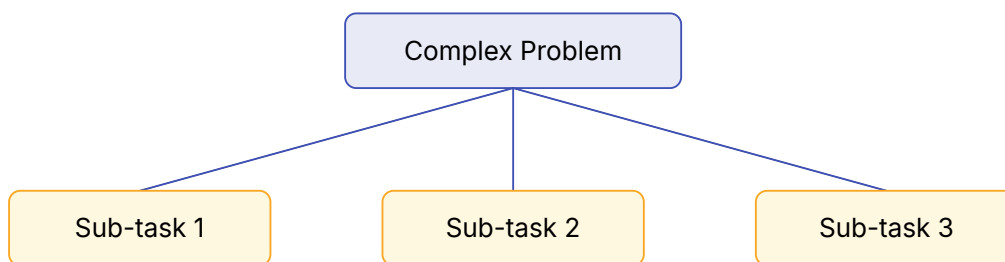
2.3.1 Decomposition ★ Golden

2.3 Algorithmic Strategies for Problem Solving. An **algorithmic strategy** is a general method that helps you design an algorithm. The three strategies in your syllabus are decomposition, pattern recognition and abstraction.

LEARN IT

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts. By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.

Each small part is called a **sub-task** (or sub-problem). You solve each sub-task separately, then join the solutions to solve the whole problem.



Decomposition: one complex problem is split into smaller sub-tasks.

WORKED EXAMPLE

Real-life example: Learning a new language

"Learn English" (or any language) is too big to do in one go. Decompose it:

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."

Vocabulary → Grammar rules → Tenses → Sentence building

WORKED EXAMPLE

Computing example: a "Student Result" program

Problem: read the marks of a student in 5 subjects and print the total, percentage and grade.

- Sub-task 1: Input marks of 5 subjects.
- Sub-task 2: Calculate total = sum of the 5 marks.
- Sub-task 3: Calculate percentage = $\text{total} \div 500 \times 100$.
- Sub-task 4: Decide the grade from the percentage.
- Sub-task 5: Print the result card.

Each sub-task is now so small that you can write it in two or three lines of code.

Why is decomposition useful?

- **Reduces complexity:** you focus on one piece at a time.
- **Helps in debugging:** you can find exactly where the mistake is (e.g., in the language example, is the error in the verb tense or the word order?).
- **Promotes reusability:** once you learn or build a small part, you can apply it elsewhere (the "calculate percentage" part can be reused in an attendance program).

REMEMBER IT

Think of eating a big *roti*: you cannot eat it in one bite, so you break it into small pieces.

Decomposition = "break the roti".

COMMON MISTAKE

Do not break a problem into random pieces. Each sub-task must be a meaningful, clear job, and all sub-tasks together must solve the *whole* problem. If one part is missing (e.g., "print result"), the solution is incomplete.

اردو میں سمجھیں

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔ مثلاً نئی زبان سیکھنے کو الفاظ، گرامر، زمانے (Tenses) اور جملہ سازی میں تقسیم کیا جا سکتا ہے۔ اس سے پیچیدگی کم ہوتی ہے، غلطی ڈھونڈنا آسان ہوتا ہے اور حصے دوبارہ استعمال ہو سکتے ہیں۔

EXAM TIP

Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task 1, 2, 3), show the language-learning table, and list the three benefits: reduces complexity, helps debugging, promotes reusability.

CHECK YOURSELF

1. What is each smaller part of a decomposed problem called?
2. Decompose "Making a school timetable" into three sub-tasks.
3. State the three benefits of decomposition.

Answers: 1. A sub-task (sub-problem). 2. Example: list subjects and teachers; decide periods per subject per week; place periods in days so no teacher has two classes at once. 3. Reduces complexity, helps in debugging, promotes reusability.

2.3.2 Pattern Recognition

LEARN IT

Pattern recognition is the process of identifying regularities, repetitions or rules in data. It helps in reducing the overall complexity of an algorithm and forms the foundation of concepts such as **loops**.

How it works — ask yourself three questions:

- What is repeating?
- What changes in a predictable way?
- Can I generalise this (write one rule that works for every case)?

WORKED EXAMPLE

Example: Triangle of stars

OUTPUT

```
*  
**  
***  
****  
*****
```

Step 1 — What repeats? Every row prints stars.

Step 2 — What changes predictably? Each row adds 1 star to the previous row's count.

Row	Stars
1	1
2	2
3	3
...	...
N	N

Step 3 — Generalise: Row N = N stars. Recognising this pattern lets us write one generic loop instead of 5 separate print statements.

Without the pattern (5 statements)

```
print("*")  
print("**")  
print("***")  
print("****")  
print("*****")
```

With the pattern (one loop)

```
n = 5  
for row in range(1, n + 1):  
    print("*" * row)
```

Both programs print exactly the same triangle. But if you want 100 rows, the loop version needs only one change ($n = 100$), while the first version needs 100 lines.

REMEMBER IT

Whenever you see "the same thing again and again, with a small change each time", think **loop**. Pattern recognition is the habit of spotting that.

COMMON MISTAKE

Do not generalise from too few cases. Check your rule on several rows (or items) before trusting it. "Row N = N stars" is correct here; a different triangle (1, 3, 5, 7 stars) follows the rule "Row N = $2N - 1$ stars".

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔ ستاروں کی تکون میں ہر قطار میں پچھلی قطار سے ایک ستارہ زیادہ ہے، یعنی قطار N میں N ستارے۔ یہ پیٹرن پہچان کر ہم پانچ الگ پرنت کے بجائے ایک لوپ (Loop) لکھ سکتے ہیں۔

CHECK YOURSELF

1. Which programming concept is based on pattern recognition?
2. How many stars will row 8 of the triangle have?
3. Find the rule: 2, 4, 6, 8, ...

Answers: 1. Loops. 2. 8 stars (Row N = N stars). 3. Term N = 2 × N (each term adds 2).

2.3.3 Abstraction

LEARN IT

Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem. It reduces complexity by ignoring what is not important (the **noise**).

When you apply abstraction you sort details into three groups:

- **Necessary steps** — the core logic that never changes.
- **Changes based on the situation** — these become **variables** (inputs) in the algorithm.
- **Abstraction items (noise)** — details you can ignore completely.

WORKED EXAMPLE

Example: applying abstraction to "Making Tea"

To make an algorithm for tea, we separate the necessary steps from the situational changes and the noise.

1. Necessary steps (core logic)	2. Changes based on situation (variables)	3. Abstraction items (ignore / noise)
1. Boil water 2. Add tea leaves/bag 3. Steep for a few minutes 4. Pour into cup 5. Serve	Ingredients (black, green, milk) Additions (sugar, lemon) Tools (stove vs kettle)	Brand of kettle Cup colour or design Where tea is made (kitchen, office)

The core abstracted logic, independent of cup colour or kettle brand:



This one algorithm now works for green tea in an office kettle and for milk tea on a kitchen stove. The middle column simply becomes inputs: `tea_type`, `addition`, `tool`.

Computers use abstraction everywhere. When you call `print("Hello")` in Python, you do not think about how the screen lights up each pixel — those details are hidden from you. A map of Karachi shows roads and landmarks but hides every tree and car: it is an abstraction of the city.

REMEMBER IT

Abstraction = "**Keep the skeleton, drop the decoration.**" Ask: "If I remove this detail, does the solution still work?" If yes, it is noise.

COMMON MISTAKE

Do not confuse abstraction with decomposition. Decomposition *splits* a problem into parts. Abstraction *removes* unimportant details. Also, do not throw away details that really matter — "boil water" can never be ignored, but kettle brand can.

اروین سمجھیں

ایبسٹریکشن (Abstraction) کا مطلب ہے غیر ضروری تفصیلات کو چھپا کر صرف ضروری باتوں پر توجہ دینا۔ چائے بنانے کی مثال میں پانی ابلنا، پتی ڈالنا، دم دینا، کپ میں ڈالنا اور پیش کرنا ضروری مراحل ہیں؛ چائے کی قسم یا چینی متغیرات (Variables) ہیں؛ جبکہ کیتلی کا برانڈ یا کپ کا رنگ فالتو باتیں (Noise) ہیں جنہیں نظر انداز کیا جاتا ہے۔

CHECK YOURSELF

1. What is "noise" in abstraction?
2. In the tea example, is "sugar or lemon" a necessary step, a variable, or noise?
3. How is abstraction different from decomposition?

Answers: 1. Unnecessary details that do not affect the solution. 2. A variable (it changes with the situation). 3. Decomposition breaks a problem into parts; abstraction hides details that are not important.

2.4 Sorting and Searching Algorithms

LEARN IT

Sorting and searching are both fundamental techniques of computer science. They work on **lists of the same data type** (for example, a list of all integers or a list of all names).

Sorting	Searching
Arranging a collection of items in an order. The order may be ascending (0–9, A–Z) or descending (9–0, Z–A). Used to organise data.	Locating a specific element within a list to find its position or check its existence (is it there or not?).
Examples: Bubble Sort, Selection Sort	Examples: Linear Search, Binary Search
Real life: merit list arranged from highest to lowest marks.	Real life: finding your roll number in the merit list.

Notation you will see

- **DATA** — the list (array). `DATA[1]` is the 1st element in the textbook algorithms (counting starts at 1).
- In **pseudocode and Python** counting starts at **0**: `DATA[0]` is the 1st element. This is called an **index**.
- **N** — the number of elements in the list.
- **PTR** (pointer) — a variable that holds the current position.
- **temp** — a temporary variable used to **swap** (exchange) two values.
- **Pass** — one complete run of the inner loop over the list.

WORKED EXAMPLE

How a swap works with temp. Suppose $A = 8$ and $B = 4$. To exchange them:

$\text{temp} = A \text{ (temp} = 8\text{)} \rightarrow A = B \text{ (} A = 4\text{)} \rightarrow B = \text{temp (} B = 8\text{)}$

Without `temp`, writing `A = B` first would destroy the 8. In Python you may also write `a, b = b, a`.

COMMON MISTAKE

Position 1 in the textbook algorithm is index 0 in Python. In exams, if a question uses `DATA[1]`, count from 1; if it uses Python, count from 0. State clearly which one you use.

اردو میں سمجھیں

سارٹنگ (Sorting) کا مطلب ہے ڈیٹا کو صعودی (چھوٹے سے بڑے) یا نزولی (بڑے سے چھوٹے) ترتیب میں رکھنا۔ سرچنگ (Searching) کا مطلب ہے فہرست میں کسی خاص چیز کو تلاش کرنا تاکہ اس کی جگہ یا موجودگی معلوم ہو سکے۔ دونوں ایک ہی قسم کے ڈیٹا کی فہرست پر کام کرتے ہیں۔

CHECK YOURSELF

1. Arrange C, A, D, B in descending order.
2. What is the purpose of the variable `temp`?

Answers: 1. D, C, B, A. 2. It temporarily holds one value so that two values can be swapped without losing either.

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

LEARN IT

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing **adjacent** (side-by-side) elements and swapping them if they are in the wrong order.

In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water. So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on. A list of N items needs **$N - 1$ passes**, and pass K makes **$N - K$ comparisons**.

Compare `DATA[j]` and `DATA[j+1]` → Left > Right? → Yes: swap → Move one step right →
End of pass: largest is at the end → Repeat $N-1$ passes

Algorithm (textbook)

Pseudocode

```

Step 1: Start
Step 2: Set DATA := {8,4,1,9,3}
Step 3: Set N := 5
Step 4: Repeat Step 5 and 6 for
        K = 1 to N-1
Step 5: Set PTR := 1
Step 6: Repeat while PTR ≤ N-K:
    a: If DATA[PTR] > DATA[PTR+1]:
        1. Set temp = DATA[PTR]
        2. Set DATA[PTR] = DATA[PTR+1]
        3. Set DATA[PTR+1] = temp
    b: Set PTR := PTR + 1
Step 7: End

```

```

Start
DATA = {8,4,1,9,3}
N = 5
For outer = 0 to N-1
    For inner = 0 to N-2-outer
        IF DATA[inner] > DATA[inner+1]
            Temp = DATA[inner]
            DATA[inner] = DATA[inner+1]
            DATA[inner+1] = Temp
        End If
    End For
End For
End

```

WORKED EXAMPLE

Worked example: sort 8, 4, 1, 9, 3 in ascending order. The pair in brackets [] is being compared. Positions are counted from 1.

Pass 1 (K = 1, 4 comparisons)

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

9 is now at its correct (final) place.

Pass 2 (K = 2, 3 comparisons)

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[4, 1], 8, 3, 9	4 > 1 Yes	Swap	1, 4, 8, 3, 9
2nd vs 3rd	1, [4, 8], 3, 9	4 > 8 No	No change	1, 4, 8, 3, 9
3rd vs 4th	1, 4, [8, 3], 9	8 > 3 Yes	Swap	1, 4, 3, 8, 9

Pass 3 (K = 3, 2 comparisons)

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[1, 4], 3, 8, 9	1 > 4 No	No change	1, 4, 3, 8, 9
2nd vs 3rd	1, [4, 3], 8, 9	4 > 3 Yes	Swap	1, 3, 4, 8, 9

Pass 4 (K = 4, 1 comparison)

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[1, 3], 4, 8, 9	1 > 3 No	No change	1, 3, 4, 8, 9

Sorted list: 1, 3, 4, 8, 9. Totals: $4 + 3 + 2 + 1 = 10$ comparisons and **6 swaps**.

Pass	List after pass	Fixed at the end
Start	8, 4, 1, 9, 3	—
1	4, 1, 8, 3, 9	9
2	1, 4, 3, 8, 9	8, 9
3	1, 3, 4, 8, 9	4, 8, 9
4	1, 3, 4, 8, 9	all

WORKED EXAMPLE

Bubble Sort in Python (follows the pseudocode exactly; Python indexes start at 0):

```
data = [8, 4, 1, 9, 3]
n = len(data)
for outer in range(n - 1):
    for inner in range(n - 1 - outer):
        if data[inner] > data[inner + 1]:
            temp = data[inner]
            data[inner] = data[inner + 1]
            data[inner + 1] = temp
    print("After pass", outer + 1, ":", data)
print("Sorted list:", data)
```

OUTPUT

```
After pass 1 : [4, 1, 8, 3, 9]
After pass 2 : [1, 4, 3, 8, 9]
After pass 3 : [1, 3, 4, 8, 9]
After pass 4 : [1, 3, 4, 8, 9]
Sorted list: [1, 3, 4, 8, 9]
```

`range(n - 1 - outer)` gives `inner = 0 ... N-2-outer`, exactly as in the pseudocode. Notice the list was already sorted after pass 3, but the plain algorithm still does pass 4.

Advantages and disadvantages

Advantages	Disadvantages
Easiest algorithm to visualise and explain.	Uses nested loops, so the time grows very fast as the list grows (about N^2 comparisons).
Does not need extra memory (in-place sorting).	Performs a massive number of swaps.
Stable (equal items stay in their original order).	Useless for real-world "Big Data".

REMEMBER IT

Bubble = neighbours. Bubble Sort only ever compares *neighbours* (j and $j+1$). Each pass pushes the biggest bubble to the end. For **descending** order, just change `>` to `<` in the comparison.

COMMON MISTAKE

The slides say the time grows "exponentially". Strictly, Bubble Sort's time grows **quadratically**: for N items it makes $N(N-1)/2$ comparisons, which is $O(N^2)$. Doubling the list makes it about 4 times slower — bad, but not exponential. Another common mistake: students forget that pass K does only $N - K$ comparisons; the sorted tail is never compared again.

اروین سمجھیں

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ موجود دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔ ہر پاس (Pass) کے بعد سب سے بڑا عدد آخر میں پہنچ جاتا ہے۔ N عناصر کے لیے $N-1$ پاس درکار ہوتے ہیں۔ یہ سمجھنے میں آسان اور مستحکم (Stable) ہے لیکن بڑے ڈیٹا کے لیے بہت سست ہے۔

EXAM TIP

The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns *Pair*, *Compare*, *Action*, *List after*. Mark the element fixed at the end after every pass. Also be ready to write the algorithm (with PTR and K) or the pseudocode (with outer/inner), and to list two advantages and two disadvantages.

CHECK YOURSELF

1. How many passes and comparisons does Bubble Sort need for 6 elements?
2. Show the list 5, 1, 4 after pass 1 of Bubble Sort.
3. Sort 2, 9, 4, 7 in descending order with Bubble Sort: give the list after each pass.

Answers: 1. 5 passes; $5 + 4 + 3 + 2 + 1 = 15$ comparisons. 2. (5,1) swap $\rightarrow$ 1, 5, 4; (5,4) swap $\rightarrow$ 1, 4, 5. 3. Pass 1: 9, 4, 7, 2; Pass 2: 9, 7, 4, 2; Pass 3: 9, 7, 4, 2.

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

LEARN IT

Selection Sort is a simple sorting algorithm that works by repeatedly **selecting the smallest element** from the unsorted portion of the list and moving it to its correct position at the front.

Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on. In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are swapped. So at the end of pass K, the K-th smallest value sits in position K.



Algorithm (textbook)

```
Step 1: Start
Step 2: Set DATA := {8,4,1,9,3}
Step 3: Set N := 5
Step 4: Repeat Step 5 and 6 for
        K = 1 to N
Step 5: Set PTR := K+1
Step 6: Repeat while PTR ≤ N:
    a: If DATA[K] > DATA[PTR]:
        1. Set temp = DATA[K]
        2. Set DATA[K] = DATA[PTR]
        3. Set DATA[PTR] = temp
    b: Set PTR := PTR + 1
Step 7: End
```

Pseudocode

```
Start
DATA = {8,4,1,9,3}
N = 5
For outer = 0 to N-1
    For inner = outer+1 to N-1
        IF DATA[outer] > DATA[inner]
            Temp = DATA[outer]
            DATA[outer] = DATA[inner]
            DATA[inner] = Temp
        End If
    End For
End For
End
```

REMEMBER IT

Teacher's tip — spot the difference: in Selection Sort the inner loop starts at `outer + 1` and compares with the *fixed* position `DATA[outer]` . In Bubble Sort the inner loop always starts at `0` and compares *neighbours* `DATA[inner]` and `DATA[inner+1]` . Bubble pushes the largest to the **end**; Selection brings the smallest to the **front**.

WORKED EXAMPLE

Worked example: sort 8, 4, 1, 9, 3 in ascending order (textbook method). The two positions in brackets are being compared.

Pass 1 — find the smallest for Position 1

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	4 > 1 Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	1 > 9 No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	1 > 3 No	No change	1 , 8, 4, 9, 3

1 is now permanently in Position 1.

Pass 2 — find the smallest for Position 2

Compare	List	Pos2 > PosX?	Action	List after
Pos2 vs Pos3	1, [8], [4], 9, 3	8 > 4 Yes	Swap	1, 4, 8, 9, 3
Pos2 vs Pos4	1, [4], 8, [9], 3	4 > 9 No	No change	1, 4, 8, 9, 3
Pos2 vs Pos5	1, [4], 8, 9, [3]	4 > 3 Yes	Swap	1, 3 , 8, 9, 4

Pass 3 — find the smallest for Position 3

Compare	List	Pos3 > PosX?	Action	List after
Pos3 vs Pos4	1, 3, [8], [9], 4	8 > 9 No	No change	1, 3, 8, 9, 4
Pos3 vs Pos5	1, 3, [8], 9, [4]	8 > 4 Yes	Swap	1, 3, 4 , 9, 8

Pass 4 — find the smallest for Position 4

Compare	List	Pos4 > PosX?	Action	List after
Pos4 vs Pos5	1, 3, 4, [9], [8]	9 > 8 Yes	Swap	1, 3, 4, 8, 9

Sorted list: 1, 3, 4, 8, 9. Totals: 4 + 3 + 2 + 1 = **10 comparisons, 6 swaps** in this textbook method.

Pass	List after pass	Fixed at the front
Start	8, 4, 1, 9, 3	—
1	1, 8, 4, 9, 3	1
2	1, 3, 8, 9, 4	1, 3
3	1, 3, 4, 9, 8	1, 3, 4
4	1, 3, 4, 8, 9	all

WORKED EXAMPLE

Selection Sort in Python (follows the textbook pseudocode):

```
data = [8, 4, 1, 9, 3]
n = len(data)
for outer in range(n - 1):
    for inner in range(outer + 1, n):
        if data[outer] > data[inner]:
            temp = data[outer]
            data[outer] = data[inner]
            data[inner] = temp
    print("After pass", outer + 1, ":", data)
print("Sorted list:", data)
```

OUTPUT

```
After pass 1 : [1, 8, 4, 9, 3]
After pass 2 : [1, 3, 8, 9, 4]
After pass 3 : [1, 3, 4, 9, 8]
After pass 4 : [1, 3, 4, 8, 9]
Sorted list: [1, 3, 4, 8, 9]
```

WORKED EXAMPLE

The "classic" Selection Sort (one swap per pass). Many books first *find the position of the smallest* value in the unsorted part, and swap only once at the end of each pass. The result is the same; only the swaps are fewer.

```
data = [8, 4, 1, 9, 3]
n = len(data)
for outer in range(n - 1):
    smallest = outer
    for inner in range(outer + 1, n):
        if data[inner] < data[smallest]:
            smallest = inner
    data[outer], data[smallest] = data[smallest], data[outer]
    print("After pass", outer + 1, ":", data)
```

OUTPUT

```
After pass 1 : [1, 4, 8, 9, 3]
After pass 2 : [1, 3, 8, 9, 4]
After pass 3 : [1, 3, 4, 9, 8]
After pass 4 : [1, 3, 4, 8, 9]
```

Here only 4 swaps are made (at most $N - 1$). This is why Selection Sort is said to perform fewer swaps than Bubble Sort. In exams, use the textbook method unless the question asks otherwise.

Advantages and disadvantages

Advantages	Disadvantages
Very easy to code.	Time grows very fast with list size (nested loops, about N^2 comparisons).
Performs fewer swaps overall compared to Bubble Sort.	Must finish every single loop — it is "blind" to an already sorted list.

Does not need extra memory.

Not stable (it might change the relative positions of equal items).

WORKED EXAMPLE

Why "not stable"? Take two equal marks, 5_a (Ali) and 5_b (Bilal), in the list $5_a, 5_b, 2$. Pass 1 compares 5_a with 5_b (no swap), then 5_a with $2 \rightarrow \text{swap} \rightarrow 2, 5_b, 5_a$. Now Bilal is before Ali, although Ali came first originally. Bubble Sort never does this, because it swaps only neighbours that are strictly out of order.

COMMON MISTAKE

In the textbook method, the comparison is always with the **fixed position** $\text{DATA}[K]$ (Pos1 in pass 1, Pos2 in pass 2), not with the neighbour. Students often mix this up with Bubble Sort and compare neighbours. Also, after a swap, the *new* value at Pos K is used for the next comparison.

ارو میں سمجھیں

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔ ہر پاس میں ایک مقررہ جگہ (Position) کا موازنہ اس کے بعد والے تمام عناصر سے ہوتا ہے۔ اس کا کوڈ آسان ہے اور ببل سارٹ سے کم تبادلے (Swaps) ہوتے ہیں، لیکن یہ مستحکم (Stable) نہیں اور ترتیب شدہ فہرست پر بھی پورے لوپ چلاتا ہے۔

EXAM TIP

Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In the difference question mention: what is compared (neighbours vs fixed position), where the inner loop starts (0 vs outer+1), where items settle (end vs front), stability, and number of swaps.

CHECK YOURSELF

1. In Selection Sort, where does the inner loop start?
2. Using the textbook method, show 7, 2, 5 after pass 1.
3. Which of the two sorts is stable?

Answers: 1. At outer + 1 (PTR = K + 1). 2. Pos1 vs Pos2: $7 > 2 \text{ swap} \rightarrow 2, 7, 5$; Pos1 vs Pos3: $2 > 5 \text{ no} \rightarrow 2, 7, 5$. 3. Bubble Sort.

2.4.2 Searching Algorithms: Linear Search

LEARN IT

Linear Search, also known as **sequential search**, works by examining each element in the list one by one, from the start, until the target is found (or the list ends). It works on both **sorted and unsorted** data.

A **flag** variable (here `ValueFound`) starts at 0 and becomes 1 when the item is found. After the loop, the flag tells us whether to print the item or "Not Found".



Algorithm (textbook)

```

Step 1: Start
Step 2: DATA := {8,3,6,1,7,2,4,5}
Step 3: N := 8
Step 4: FindValue := 7
Step 5: ValueFound := 0
Step 6: Repeat for PTR = 1 to N
Step 7:   If DATA[PTR] = FindValue:
           a. ValueFound = 1
           b. Goto Step 8
Step 8: If ValueFound = 1:
           Print DATA[PTR]
       Else:
           Print "Not Found"
Step 9: End
  
```

Pseudocode

```

Start
DATA = {8, 3, 6, 1, 7, 2, 4, 5}
N = 8
FindValue = 7
ValueFound = 0
For PTR = 0 to N-1
    If DATA[PTR] = FindValue then
        ValueFound = 1
        Exit Loop
    End If
End For
If ValueFound = 1 then
    Print DATA[PTR]
Else
    Print "Data Not Found"
End If
End
  
```

WORKED EXAMPLE

Worked example. Data list: 8, 3, 6, 1, 7, 2, 4, 5. Searching item: 7.

Step	Element checked	Condition (== 7)	Result
1	8	8 == 7 ?	Not found
2	3	3 == 7 ?	Not found
3	6	6 == 7 ?	Not found
4	1	1 == 7 ?	Not found
5	7	7 == 7 ?	Found! (Exit loop)

7 is found at position 5 (index 4) after 5 comparisons. The remaining items 2, 4, 5 are never checked.

If the item is missing — e.g., search for 9 — all 8 elements are checked, the flag stays 0, and "Data Not Found" is printed. This is the **worst case**: N comparisons.

WORKED EXAMPLE

Linear Search in Python

```
data = [8, 3, 6, 1, 7, 2, 4, 5]
find_value = 7
value_found = 0
for ptr in range(len(data)):
    print("Step", ptr + 1, ": checking", data[ptr])
    if data[ptr] == find_value:
        value_found = 1
        break
if value_found == 1:
    print("Found", data[ptr], "at position", ptr + 1)
else:
    print("Data Not Found")
```

OUTPUT

```
Step 1 : checking 8
Step 2 : checking 3
Step 3 : checking 6
Step 4 : checking 1
Step 5 : checking 7
Found 7 at position 5
```

`break` is Python's "Exit Loop".

Advantages and disadvantages

Advantages	Disadvantages
Does not care whether data is scrambled or sorted.	If the target is at the end of 1 million items, it does 1 million checks.
Extremely simple code (one loop, one if statement).	Time grows linearly (double the list = double the time).
Faster for very small lists (no setup time, such as sorting).	Inefficient if you need to search the same list thousands of times.
Does not require any extra space.	

REMEMBER IT

Linear Search is like looking for your friend in a queue by checking every face from the front. Best case: 1 check (first person). Worst case: N checks (last person or not there).

COMMON MISTAKE

Do not print "Not Found" inside the loop after every mismatch — that would print it many times. Decide only **after** the loop ends, using the flag. And remember to **exit the loop** once the item is found, or you waste comparisons.

لینیئر سرچ (Linear Search) کو ترتیب وار تلاش (Sequential Search) بھی کہتے ہیں۔ اس میں فہرست کے ہر عنصر کو شروع سے ایک ایک کر کے چیک کیا جاتا ہے جب تک مطلوبہ چیز نہ مل جائے۔ یہ ترتیب شدہ اور غیر ترتیب شدہ دونوں قسم کے ڈیٹا پر کام کرتی ہے اور بہت سادہ ہے، لیکن بڑی فہرست میں بہت وقت لیتی ہے۔

CHECK YOURSELF

1. What is another name for Linear Search?
2. In 8, 3, 6, 1, 7, 2, 4, 5, how many comparisons are needed to find 2?
3. What is the purpose of the variable ValueFound?

Answers: 1. Sequential search. 2. 6 comparisons (2 is in position 6). 3. It is a flag: 0 means not found yet, 1 means found; it decides what to print after the loop.

2.4.2 Searching Algorithms: Binary Search ★ Golden

LEARN IT

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half. **Crucial limitation: it only works on a SORTED list.**

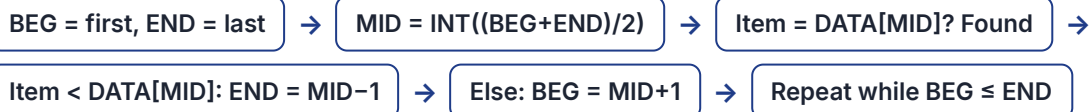
Three variables mark the part of the list still being searched:

- **BEG** (low) — first position of the current part.
- **END** (high) — last position of the current part.
- **MID** = $\text{INT}((\text{BEG} + \text{END}) / 2)$ — the middle position. INT means "drop the decimal part".

At each step compare the item with `DATA[MID]`:

- Equal → found, stop.
- Item < `DATA[MID]` → the item can only be in the **left** half: set `END = MID - 1`.
- Item > `DATA[MID]` → the item can only be in the **right** half: set `BEG = MID + 1`.

If `BEG` becomes greater than `END`, the part is empty and the item is **not in the list**.



Algorithm (textbook summary)

Pseudocode

```

Step 1: Start
Step 2: DATA := {1,2,3,4,5,6,7,8,9}
        (Sorted)
Step 3: BEG = 1, END = 9
Step 4: MID = INT((BEG+END)/2)
Step 5: Item = 7
Step 6: LOC = 0
Step 7: Repeat while BEG ≤ END AND
        DATA[MID] ≠ Item
Step 8:   If Item < DATA[MID]:
            END = MID - 1
        Else:
            BEG = MID + 1
Step 9:   MID = INT((BEG+END)/2)
Step 10: [End loop]
Step 11: If DATA[MID] = Item:
            LOC = MID
        Else:
            LOC = 0
Step 12: Print result based on LOC
Step 13: End

```

```

Start
DATA = {1,2,3,4,5,6,7,8,9}
BEG = 1, END = 9
MID = INT((BEG+END)/2)
Item = 7
LOC = NULL
While BEG ≤ END AND DATA[MID] ≠ Item
    IF Item < DATA[MID] then
        END = MID - 1
    Else
        BEG = MID + 1
    End If
    MID = INT((BEG+END)/2)
End While
If DATA[MID] = Item then
    LOC = MID
Else
    LOC = NULL
End If
If LOC ≠ NULL then
    Print DATA[MID]
Else
    Print "Not Found"
End If
End

```

COMMON MISTAKE

The slide's pseudocode writes the loop condition as $BEG \leq END \text{ OR } DATA[MID] \neq Item$. That is a slip: with OR the loop would never stop once the item is found while $BEG \leq END$. The textbook algorithm (Step 7) correctly uses **AND** — keep going only while the part is not empty *and* the item is not yet found. Likewise, the slide sets $LOC = NULL$ but then tests " $LOC \geq 0$ "; the check should simply be " LOC is not NULL" (or, with $LOC = 0$ meaning not found, " $LOC > 0$ "). The pseudocode above shows the corrected form.

WORKED EXAMPLE

Worked example 1 (textbook algorithm). DATA = 1, 2, 3, 4, 5, 6, 7, 8, 9; Item = 7. Positions 1 to 9.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	$INT(10/2) = 5$	5	$7 > 5$	BEG = 6 (go right)
2	6	9	$INT(15/2) = 7$	7	$7 = 7$	Found, LOC = 7

Only **2 comparisons**. Linear Search would need 7.

WORKED EXAMPLE

Worked example 2 (slide example). Sorted list: 1, 2, 3, 4, 5, 6, 7, 8. Searching item: 3.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	8	$\text{INT}(9/2) = 4$	4	$3 < 4$	END = 3 (go left)
2	1	3	$\text{INT}(4/2) = 2$	2	$3 > 2$	BEG = 3 (go right)
3	3	3	$\text{INT}(6/2) = 3$	3	$3 = 3$	FOUND at position 3

Picture the list shrinking (the highlighted cell is MID):

Step	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	1	2	3					
3			3					

The slide describes the same steps using the size of each sub-list: Step 1 $\text{MID} = 8/2 = 4$ (value 4), $3 < 4$, go left; Step 2 sub-list [1, 2, 3], middle = $3/2 = 1.5 \rightarrow 2$ nd item (value 2), $3 > 2$, go right; Step 3 sub-list [3], middle = 1st item (value 3), FOUND. Both ways land on the same values $4 \rightarrow 2 \rightarrow 3$.

WORKED EXAMPLE

Binary Search in Python (index starts at 0, so positions 1–8 become indexes 0–7):

```
data = [1, 2, 3, 4, 5, 6, 7, 8]
item = 3
beg = 0
end = len(data) - 1
loc = -1
step = 0
while beg ≤ end:
    mid = (beg + end) // 2
    step = step + 1
    print("Step", step, ": beg =", beg, "mid =", mid, "end =", end, "value =", data[mid])
    if data[mid] == item:
        loc = mid
        break
    elif item < data[mid]:
        end = mid - 1
    else:
        beg = mid + 1
if loc ≥ 0:
    print("Found", item, "at index", loc)
else:
    print("Not Found")
```

OUTPUT

```
Step 1 : beg = 0 mid = 3 end = 7 value = 4
Step 2 : beg = 0 mid = 1 end = 2 value = 2
Step 3 : beg = 2 mid = 2 end = 2 value = 3
Found 3 at index 2
```

// is integer division, the same as INT(). Index 2 is position 3. Here `loc = -1` means "not found", so the test `loc ≥ 0` is correct.

Why is it so fast?

Every comparison throws away half of the remaining list. $1,000,000 \rightarrow 500,000 \rightarrow 250,000 \rightarrow \dots \rightarrow 1$ takes about 20 halvings, because $2^{20} = 1,048,576$. So Binary Search finds an item among 1 million elements in at most about **20 steps**, while Linear Search may need 1,000,000.

Advantages and disadvantages

Advantages	Disadvantages
Extremely fast for "Big Data": finds an item in 1M elements in just ~20 steps.	The list must be sorted first; it fails on scrambled data.
Massive time savings for static lists that are searched repeatedly.	Requires random access to elements (works on arrays, not easily on linked lists).
	More complex code, prone to logic errors (off-by-one).

REMEMBER IT

Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. **"Sorted first, then split."**

COMMON MISTAKE

Common errors: (1) using Binary Search on an unsorted list — the answer will be wrong; (2) setting $END = MID$ instead of $MID - 1$ (or $BEG = MID$ instead of $MID + 1$), which can cause an infinite loop — this is the "off-by-one" error; (3) forgetting INT , so MID becomes a decimal like 4.5.

اروین سمجھیں

بائنری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔ یہ صرف ترتیب شدہ (Sorted) فہرست پر کام کرتا ہے۔ درمیانی عنصر (MID) سے موازنہ کیا جاتا ہے: اگر مطلوبہ چیز چھوٹی ہو تو بائیں آدھے حصے میں، اور بڑی ہو تو دائیں آدھے حصے میں تلاش جاری رہتی ہے۔ دس لاکھ عناصر میں بھی تقریباً بیس قدموں میں نتیجہ مل جاتا ہے۔

EXAM TIP

You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with **BEG, END, MID, DATA[MID], comparison, action**. Show the INT calculation for MID . If the item is absent, continue until $BEG > END$ and write "Not Found". Also learn: "Why must the list be sorted?" and "Compare Linear and Binary Search".

CHECK YOURSELF

1. What is the crucial condition for Binary Search to work?
2. $BEG = 6$, $END = 9$. What is MID ?
3. In 1, 2, 3, 4, 5, 6, 7, 8 (positions 1–8), list the MID values visited when searching for 8.

Answers: 1. The list must be sorted. 2. $INT(15/2) = 7$. 3. $MID = 4$ (value 4, go right, $BEG = 5$) $\rightarrow MID = INT(13/2) = 6$ (value 6, go right, $BEG = 7$) $\rightarrow MID = INT(15/2) = 7$ (value 7, go right, $BEG = 8$) $\rightarrow MID = 8$ (value 8, found). Four comparisons.

2.5 Evaluation Criteria of Algorithm

LEARN IT

Many algorithms can solve the same problem. To judge which is better, we evaluate them on three criteria: **correctness**, **efficiency** and **clarity**.

2.5.1 Correctness

An algorithm is **correct** if:

- its output matches the expected outcomes;
- it handles **edge cases** gracefully (unusual inputs such as an empty list, 0, a negative number, or an item that is not present);
- it is guaranteed to finish — no infinite loops.

We test correctness with a **trace table** (dry run): we feed sample inputs and record every value by hand, then compare the result with what we expect.

WORKED EXAMPLE

Trace table example: Even/Odd checker. Rule: divide the number by 2; if the remainder is 0 it is Even, otherwise Odd.

Input	Operation (divide by 2, find remainder)	Remainder == 0?	Result
3	3 / 2 (Quotient = 1, Remainder = 1)	No	Odd
6	6 / 2 (Quotient = 3, Remainder = 0)	Yes	Even
15	15 / 2 (Quotient = 7, Remainder = 1)	No	Odd

All three results match what we expect, so the algorithm is correct for these inputs. The same check in Python:

```
for number in [3, 6, 15]:
    remainder = number % 2
    if remainder == 0:
        print(number, "remainder", remainder, "→ Even")
    else:
        print(number, "remainder", remainder, "→ Odd")
```

OUTPUT

```
3 remainder 1 → Odd
6 remainder 0 → Even
15 remainder 1 → Odd
```

A good tester would also try the edge case 0 (remainder 0 → Even, which is correct).

2.5.2 Efficiency

We measure algorithm efficiency through **time** (speed) and **space** (memory). The key question is: how fast does it run as the input size grows — does it **scale** well?

Computer scientists describe growth with **Big O notation**. It counts roughly how many steps are needed for N items:

- **O(1)** — constant: same time, whatever N is.
- **O(log N)** — logarithmic: grows very slowly (Binary Search).
- **O(N)** — linear: double N, double the time (Linear Search).
- **O(N²)** — quadratic: double N, four times the time (Bubble and Selection Sort).

WORKED EXAMPLE

How the worst-case number of comparisons grows

N (items)	Linear Search (N)	Binary Search (about $\log_2 N$)	Bubble/Selection Sort $N(N-1)/2$
10	10	4	45
100	100	7	4,950
1,000	1,000	10	499,500
1,000,000	1,000,000	20	499,999,500,000

This is why Binary Search is called excellent for Big Data and the simple sorts are called useless for it.

2.5.3 Clarity (Readability)

An algorithm must be easy to read, understand and modify. It should use **clear steps**, **meaningful variable names** and **good indentation**.

Unclear

```
a=[3,6,15]
for x in a:
    if x%2==0:print(x,"E")
    else:print(x,"O")
```

Clear

```
numbers = [3, 6, 15]
for number in numbers:
    if number % 2 == 0:
        print(number, "Even")
    else:
        print(number, "Odd")
```

Both work, but the second is far easier for another person (or you, a month later) to understand and change.

REMEMBER IT

Remember **C-E-C**: **C**orrect (right answer, always finishes), **E**fficient (fast and light), **C**lear (easy to read).

COMMON MISTAKE

Testing only "normal" inputs is not enough to prove correctness. Always include edge cases in your trace table, e.g., the first item, the last item, a missing item, or an empty list in a search.

اردو میں سمجھیں

کسی الگورتھم کو تین معیارات پر پرکھا جاتا ہے: درستگی (Correctness) یعنی درست نتیجہ دے، غیر معمولی صورتوں کو سنبھالے اور ہمیشہ ختم ہو؛ کارکردگی (Efficiency) یعنی کم وقت اور کم میموری استعمال کرے؛ اور وضاحت (Clarity) یعنی پڑھنے، سمجھنے اور تبدیل کرنے میں آسان ہو۔ درستگی جانچنے کے لیے ٹریس ٹیبل (Trace Table) بنایا جاتا ہے۔

CHECK YOURSELF

1. Name the three evaluation criteria of an algorithm.
2. Give one example of an edge case for a search algorithm.
3. Efficiency is measured in terms of which two resources?

Answers: 1. Correctness, efficiency, clarity. 2. The item is not in the list (or the list is empty, or the item is the last element). 3. Time (speed) and space (memory).

2.6 How to Decide Which Algorithm to Use

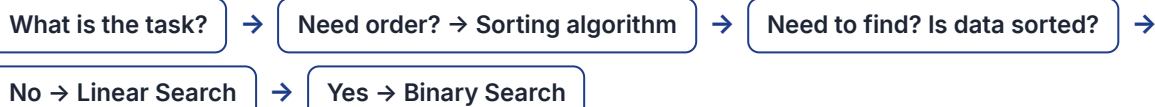
LEARN IT

Before choosing an algorithm, ask three questions:

- **What is the input size?** (10 items or 10 million?)
- **Is the data sorted?**
- **Do we need speed or memory efficiency?**

Simple decision guide:

- If searching on **unsorted** data → use **Linear Search**.
- If searching on **sorted** data → use **Binary Search**.
- If data needs **ordering** → use a **sorting** algorithm.



Comparison of the four algorithms

Algorithm	Type	Best case	Worst case	Extra memory	Needs sorted data?	Stable?
Bubble Sort	Sorting	$O(N^2)$ as written; $O(N)$ with an early-stop check	$O(N^2)$	None — $O(1)$	—	Yes
Selection Sort	Sorting	$O(N^2)$	$O(N^2)$	None — $O(1)$	—	No
Linear Search	Searching	$O(1)$ (first item)	$O(N)$ (last item / absent)	None — $O(1)$	No	—
Binary Search	Searching	$O(1)$ (item at first MID)	$O(\log N)$	None — $O(1)$	Yes	—

Point	Bubble Sort	Selection Sort
What is compared	Adjacent elements (j and $j+1$)	Fixed position K with every later element
Inner loop starts at	0	outer + 1
After each pass	Largest moves to the end	Smallest moves to the front

Swaps	Many	Fewer
Stable	Yes	No

Point	Linear Search	Binary Search
Method	Checks one by one	Splits the list in half each step
Data required	Sorted or unsorted	Sorted only
Worst case (1M items)	1,000,000 checks	About 20 checks
Code	Very simple	More complex, off-by-one risk
Best for	Small or unsorted lists, one-off searches	Large, sorted lists searched many times

WORKED EXAMPLE

Choosing in practice

- **A shopkeeper's list of 15 items, written in random order, searched once:** Linear Search — the list is unsorted and tiny; sorting first would waste time.
- **NADRA looking up a CNIC number among millions of sorted records, thousands of times a day:** Binary Search — data is sorted, huge, and searched repeatedly.
- **A teacher arranging 40 students' marks into a merit list:** a sorting algorithm; for 40 items even Bubble or Selection Sort is acceptable.
- **An unsorted list of 1 million records that will be searched thousands of times:** sort it once, then use Binary Search for every search.

REMEMBER IT

Two keywords decide most exam answers: "**sorted?**" and "**how big?**". Unsorted or small → Linear. Sorted and big → Binary.

COMMON MISTAKE

"Binary Search is always better" is wrong. On unsorted data it does not work at all, and for a single search on a small unsorted list, sorting first costs more than a simple Linear Search.

اردو میں سمجھیں

الگورتھم چننے سے پہلے دیکھیں کہ ڈیٹا کتنا بڑا ہے، ترتیب شدہ ہے یا نہیں، اور ہمیں رفتار چاہیے یا میموری کی بچت۔ غیر ترتیب شدہ ڈیٹا میں تلاش کے لیے لینیئر سرچ، ترتیب شدہ ڈیٹا میں تلاش کے لیے بائنری سرچ، اور ڈیٹا کو ترتیب دینے کے لیے سارٹنگ الگورتھم استعمال کریں۔

CHECK YOURSELF

1. Which search would you use on an unsorted list of phone contacts?
2. What is the worst-case time complexity of Binary Search?
3. Which sort has the same time complexity in its best and worst case?

Answers: 1. Linear Search. 2. $O(\log N)$. 3. Selection Sort ($O(N^2)$ in both).

Chapter 2 Review

Chapter summary

- **Computational thinking** is systematic problem solving using decomposition, pattern recognition, abstraction and algorithms.
- An **algorithm** is a step-by-step logical procedure to solve a problem; it is conceptual and language-independent.
- Algorithms give structure, optimise time and memory, are reusable, ensure accuracy and act as a blueprint for code.
- **Pseudocode** is a human-readable, structured representation of an algorithm using keywords like IF, FOR and WHILE; it is not executable.
- An algorithm defines *what* to do; pseudocode shows *how* to implement it.
- **Decomposition** breaks a big problem into smaller sub-tasks; it reduces complexity, helps debugging and promotes reusability.
- **Pattern recognition** finds repetitions and rules (Row N = N stars) and is the foundation of loops.
- **Abstraction** hides noise and keeps only essential steps; situational details become variables.
- Sorting arranges data (ascending or descending); searching locates an element's position or existence.
- **Bubble Sort** compares and swaps adjacent elements; the largest value moves to the end each pass; N – 1 passes; stable; $O(N^2)$.
- **Selection Sort** brings the smallest remaining value to the front each pass; inner loop starts at outer + 1; fewer swaps; not stable; $O(N^2)$.
- **Linear Search** checks items one by one; works on unsorted data; $O(N)$ worst case.
- **Binary Search** halves the search range using BEG, MID and END; needs sorted data; $O(\log N)$ — about 20 steps for 1 million items.
- Algorithms are evaluated on **correctness** (checked with trace tables and edge cases), **efficiency** (time and space) and **clarity**.
- Choose by asking: input size, sorted or not, speed or memory? Unsorted → Linear; sorted → Binary; needs order → sort.

Key terms

Term	Meaning	اردو
Computational thinking	A systematic approach to solving problems using decomposition, patterns, abstraction and algorithms.	کمپیوٹیشنل سوچ
Algorithm	A step-by-step logical procedure to solve a problem, independent of any language.	الگورتھم، مرحلہ وار طریقہ حل
Pseudocode	A human-readable, structured but non-executable representation of an algorithm.	سوڈو کوڈ، فرضی کوڈ

Decomposition	Breaking a complex problem into smaller, manageable parts.	تقسیم کاری
Sub-task	One small part of a decomposed problem.	ذیلی کام
Pattern recognition	Identifying regularities, repetitions or rules in data.	نمونے کی پہچان
Abstraction	Hiding unnecessary details and focusing on essentials.	تجربہ، غیر ضروری تفصیل چھپانا
Noise	Details that do not affect the solution and can be ignored.	فالتو تفصیلات
Sorting	Arranging items in ascending or descending order.	ترتیب دینا
Searching	Locating an element in a list to find its position or existence.	تلاش کرنا
Ascending order	Smallest to largest (0–9, A–Z).	صعودی ترتیب
Descending order	Largest to smallest (9–0, Z–A).	نزولی ترتیب
Bubble Sort	Sort that repeatedly compares and swaps adjacent elements.	ببل سارٹ
Selection Sort	Sort that repeatedly selects the smallest element and puts it at the front.	سلیکشن سارٹ
Pass	One full run of the inner loop through the list.	ایک چکر
Swap	Exchanging the values of two positions, usually with a temp variable.	جگہ کا تبادلہ
In-place sorting	Sorting without needing extra memory for another list.	اضافی میموری کے بغیر ترتیب
Stable sort	A sort that keeps equal items in their original order.	مستحکم ترتیب
Linear (sequential) search	Checking each element one by one until the target is found.	ترتیب وار تلاش
Binary search	Search that halves a sorted list at each step.	دو حصی تلاش
BEG / MID / END	First, middle and last positions of the part being searched in Binary Search.	آغاز / درمیان / اختتام
Flag	A variable (like ValueFound) that records whether something has happened.	نشان دہی کرنے والا متغیر
Correctness	Giving expected output, handling edge cases and always finishing.	درستگی

Efficiency	How well an algorithm uses time and memory as input grows.	کارکردگی
Clarity	How easy an algorithm is to read, understand and modify.	وضاحت
Trace table	A table recording values step by step during a dry run.	ٹریس ٹیبل، جانچ جدول
Edge case	An unusual or extreme input used to test an algorithm.	انتہائی صورت
Time complexity (Big O)	A measure of how the number of steps grows with input size, e.g. $O(N)$, $O(\log N)$.	وقتی پیچیدگی

Practice: multiple choice

- An algorithm is best defined as:
 - a programming language
 - a step-by-step logical procedure to solve a problem
 - a type of computer hardware
 - compiled machine code
- Pseudocode is:
 - executable machine code
 - a symbol used in flowcharts
 - a Python library
 - a human-readable representation of an algorithm using plain English and keywords
- Breaking a complex problem into smaller, manageable parts is called:
 - decomposition
 - abstraction
 - pattern recognition
 - compilation
- In the "making tea" example, ignoring the brand of the kettle is an example of:
 - decomposition
 - sorting
 - abstraction
 - searching
- Pattern recognition forms the foundation of:
 - variables
 - loops
 - comments
 - input devices
- Bubble Sort works by comparing:
 - the first and last elements
 - every element with the middle element
 - adjacent elements
 - random pairs of elements
- After the first pass of Bubble Sort (ascending) on 8, 4, 1, 9, 3, the list is:
 - 1, 3, 4, 8, 9
 - 4, 8, 1, 9, 3
 - 1, 8, 4, 9, 3
 - 4, 1, 8, 3, 9
- Binary Search works only on a list that is:
 - sorted
 - unsorted
 - stored in a linked list
 - of strings only

9. Linear Search is also known as:
- a. binary search
 - b. sequential search
 - c. bubble search
 - d. random search
10. How many passes does Bubble Sort need to sort N elements?
- a. N
 - b. $N + 1$
 - c. $N - 1$
 - d. $2N$
11. In Selection Sort pseudocode, the inner loop starts at:
- a. 0
 - b. $N - 1$
 - c. outer - 1
 - d. outer + 1
12. The worst-case time complexity of Binary Search is:
- a. $O(\log N)$
 - b. $O(N)$
 - c. $O(N^2)$
 - d. $O(1)$
13. About how many steps does Binary Search need, at most, to find an item among 1 million sorted elements?
- a. 2
 - b. 20
 - c. 1,000
 - d. 1,000,000
14. Which sorting algorithm is NOT stable?
- a. Bubble Sort
 - b. Both are stable
 - c. Selection Sort
 - d. Neither sorts data
15. Which of the following is NOT an evaluation criterion of an algorithm?
- a. Correctness
 - b. Efficiency
 - c. Clarity
 - d. Popularity

Practice: short questions

1. Define computational thinking and name its four components.
2. Write any four points on the importance of algorithms.
3. Give three differences between an algorithm and pseudocode.
4. What is decomposition? Explain it with the example of learning a new language.
5. What three questions should you ask during pattern recognition? Apply them to the triangle of stars.
6. Define abstraction. In the "making tea" example, name two variables and two noise items.
7. Differentiate between sorting and searching.
8. Why does Binary Search require a sorted list? State two disadvantages of Binary Search.
9. Give two differences between Bubble Sort and Selection Sort.
10. Name the three evaluation criteria of an algorithm. What is a trace table used for?

Practice: long / problem questions

1. Sort the list **5, 2, 7, 1, 6** in ascending order using Bubble Sort. Show every comparison of every pass in a table, and state the total number of comparisons and swaps. Then write the Python code for Bubble Sort.

- Sort the list **6, 2, 9, 5, 1** in ascending order using Selection Sort (textbook method). Show every comparison of every pass and count the swaps. Also state why Selection Sort is called "blind" to an already sorted list.
- The sorted list is **3, 8, 12, 17, 23, 31, 40, 52, 66** (Python indexes 0–8). Using Binary Search, show a low / mid / high table to search for (a) 40 and (b) 20. For each, state how many comparisons Linear Search would need.
- Write the algorithm and pseudocode for Linear Search. Dry-run it on **8, 3, 6, 1, 7, 2, 4, 5** for FindValue = 2 and for FindValue = 9, and give two advantages and two disadvantages of Linear Search.
- A school wants a program that reads the marks of 5 subjects for every student and prints each student's total, percentage and grade. Apply decomposition, pattern recognition and abstraction to this problem, and write a short algorithm.
- A library has 50,000 book IDs stored in sorted order, and the librarian searches it hundreds of times a day. (a) Which search algorithm should be used and why? (b) What is the maximum number of comparisons for Linear and for Binary Search? (c) Explain the three evaluation criteria you would use to judge the chosen program.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. c | 7. d | 10. c | 13. b |
| 2. d | 5. b | 8. a | 11. d | 14. c |
| 3. a | 6. c | 9. b | 12. a | 15. d |

Short questions — model answers

- Computational thinking is a systematic approach to solving problems. Its four components are decomposition (breaking down), pattern recognition (finding repetitions), abstraction (hiding unnecessary details) and algorithms (writing step-by-step solutions).
- Algorithms (1) provide a structured approach that makes complex tasks manageable; (2) optimise time and memory usage, crucial for large applications; (3) can be reused for similar problems; (4) ensure accuracy and consistency, reducing human error; (5) act as a blueprint before coding. (Any four.)
- (1) An algorithm is a step-by-step logical procedure; pseudocode is a human-readable representation of it using English and keywords. (2) An algorithm is conceptual and language-independent; pseudocode is semi-formal and structured but not executable. (3) An algorithm has no fixed format and is for anyone; pseudocode uses IF/FOR/WHILE, follows conventions and is mainly for programmers.
- Decomposition is breaking a complex task into smaller, manageable parts. Learning a language is decomposed into: vocabulary (words like "I", "eat", "apple"), grammar rules (Subject → Verb → Object), tenses ("I eat" vs "I ate") and sentence building (combining words with rules).
- Ask: What is repeating? What changes in a predictable way? Can I generalise this? For the star triangle: stars repeat in every row; each row has one more star than the previous; generalisation: Row N = N stars, so one loop can print it.
- Abstraction means hiding unnecessary features and focusing on the essential ones. Variables: type of tea (black, green, milk), additions (sugar, lemon), tool (stove or kettle) — any two. Noise: brand of kettle, cup colour or design, where tea is made — any two.
- Sorting arranges a collection of items in ascending or descending order to organise data (e.g., Bubble Sort, Selection Sort). Searching locates a specific element in a list to find its position or whether it exists (e.g., Linear Search, Binary Search).
- Binary Search decides to go left or right by comparing with the middle value; this decision is only valid if all smaller values are on the left and larger on the right, i.e., the list is sorted. Disadvantages: needs sorted

data; needs random access (arrays, not linked lists); code is more complex and prone to off-by-one errors. (Any two.)

9. (1) Bubble Sort compares adjacent elements; Selection Sort compares a fixed position with all later elements. (2) Bubble Sort moves the largest to the end each pass; Selection Sort moves the smallest to the front. Also: Bubble Sort is stable, Selection Sort is not; Selection Sort makes fewer swaps.
10. Correctness, efficiency (time and space) and clarity (readability). A trace table records the values of variables step by step during a dry run, so we can check that the output matches the expected result and so prove correctness.

Long questions — model answers

1. Bubble Sort on 5, 2, 7, 1, 6

Pass	Pair	Left > Right?	List after
1	[5, 2]	Yes, swap	2, 5, 7, 1, 6
1	[5, 7]	No	2, 5, 7, 1, 6
1	[7, 1]	Yes, swap	2, 5, 1, 7, 6
1	[7, 6]	Yes, swap	2, 5, 1, 6, 7
2	[2, 5]	No	2, 5, 1, 6, 7
2	[5, 1]	Yes, swap	2, 1, 5, 6, 7
2	[5, 6]	No	2, 1, 5, 6, 7
3	[2, 1]	Yes, swap	1, 2, 5, 6, 7
3	[2, 5]	No	1, 2, 5, 6, 7
4	[1, 2]	No	1, 2, 5, 6, 7

Total comparisons = $4 + 3 + 2 + 1 = 10$; total swaps = $3 + 1 + 1 + 0 = 5$. Sorted list: 1, 2, 5, 6, 7.

```
data = [5, 2, 7, 1, 6]
n = len(data)
for outer in range(n - 1):
    for inner in range(n - 1 - outer):
        if data[inner] > data[inner + 1]:
            temp = data[inner]
            data[inner] = data[inner + 1]
            data[inner + 1] = temp
print(data)
```

OUTPUT
[1, 2, 5, 6, 7]

2. Selection Sort (textbook method) on 6, 2, 9, 5, 1

Pass	Compare	Values	Action	List after
1	Pos1 vs Pos2	$6 > 2$	Swap	2, 6, 9, 5, 1
1	Pos1 vs Pos3	$2 > 9$	No	2, 6, 9, 5, 1
1	Pos1 vs Pos4	$2 > 5$	No	2, 6, 9, 5, 1
1	Pos1 vs Pos5	$2 > 1$	Swap	1, 6, 9, 5, 2
2	Pos2 vs Pos3	$6 > 9$	No	1, 6, 9, 5, 2

2	Pos2 vs Pos4	6 > 5	Swap	1, 5, 9, 6, 2
2	Pos2 vs Pos5	5 > 2	Swap	1, 2 , 9, 6, 5
3	Pos3 vs Pos4	9 > 6	Swap	1, 2, 6, 9, 5
3	Pos3 vs Pos5	6 > 5	Swap	1, 2, 5 , 9, 6
4	Pos4 vs Pos5	9 > 6	Swap	1, 2, 5, 6, 9

10 comparisons, 7 swaps. Sorted list: 1, 2, 5, 6, 9. (With the classic one-swap-per-pass method the passes give 1, 2, 9, 5, 6 → 1, 2, 9, 5, 6 → 1, 2, 5, 9, 6 → 1, 2, 5, 6, 9.)

Selection Sort is "blind" to a sorted list because its loops have fixed limits: it always makes all $N(N-1)/2$ comparisons, even if the list is already in order. It has no way to notice that no swaps were needed and stop early.

3. (a) Search for 40

Step	low	high	mid	data[mid]	Compare	Action
1	0	8	$(0+8)//2 = 4$	23	40 > 23	low = 5
2	5	8	$(5+8)//2 = 6$	40	40 = 40	Found at index 6

Binary: 2 comparisons. Linear Search: 40 is the 7th item, so 7 comparisons.

(b) Search for 20

Step	low	high	mid	data[mid]	Compare	Action
1	0	8	4	23	20 < 23	high = 3
2	0	3	$(0+3)//2 = 1$	8	20 > 8	low = 2
3	2	3	$(2+3)//2 = 2$	12	20 > 12	low = 3
4	3	3	3	17	20 > 17	low = 4
—	4	3	low > high, so the loop stops: Not Found			

Binary: 4 comparisons. Linear Search must check all 9 items before saying "Not Found".

4. Algorithm and pseudocode: as in section 2.4.2 (Start; DATA, N, FindValue, ValueFound := 0; repeat for PTR = 1 to N: if DATA[PTR] = FindValue then ValueFound = 1 and exit; after the loop print DATA[PTR] if ValueFound = 1, else "Not Found"; End).

FindValue = 2:

Step	1	2	3	4	5	6
Element	8	3	6	1	7	2
== 2?	No	No	No	No	No	Yes — exit

Found at position 6 after 6 comparisons; ValueFound = 1; prints 2.

FindValue = 9: 8, 3, 6, 1, 7, 2, 4, 5 are all checked (8 comparisons), none equals 9, ValueFound stays 0, so "Data Not Found" is printed.

Advantages: works on sorted and unsorted data; very simple code; fast for small lists; no extra space. Disadvantages: up to N checks (1 million for 1 million items); time grows linearly; inefficient if the same list is searched thousands of times. (Any two of each.)

5. **Decomposition:** (1) input a student's 5 marks; (2) calculate total; (3) calculate percentage = $\text{total} \div 500 \times 100$; (4) decide grade; (5) print result; (6) repeat for the next student.

Pattern recognition: the same steps repeat for every student, and the same "add a mark" step repeats for every subject — so use a loop over students and a loop over subjects.

Abstraction: keep only roll number, marks, total, percentage and grade; ignore noise such as the student's address, uniform or the colour of the result card. The number of students can be a variable.

Algorithm: Step 1: Start. Step 2: Read number of students S. Step 3: Repeat Steps 4–8 for each student. Step 4: TOTAL = 0. Step 5: Repeat 5 times: read MARK, TOTAL = TOTAL + MARK. Step 6: PER = TOTAL / 500×100 . Step 7: If PER ≥ 80 grade "A1", else if PER ≥ 70 "A", else if PER ≥ 60 "B", else if PER ≥ 50 "C", else "Fail" (school's own grading scale). Step 8: Print TOTAL, PER, GRADE. Step 9: End.

6. (a) **Binary Search**, because the data is already sorted, it is large (50,000 items) and it is searched many times a day, so the time saved per search adds up; Linear Search would be far slower.

(b) Linear Search worst case: 50,000 comparisons. Binary Search worst case: 16 comparisons, because $2^{15} = 32,768 < 50,000 \leq 2^{16} = 65,536$, so 16 halvings are enough to reduce 50,000 items to one.

(c) **Correctness:** it must return the right position for IDs that exist, say "Not Found" for missing IDs (edge cases: first ID, last ID, missing ID), and always stop. **Efficiency:** time $O(\log N)$ and no extra memory ($O(1)$).

Clarity: meaningful names like low, mid, high, clear steps and good indentation so another programmer can maintain it.

Self-assessment: I can...

- | | |
|---|---|
| ☐ define computational thinking and name its four parts. | ☐ define an algorithm and give five reasons it is important. |
| ☐ draw a table comparing algorithm and pseudocode on at least five points. | ☐ decompose a real-life problem into sub-tasks and explain three benefits. |
| ☐ find a pattern in data and turn it into a loop. | ☐ separate necessary steps, variables and noise using abstraction. |
| ☐ dry-run Bubble Sort pass by pass and count comparisons and swaps. | ☐ dry-run Selection Sort pass by pass and explain why it is not stable. |
| ☐ trace Linear Search with a step table, including the "not found" case. | ☐ trace Binary Search with a BEG / MID / END table and explain why it needs sorted data. |
| ☐ state best and worst time complexity for all four algorithms. | ☐ evaluate an algorithm for correctness, efficiency and clarity, and choose the right algorithm for a situation. |

Programming Fundamentals

In this chapter you start writing real programs in Python. You will learn how programs store data, take input, make decisions, repeat work, use ready-made libraries, and how to find and fix your own mistakes.

اس باب میں آپ پائتھن (Python) میں حقیقی پروگرام لکھنا سیکھیں گے۔ آپ جانیں گے کہ پروگرام ڈیٹا کیسے محفوظ کرتا ہے، فیصلے کیسے کرتا ہے، کام کو کیسے دہراتا ہے، اور غلطیاں کیسے تلاش کی جاتی ہیں۔

What you will learn

- Define a program and a programming language, and compare low-, mid- and high-level languages.
- Describe the features of an IDE and the main parts of the VS Code window.
- Take input with `input()`, convert it with `int()` / `float()`, and display output with `print()` and f-strings.
- Use sequence, selection (`if`, `if-else`, `if-elif-else`, nested `if`) and repetition (`for`, `while`, `range()`, `break`, `continue`, nested loops).
- Identify syntax, runtime and logical errors and read a traceback to fix a bug.
- Name career areas where Python is used.
- Write Python programs that use comments, correct indentation, valid variable names and the main data types.
- Evaluate expressions with arithmetic, assignment, relational, logical, membership and bitwise operators, including binary working.
- Import and use built-in libraries (`math`, `random`, `datetime`) and install third-party libraries with `pip`.

3.1 Programming

LEARN IT

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own. They need exact, step-by-step **instructions**.

- A set of instructions given to a computer is called a **program**.
- The language used to write these instructions is called a **programming language**.
- Writing programs is called **programming**, and the person who writes them is a **programmer**.

Think of a recipe. A cook follows the recipe line by line: "boil water, add rice, wait 15 minutes". A computer follows a program in the same way, one instruction at a time, without guessing what you meant.

WORKED EXAMPLE

A tiny program. Here is a complete Python program with one instruction. It tells the computer to show a message on the screen.

```
print("Hello")
```

OUTPUT
Hello

The word `print` is the instruction; `"Hello"` is the data it works on.

REMEMBER IT

Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix it for you.

COMMON MISTAKE

Students often think the computer "understands" what they want. It does not. If you type `Print("Hello")` with a capital P, Python stops with an error, because it only knows `print`. Be exact.

اروین سمجھیں

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔ اسے کام کرنے کے لیے ہدایات چاہئیں۔ ہدایات کے مجموعے کو پروگرام (Program) کہتے ہیں، اور جس زبان میں یہ ہدایات لکھی جاتی ہیں اسے پروگرامنگ لینگویج (Programming Language) کہتے ہیں۔

CHECK YOURSELF

1. What is a program?
2. Why do we say computers are "dumb" machines?

Answers: 1. A set of instructions given to a computer to perform a task. 2. Because they cannot think or do anything without instructions; they only follow exactly what they are told.

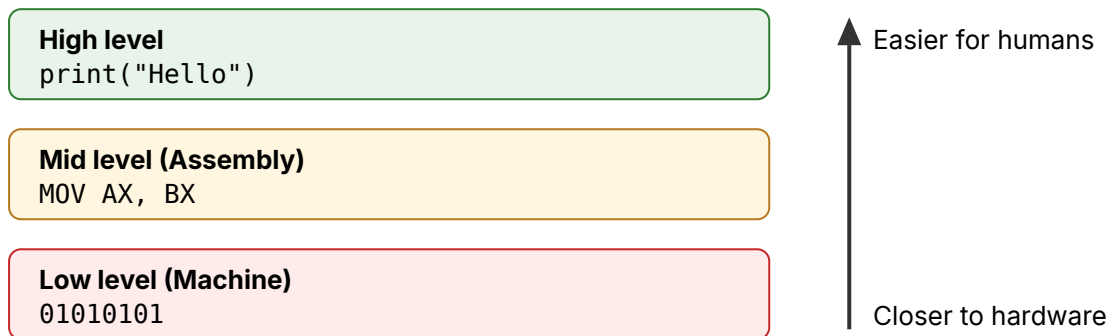
3.1.1 Programming Language Types

LEARN IT

Programming languages are classified into **three main types**, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)

Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. <code>MOV</code> , <code>ADD</code>).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



The three levels of programming languages, with a sample line from each.

Common high-level languages

Language	Key idea	Used for
C++	Powerful and closer to hardware.	System software, operating systems, high-performance computer games.
Java	"Write once, run anywhere."	Large enterprise applications, Android mobile apps, server programming.
Python	Simple, readable syntax.	Data Science, Artificial Intelligence, Web Development, and learning to program.

WORKED EXAMPLE

Same task, three levels. Suppose you want to copy a value.

- Machine language: a pattern like `01010101` — you would need a chart to know what it means.
- Assembly: `MOV AX, BX` — "move the value in register BX into register AX". Shorter words, but you still need to know the hardware.
- Python: `a = b` — you simply say "a gets the value of b".

The higher the level, the less you need to know about the hardware and the faster you can write the program.

REMEMBER IT

"Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

COMMON MISTAKE

"Low level" does not mean "bad" or "weak". Low-level code runs very fast and controls hardware directly. It is just harder for humans to write.

اردو میں سمجھیں

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو ہائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الفاظ (Mnemonics) استعمال کرتی ہے، اور ہائی لیول (High Level) جیسے پائتھن، جاوا اور سی پلس۔ سی پلس جو انگریزی جیسے الفاظ استعمال کرتی ہیں اور انسانوں کے لیے آسان ہیں۔

CHECK YOURSELF

1. Which level is `MOV AX, BX`?
2. Which high-level language is famous for "write once, run anywhere"?
3. Give one use of C++.

Answers: 1. Mid level (assembly language). 2. Java. 3. Any one: system software, operating systems, high-performance games.

3.2 Career Growth with Python

LEARN IT

Python is one of the most in-demand languages in the world. Learning it opens many career paths:

- **Software Development:** build desktop and web applications.
- **Data Science and Data Analysis:** analyse data, create reports and make predictions.
- **Artificial Intelligence and Machine Learning:** develop intelligent systems, chatbots and recommendation algorithms (like "videos you may like").
- **Web Development:** use frameworks such as **Django** and **Flask** to build websites.
- **Automation and Scripting:** automate repetitive office or lab tasks (renaming 500 files, filling spreadsheets).
- **Cybersecurity and Networking:** write security tools and analyse network data.

Why Python? Python lets you focus on **problem-solving and logic** instead of memorising complex syntax rules. Compare printing a message in Python with C++:

```
# Python
print("Hello")
```

```
// C++
#include <iostream>
int main() {
    std::cout << "Hello";
    return 0;
}
```

WORKED EXAMPLE

Automation example. A school clerk must write "Roll No. 1" to "Roll No. 5" on a list. Python does it in two lines:

```
for n in range(1, 6):  
    print("Roll No.", n)
```

OUTPUT

```
Roll No. 1  
Roll No. 2  
Roll No. 3  
Roll No. 4  
Roll No. 5
```

Change 6 to 501 and it writes 500 lines just as fast. (You will learn loops in 3.5.3.)

REMEMBER IT

Six careers — remember **"S-D-A-W-A-C"**: Software, Data, AI, Web, Automation, Cybersecurity.

COMMON MISTAKE

Django and Flask are **web frameworks** (for websites), not AI tools. AI/ML uses libraries such as TensorFlow and PyTorch (see 3.6.2).

اروین سمجھیں

پائتھن سیکھ کر آپ سافٹ ویئر ڈیولپمنٹ، ڈیٹا سائنس، مصنوعی ذہانت (AI)، ویب ڈیولپمنٹ، آٹومیشن اور سائبر سیکیورٹی جیسے شعبوں میں کیریئر بنا سکتے ہیں۔ پائتھن کا آسان انداز آپ کو مشکل اصول یاد کرنے کے بجائے مسئلہ حل کرنے پر توجہ دینے دیتا ہے۔

CHECK YOURSELF

1. Name two Python web frameworks.
2. Which career area builds chatbots and recommendation systems?

Answers: 1. Django and Flask. 2. Artificial Intelligence and Machine Learning.

3.3 Integrated Development Environment (IDE)

LEARN IT

An **IDE (Integrated Development Environment)** is a software tool that helps programmers **write, run and manage code in one place**. It is a complete workspace for programming. "Integrated" means many tools are joined together in one window.

Key features of an IDE:

- **Code Editor:** where you write and edit code.
- **Syntax Highlighting:** colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- **Auto-completion:** suggests code while you type (type `pri` and it offers `print`).
- **Run/Execute Tool:** runs the program with one click.
- **Debugger:** helps you find and fix errors, for example by pausing the program at a chosen line.

WORKED EXAMPLE

Without vs with an IDE. Without an IDE you would write code in Notepad, save it, open a terminal and type `python file.py`, then go back to Notepad to fix mistakes. With an IDE you write, run (press the ► Run button) and debug in the same window, and errors are underlined while you type.

REMEMBER IT

IDE features — "**E-S-A-R-D**": **E**ditor, **S**yntax highlighting, **A**uto-completion, **R**un, **D**ebugger.

COMMON MISTAKE

An IDE does not write the program for you and does not catch logical mistakes. It only makes writing, running and debugging easier.

اروین سمجھیں

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔ اس کی اہم خصوصیات میں کوڈ ایڈیٹر، رنگین کوڈ (Syntax Highlighting)، خودکار تجاویز (Auto completion)، رن بٹن اور ڈیبیگر شامل ہیں۔

CHECK YOURSELF

1. What does "integrated" mean in IDE?
2. Which IDE feature suggests code while you type?

Answers: 1. Many programming tools (editor, runner, debugger) are combined in one program. 2. Auto-completion.

3.3.1 Visual Studio Code (VS Code)

LEARN IT

VS Code is a **free, lightweight and powerful code editor** made by Microsoft and widely used for Python. By default it is **not a full IDE**, but it becomes one when you install **extensions** (add-ons), such as the **Python extension by Microsoft**.

VS Code interface elements

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

WORKED EXAMPLE

Setting up and running your first program in VS Code:

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) →

Explorer: new file `hello.py` → Type code → Click ► Run

```
print("Hello, World!")
```

OUTPUT

Hello, World!

The output appears in the **Terminal** panel at the bottom of the window.

REMEMBER IT

Extensions are the **"App Store"** of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

COMMON MISTAKE

Always save Python files with the `.py` extension (e.g. `hello.py`). Without it, VS Code does not know the file is Python, so there is no syntax highlighting and no Run button.

اردو میں سمجھیں

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔ یہ اپنے آپ میں مکمل آئی ڈی ای نہیں، لیکن ایکسٹینشنز (Extensions) لگانے سے آئی ڈی ای بن جاتا ہے۔ اس کے اہم حصے مینو بار، سائیڈ بار، ایکسپلورر، سرچ، سورس کنٹرول، رن اینڈ ڈیبگ اور ایکسٹینشنز ہیں۔

CHECK YOURSELF

1. Which VS Code element connects to Git?
2. Is VS Code a full IDE by default? How does it become one?
3. Where do you see and manage your project's files?

Answers: 1. Source Control. 2. No; it becomes an IDE by installing extensions (e.g. the Python extension). 3. Explorer.

3.4 Fundamentals of Python Programming ★ Golden

LEARN IT

The basic structure of a Python program is simple and clear. A program takes data, **processes it using statements**, and finally **shows output**. Python runs the statements **top to bottom, one after another** (sequentially).

A **statement** is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

1. **Comments (#)** — Comments are notes written in the code **for humans**. They start with the `#` symbol. The Python interpreter completely **ignores** them. Use them to explain what your code does.
2. **Indentation (spaces)** — C++ and Java use curly brackets `{ }` to mark a **block** of code (a group of statements that belong together). Python instead uses **indentation**: spaces at the beginning of a line. Indentation is **mandatory** in Python. The standard is **4 spaces** (or 1 tab).

```
# This is a comment. It explains the code below.
print("Hello, World!")
if 5 > 2:
    # The space before 'print' is the indentation
    print("Five is greater than two!")
```

OUTPUT

```
Hello, World!
Five is greater than two!
```

WORKED EXAMPLE

Reading the program line by line:

1. Line 1 starts with #, so Python skips it.
2. Line 2 prints Hello, World!.
3. Line 3 asks a question: is 5 greater than 2? The colon : says "a block follows".
4. Line 4 is a comment inside the block (ignored).
5. Line 5 is indented by 4 spaces, so it belongs to the if block. Because $5 > 2$ is true, it runs.

What happens if you forget the indentation?

```
if 5 > 2:
print("Five is greater than two!")
```

OUTPUT

```
File "indent.py", line 2
  print("Five is greater than two!")
  ^
```

```
IndentationError: expected an indented block after 'if' statement on line 1
```

Python refuses to run the program and points to line 2. Add 4 spaces before print to fix it.

REMEMBER IT

Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A colon : at the end of a line always means "indent the next line".

COMMON MISTAKE

IndentationError? Check your spaces. **Mixing tabs and spaces** can break a Python program even when the code looks lined up. Choose 4 spaces and use them everywhere (VS Code inserts 4 spaces when you press Tab in a .py file).

اردو میں سمجھیں

پائتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔ # سے شروع ہونے والی لائن تبصرہ (Comment) ہوتی ہے جسے پائتھن نظر انداز کر دیتا ہے۔ پائتھن میں کوڈ کے بلاک کو بریکٹ کے بجائے انڈینٹیشن (Indentation) یعنی لائن کے شروع میں چار خالی جگہوں سے ظاہر کیا جاتا ہے، اور یہ لازمی ہے۔

EXAM TIP

Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write: comments start with # and are ignored by the interpreter; Python uses indentation (4 spaces) instead of { }; wrong indentation gives an IndentationError. Always give a 3-4 line code example.

CHECK YOURSELF

1. Predict the output:

```
# print("A")
print("B")
if 3 > 7:
    print("C")
print("D")
```

2. What is the standard amount of indentation in Python?

Answers: 1. B then D on separate lines (line 1 is a comment; 3 > 7 is false so C is skipped). 2. 4 spaces (or 1 tab).

3.4.1 Variables ★ Golden

LEARN IT

A **variable** is a **named space in computer memory used to store data**. Think of it as a labelled container (a box with a name written on it). You put a value in it with the **assignment operator** `=`, which means "store", not "equals".

In Python you **do not need to declare the type** of a variable. Python figures it out automatically from the value. This is called **dynamic typing**: the same variable can later hold a different type.

Rules for naming variables:

- Can contain **letters, numbers and underscores** (`_`) only — no spaces or symbols like `-`, `@`, `$`.
- **Must start with a letter or underscore**, NOT a number (`1name` is invalid).
- Are **case-sensitive**: `age` and `Age` are two completely different variables.
- Cannot be a Python **keyword** (a reserved word such as `if`, `for`, `while`, `else`, `True`).

Valid names ✓

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
```

Invalid names ✗

```
1name = "Ali"    # starts with a number
my height = 5.7  # contains a space
if = 100         # keyword used
```

WORKED EXAMPLE

Using variables and case-sensitivity:

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
print(name, age, my_height, score1)

Age = 17          # a different variable from age
print(age, Age)
```

OUTPUT

```
Ali 16 5.7 100
16 17
```

Dynamic typing — the type follows the value. `type()` tells you the type:

```
x = 5
print(x, type(x))
x = "five"
print(x, type(x))
```

OUTPUT

```
5 <class 'int'>
five <class 'str'>
```

What Python says about an invalid name:

```
1name = "Ali"
```

OUTPUT

```
File "names.py", line 1
  1name = "Ali"
    ^
SyntaxError: invalid decimal literal
```

REMEMBER IT

Choose **meaningful names**: `total_marks` is far clearer than `tm` or `x`. Use lowercase words joined by underscores (this style is called **snake_case**).

COMMON MISTAKE

`=` is **not** "equals" in the maths sense. `x = x + 1` is not a false equation; it means "take the value of `x`, add 1, and store the result back in `x`". To **compare** two values you use `==` (see 3.4.3).

اردو میں سمجھیں

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈبے کی طرح۔ پائتھن میں ٹائپ بتانے کی ضرورت نہیں ہوتی۔ نام حرف یا انڈرسکور سے شروع ہونا چاہیے، اس میں خالی جگہ نہیں ہو سکتی، یہ کیس سینسیٹو (Case-sensitive) ہوتا ہے اور کوئی کی ورڈ (Keyword) نہیں ہو سکتا۔

EXAM TIP

A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place — invalid, starts with a digit". Remember that `While` (capital W) is **valid** because keywords are lowercase and Python is case-sensitive.

CHECK YOURSELF

1. Valid or invalid? `_total`, `roll-no`, `class`, `marks2026`, `2marks`
2. Predict the output:

```
city = "Karachi"
City = "Hyderabad"
print(city)
```

Answers: 1. `_total` valid; `roll-no` invalid (hyphen); `class` invalid (keyword); `marks2026` valid; `2marks` invalid (starts with a digit). 2. `Karachi` — `city` and `City` are different variables.

3.4.1 Data Types in Python ★ Golden

LEARN IT

A **data type** specifies the **kind of data** a variable holds — a whole number, a decimal, text, True/False, or a collection of items. The type decides what you can do with the data (you can divide numbers, but not names). Python has several built-in data types:

Name	Type	Description	Example
Integer	<code>int</code>	Whole numbers without decimals (positive, negative or zero).	<code>age = 16</code>
Floating point	<code>float</code>	Numbers with a decimal point.	<code>height = 5.7</code>
String	<code>str</code>	Text data, enclosed in quotes (" " or ' ').	<code>name = "Sammy"</code> or <code>year = "2026"</code>
Boolean	<code>bool</code>	Logical value: <code>True</code> or <code>False</code> (capital T / F).	<code>is_student = True</code>
List	<code>list</code>	Ordered, changeable collection in square brackets [].	<code>marks = [85, 90, 75]</code>
Tuple	<code>tuple</code>	Ordered, unchangeable sequence in parentheses ().	<code>coords = (10, 20)</code>
Dictionary	<code>dict</code>	Key : value pairs in curly braces { }.	<code>student = {"name": "Ahmed", "age": 16}</code>
Set	<code>set</code>	Unordered collection of unique items (no duplicates).	<code>vowels = {'a', 'e', 'i'}</code>

Note: the slide abbreviates tuple as "tup"; the real Python type name is `tuple`, as the program below shows. The slides describe dictionaries as unordered; since Python 3.7 a dictionary remembers the order in which keys were added, but you always look items up **by key**, not by position.

WORKED EXAMPLE

Checking every type with `type()` :

```
age = 16
height = 5.7
name = "Sammy"
year = "2026"
is_student = True
marks = [85, 90, 75]
coords = (10, 20)
student = {"name": "Ahmed", "age": 16}
vowels = {'a', 'e', 'i'}

print(type(age))
print(type(height))
print(type(name), type(year))
print(type(is_student))
print(type(marks))
print(type(coords))
print(type(student))
print(type(vowels))
```

OUTPUT

```
<class 'int'>
<class 'float'>
<class 'str'> <class 'str'>
<class 'bool'>
<class 'list'>
<class 'tuple'>
<class 'dict'>
<class 'set'>
```

Notice that `year = "2026"` is a **string**, not a number, because it is inside quotes.

Working with the collection types:

```
marks = [85, 90, 75]
marks[0] = 88           # lists are changeable
marks.append(60)
print(marks)

coords = (10, 20)
print(coords[0])        # read item at position 0

student = {"name": "Ahmed", "age": 16}
print(student["name"])   # look up by key

nums = {1, 2, 2, 3, 3, 3} # duplicates are removed
print(nums)
print(len(nums))
```

OUTPUT

```
[88, 90, 75, 60]
10
Ahmed
{1, 2, 3}
3
```

Positions (called **indexes**) start at **0**, so `coords[0]` is the first item. Trying to change a tuple causes an error:

```
coords = (10, 20)
coords[0] = 50
```

OUTPUT

```
Traceback (most recent call last):
  File "tuple_test.py", line 2, in <module>
    coords[0] = 50
    ~~~~~^
TypeError: 'tuple' object does not support item assignment
```

REMEMBER IT

Brackets decide the collection: `[ ]` List (changeable), `( )` Tuple (locked), `{ }` with **key:value** = Dictionary, `{ }` with single items = Set (unique). Think: a **list** is a shopping list you can edit; a **tuple** is your date of birth — it never changes.

COMMON MISTAKE

`True` and `False` must start with a capital letter. `true` is not a Boolean; Python treats it as an unknown variable name and gives a `NameError`. Also, `"16"` (string) and `16` (int) are different: `"16" + "1"` gives `"161"`, while `16 + 1` gives `17`.

اروین سمجھیں

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔ `int` مکمل اعداد، `float` اعشاریہ والے اعداد، `str` متن، اور `bool` صرف `True` یا `False` کے لیے ہے۔ لسٹ (List) تبدیل ہو سکتی ہے، ٹپل (Tuple) تبدیل نہیں ہو سکتا، ڈکشنری (Dictionary) میں کی اور ویلیو کے جوڑے ہوتے ہیں، اور سیٹ (Set) میں کوئی چیز دہرائی نہیں جاتی۔

EXAM TIP

Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — `[ ]`, changeable, e.g. `marks = [85, 90]`; tuple — `( )`, unchangeable, e.g. `coords = (10, 20)`.

CHECK YOURSELF

1. What is the data type of each? `3.0`, `"3"`, `3`, `False`, `[3]`
2. Predict the output: `print(len({5, 5, 6, 6, 6}))`
3. Which type should store a student's CNIC/B-Form number such as `"42101-1234567-1"` ? Why?

Answers: 1. float, str, int, bool, list. 2. 2 (the set keeps only 5 and 6). 3. str, because it contains hyphens and you never do arithmetic on it.

3.4.2 Input and Output Handling

LEARN IT

Output — the `print()` function. A **function** is a named, ready-made piece of code that does a job when you call it with brackets. `print()` displays information on the screen.

- Items separated by commas are printed with one space between them.
- An **f-string** (a string with `f` before the quote) lets you put variables directly inside `{ }`. It is the fastest and most readable way to format output.

Input — the `input()` function. It shows a message (a **prompt**), **pauses the program** and waits for the user to type something and press Enter. Whatever the user types is **always a string** by default.

Type casting means converting a value from one type to another: `int()` converts to an integer, `float()` to a decimal number, `str()` to a string.

```
# 1. Print only a string
print("Hello Class!")

# 2. Print a string and a variable (comma separates them)
name = "Ali"
print("Welcome,", name)

# 3. f-strings (fastest and most readable way to format)
age = 16
print(f"My name is {name} and I am {age} years old.")
```

OUTPUT

```
Hello Class!
Welcome, Ali
My name is Ali and I am 16 years old.
```

Two useful extras: `sep` changes the separator between items, and `end` changes what is printed at the end (normally a new line).

```
print("10", "09", "2026", sep="-")
print("Loading", end="...")
print("done")
print(f"Total: {7 * 3}")
```

OUTPUT

```
10-09-2026
Loading... done
Total: 21
```

WORKED EXAMPLE

Taking input with type casting.

```
# Taking text input
user_name = input("Enter your name: ")
# Type casting: converting string input to an integer (int)
age = int(input("Enter your age: "))
# Type casting: converting string input to a decimal (float)
height = float(input("Enter your height: "))
print(f"Hello {user_name}! Next year you will be {age + 1}.")
print("Height:", height, "feet")
```

Sample run (the user types Sara, 16 and 5.4):

OUTPUT

```
Enter your name: Sara
Enter your age: 16
Enter your height: 5.4
Hello Sara! Next year you will be 17.
Height: 5.4 feet
```

Why type casting matters. Without `int()`, the numbers stay as text and `+` joins them:

```
a = input("First number: ")
b = input("Second number: ")
print(a + b)
print(int(a) + int(b))
```

OUTPUT

```
First number: 5
Second number: 3
53
8
```

"5" + "3" joins two strings into "53" (called **concatenation**). 5 + 3 adds numbers to give 8.

REMEMBER IT

"input() always gives a string." If you plan to do maths, wrap it: `int(input(...))` for whole numbers, `float(input(...))` for decimals. Read it from the inside out: first `input`, then convert.

COMMON MISTAKE

If the user types letters where a number is expected, `int()` crashes with `ValueError: invalid literal for int() with base 10` (see 3.7). Also, `int("5.4")` fails — use `float()` for decimal input.

`print()` اسکرین پر معلومات دکھاتا ہے اور `input()` صارف سے معلومات لیتا ہے۔ یاد رکھیں کہ `input()` ہمیشہ اسٹرنگ (String) دیتا ہے، اس لیے حساب کے لیے اسے `int()` یا `float()` سے تبدیل کرنا ضروری ہے۔ اسے ٹائپ کاسٹنگ (Casting) کہتے ہیں۔ ایف اسٹرنگ (f-string) سے ویری ایبلز کو جملے کے اندر آسانی سے لکھا جا سکتا ہے۔

CHECK YOURSELF

1. Predict the output:

```
a = "3"
b = 4
print(a * b)
print(int(a) * b)
```

2. Find the error: the user types 21 but the output is 2121, not 42.

```
num = input("Enter a number: ")
print(num * 2)
```

Answers: 1. 3333 then 12 (a string times a number repeats the string). 2. `input()` returns the string "21", and a string times 2 repeats it. Fix: `num = int(input("Enter a number: "))`.

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

LEARN IT

An **operand** is the value or variable being worked on. An **operator** is the symbol that performs the action. Together they form an **expression**, which Python evaluates to a single value.

In `10 + 20`: **10** is an operand, **+** is the operator, **20** is an operand, and the result is **30**.

1. **Arithmetic operators** do maths:

Operator	Purpose	Example	Result
+	Addition	<code>5 + 3</code>	8
-	Subtraction	<code>10 - 4</code>	6
*	Multiplication	<code>4 * 2</code>	8
/	Division (result is always a float)	<code>9 / 2</code>	4.5
//	Floor division (drops the decimal part)	<code>9 // 2</code>	4
%	Modulus (remainder)	<code>9 % 2</code>	1
**	Exponentiation (power)	<code>2 ** 3</code>	8

```

print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)

```

OUTPUT

```

8
6
8
4.5
4
1
8
4.0

```

Notice the last line: `8 / 2` gives `4.0`, not `4`, because `/` always gives a float.

Order of operations (precedence). Python follows the same order as maths (like BODMAS): brackets first, then `**`, then `*` `/` `//` `%` (left to right), then `+` `-` (left to right). `**` is the exception: it works right to left.

```

print(2 + 3 * 4)
print((2 + 3) * 4)
print(2 ** 3 ** 2)
print(10 - 4 - 3)
print(-9 // 2)
print(17 % 5, 17 // 5)

```

OUTPUT

```

14
20
512
3
-5
2 3

```

`2 ** 3 ** 2 = 2 ** 9 = 512` (right to left). `-9 // 2` is `-5` because floor division always rounds **down** to the smaller whole number (`-4.5 → -5`).

2. Assignment operators store a value in a variable, often updating it. The short forms are called **compound assignment**.

Operator	Example	Equivalent to
<code>=</code>	<code>x = 5</code>	Assign 5 to x
<code>+=</code>	<code>x += 2</code>	<code>x = x + 2</code>
<code>-=</code>	<code>x -= 1</code>	<code>x = x - 1</code>
<code>*=</code>	<code>x *= 3</code>	<code>x = x * 3</code>
<code>/=</code> , <code>/=</code> , <code>%=</code> , <code>**=</code>	<code>x /= 2</code>	<code>x = x / 2</code> (and so on)

WORKED EXAMPLE

Tracing assignment operators. Follow the value of `x` line by line:

```
x = 5
x += 2      # x = 5 + 2
print(x)
x -= 1      # x = 7 - 1
print(x)
x *= 3      # x = 6 * 3
print(x)
x /= 2      # x = 18 / 2
print(x)
```

OUTPUT

```
7
6
18
9.0
```

Using `%` and `//` to split digits. `% 10` gives the last digit; `// 10` removes it.

```
n = 357
last = n % 10      # remainder after dividing by 10
rest = n // 10     # drop the last digit
print("Last digit:", last)
print("Remaining:", rest)
```

OUTPUT

```
Last digit: 7
Remaining: 35
```

$357 \div 10 = 35$ remainder 7. So `357 // 10 = 35` and `357 % 10 = 7`.

REMEMBER IT

Three kinds of division: `/` = exact answer (float), `//` = quotient only ("floor"), `%` = remainder only. A number is **even** when `n % 2 == 0`.

COMMON MISTAKE

`^` is **not** power in Python — it is bitwise XOR (see bitwise operators). `2 ^ 3` gives 1, not 8. For power, always use `**`.

ارو میں سمجھیں

آپرینڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔ `/` ہمیشہ اعشاریہ والا جواب دیتا ہے، `//` صرف حاصل تقسیم دیتا ہے، `%` باقی دیتا ہے اور `**` طاقت (Power) نکالتا ہے۔ `x += 2` کا مطلب ہے

`x = x + 2`

EXAM TIP

You will be asked to evaluate expressions such as `17 // 5`, `17 % 5`, `2 ** 4` or to trace compound assignment. Show each step and write the final value with its type (e.g. `9.0` is a float, so write the `.0`).

CHECK YOURSELF

1. Predict the output:

```
x = 7
y = 2
print(x // y, x % y, x / y)
print(x ** y)
```

2. Predict the output:

```
x = 10
x += 5
x /= 4
print(x)
```

Answers: 1. `3 1 3.5` then `49` . 2. `3` (`10 + 5 = 15`; `15 // 4 = 3`).

3.4.3 Relational, Logical and Membership Operators ★ Golden

LEARN IT

3. Relational (comparison) operators compare two values and return a **Boolean** (`True` or `False`). They are used in conditions.

Op	Description	Example (<code>a = 10, b = 5</code>)	Result
<code>=</code>	Equal to	<code>a == b</code>	<code>False</code>
<code>≠</code>	Not equal to	<code>a != b</code>	<code>True</code>
<code>></code>	Greater than	<code>a > b</code>	<code>True</code>
<code><</code>	Less than	<code>a < b</code>	<code>False</code>
<code>≥</code>	Greater than or equal to	<code>a ≥ 10</code>	<code>True</code>
<code>≤</code>	Less than or equal to	<code>a ≤ b</code>	<code>False</code>

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

OUTPUT

False
True
True
False
True
False

4. Logical operators combine several conditions:

A	B	A and B	A or B	not A
False	False	False	False	True
False	True	False	True	True
True	False	False	True	False
True	True	True	True	False

- **and** — True **only if both** conditions are true.
- **or** — True if **at least one** condition is true.
- **not** — **reverses** the result (True becomes False, False becomes True).

```
a = 10
b = 5
print(a > 5 and b > 5)
print(a > 5 or b > 5)
print(not a > 5)
```

OUTPUT

False
True
False

5. Membership operators check whether a value exists inside a sequence (such as a string or list): **in** and **not in**.

```
name = "Python"
print("P" in name)
print("z" not in name)
print("p" in name)
marks = [85, 90, 75]
print(90 in marks)
```

OUTPUT

True
True
False
True

"p" in name is False because strings are case-sensitive: the word has a capital P.

WORKED EXAMPLE

Real-life conditions. A person can apply for a CNIC if they are 18 or older **and** have the documents. Marks are invalid if they are below 0 **or** above 100.

```
age = 20
has_cnic = True
print(age ≥ 18 and has_cnic)
marks = 45
print(marks < 0 or marks > 100)
```

OUTPUT

```
True
False
```

Step by step: $20 \geq 18$ is True, `has_cnic` is True, True and True = **True**. For marks: $45 < 0$ is False, $45 > 100$ is False, False or False = **False** (the marks are valid).

REMEMBER IT

AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash **or** card. Order of evaluation: `not` first, then `and`, then `or`.

COMMON MISTAKE

Do not confuse `=` (assignment, "store") with `==` (comparison, "is it equal?"). Writing `if x = 5:` is a **syntax error**; the correct form is `if x == 5:`.

اردو میں سمجھیں

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔ لاجیکل آپریٹرز میں `and` تب True دیتا ہے جب دونوں شرطیں درست ہوں، `or` تب جب کم از کم ایک درست ہو، اور `not` نتیجہ الٹ دیتا ہے۔ `in` اور `not in` دیکھتے ہیں کہ کوئی چیز اسٹرنگ یا لسٹ میں موجود ہے یا نہیں۔

EXAM TIP

Typical: "Evaluate `a > b` and `b > 3` for `a = 10`, `b = 5`" or "Write the truth table of `and` / `or`". Write each sub-condition's value first, then combine. Remember: `==` compares, `=` assigns.

CHECK YOURSELF

1. Predict the output:

```
print(True and False or True)
print(not (5 > 3) or 2 == 2.0)
```

2. Find the error:

```
x = 5
if x = 5:
    print("five")
```

Answers: 1. True and True. (True and False = False; False or True = True. not True = False; 2 == 2.0 is True; False or True = True.) 2. = is assignment; a comparison needs ==. Python reports `SyntaxError: invalid syntax`. Maybe you meant '=' or '==' instead of '='? Fix: `if x == 5:`

3.4.3 Bitwise Operators ★ Golden

LEARN IT

6. Bitwise operators treat numbers as **binary digits** (bits: 0s and 1s) and work on them **bit by bit**, column by column. Think of each bit as a switch that is ON (1) or OFF (0). Python's `bin()` function shows the binary form of a number (it starts with `0b`).

- **AND (&)** — gives 1 **only if both** bits are 1, otherwise 0.
- **OR (|)** — gives 1 if **at least one** bit is 1.
- **XOR (^)** — gives 1 if the bits are **different**, 0 if they are the **same**.
- **NOT (~)** — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: `~x = -(x + 1)`. So `~10` gives `-11`.
- **Left shift (<<)** — moves bits left and adds 0s on the right. Each shift **multiplies by 2**.
- **Right shift (>>)** — moves bits right and removes the rightmost bits. Each shift **divides by 2** (floor division).

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

WORKED EXAMPLE

Worked example: a = 10, b = 6. First convert to binary using place values 8, 4, 2, 1: 10 = 8 + 2 = **1010**; 6 = 4 + 2 = **0110**. Then work column by column.

	8	4	2	1	Decimal
a = 10	1	0	1	0	10
b = 6	0	1	1	0	6
a & b	0	0	1	0	2
a b	1	1	1	0	14
a ^ b	1	1	0	0	12

- AND: only the "2" column has 1 in both → 0010 = **2**.
- OR: columns 8, 4 and 2 have at least one 1 → 1110 = 8 + 4 + 2 = **14**.
- XOR: columns 8 and 4 are different; column 2 is the same (both 1) → 1100 = 8 + 4 = **12**.
- NOT: $\sim 10 = -(10 + 1) = \mathbf{-11}$.

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

OUTPUT

```
0b1010 0b110
2
14
12
-11
```

Worked example: shifts. 5 in 8 bits is **0000 0101**.

Expression	Binary	Decimal	What happened
5	0000 0101	5	start
5 << 1	0000 1010	10	add one 0 on the right (5 × 2)
5 << 2	0001 0100	20	add 00 on the right (5 × 4)
20 >> 1	0000 1010	10	remove the rightmost bit (20 ÷ 2)
20 >> 2	0000 0101	5	remove two bits (20 ÷ 4)

```

a = 5          # Binary: 0000 0101
print(a << 1)   # 0000 1010
print(a << 2)   # 0001 0100
b = 20         # Binary: 0001 0100
print(b >> 1)   # 0000 1010
print(b >> 2)   # 0000 0101
print(bin(20 >> 2))

```

OUTPUT

```

10
20
10
5
0b101

```

REMEMBER IT

Shortcuts: $x \ll n = x \times 2^n$, $x \gg n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

COMMON MISTAKE

Do not mix up the logical and bitwise operators: `and` / `or` / `not` work on True/False conditions, while `&` / `|` / `~` work on the bits of integers. Also remember `bin(6)` prints `0b110`, without the leading zero — add zeros yourself when lining up columns.

اردو میں سمجھیں

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائنری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔ $\&$ تب 1 دیتا ہے جب دونوں بٹ 1 ہوں، $|$ تب جب کوئی ایک بھی 1 ہو، اور $\wedge$ تب جب دونوں بٹ مختلف ہوں۔ $\sim$ تمام بٹ الٹ دیتا ہے، اس لیے ~ 10 کا جواب 11 ہے۔ لیفٹ شفٹ عدد کو 2 سے ضرب دیتا ہے اور رائٹ شفٹ 2 سے تقسیم کرتا ہے۔

EXAM TIP

Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 \gg 1$ showing binary working". Marks are given for the binary conversion and column working, not just the final number. Always draw the column table.

CHECK YOURSELF

1. Predict the output (show the binary working):

```

print(12 & 10, 12 | 10, 12 ^ 10)
print(~5, 7 >> 1, 3 << 3)

```

2. What does `bin(13)` print?

Answers: 1. 8 14 6 and -6 3 24. $12 = 1100$, $10 = 1010$: AND = $1000 = 8$, OR = $1110 = 14$, XOR = $0110 = 6$. $\sim 5 = -(5 + 1) = -6$. $7 = 111 \rightarrow 11 = 3$. $3 = 11 \rightarrow 11000 = 24$ (3×8). 2. `0b1101` ($8 + 4 + 1$).

3.5 Control Structures ★ Golden

LEARN IT

Control structures decide the **order in which statements are executed** in a program (the "flow of control"). Almost every program is built from just three basic structures:

1. **Sequence** — do the steps one after another, top to bottom.
2. **Selection** — choose a path depending on a condition (`if` , `else`).
3. **Repetition** — repeat steps (loops: `for` , `while`).

WORKED EXAMPLE

Morning routine as control structures:

- Sequence: wake up → brush teeth → have breakfast.
- Selection: **if** it is raining, take an umbrella; **else** take a cap.
- Repetition: **while** the bus has not come, keep waiting.

REMEMBER IT

Remember **"S-S-R"**: **S**equence, **S**election, **R**epetition — "Step, Select, Repeat".

COMMON MISTAKE

Selection and repetition both use conditions, but they are different: selection runs a block **at most once**; repetition can run a block **many times**.

اردو میں سمجھیں

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔ تین بنیادی اسٹرکچر ہیں: سیکوئنس (ترتیب وار)، سلیکشن (شرط کی بنیاد پر انتخاب) اور ریپیٹیشن (دہرانا یعنی لوپ)۔

EXAM TIP

"Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

CHECK YOURSELF

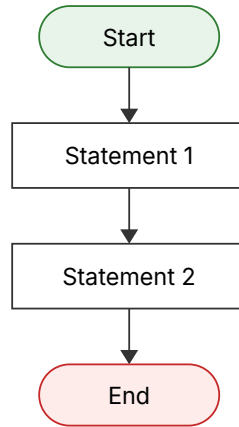
1. Name the three basic control structures.
2. Which structure is used for "keep asking for the password until it is correct"?

Answers: 1. Sequence, selection, repetition. 2. Repetition (a `while` loop).

3.5.1 Sequence

LEARN IT

Sequence means statements are executed **one after another, strictly from top to bottom**, in the exact order they are written. This is the **default behaviour** of Python. No statement is skipped or repeated.



Sequence: control flows straight down from one statement to the next.

WORKED EXAMPLE

Input → Process → Output in sequence.

```

print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
  
```

Sample run (user types 7 and 5):

OUTPUT

```

Enter two numbers
Enter first number: 7
Enter second number: 5
Sum = 12
  
```

Each line runs exactly once, in order. If you moved line 4 (`total = a + b`) above the input lines, Python would stop with a `NameError` because `a` and `b` would not exist yet.

REMEMBER IT

Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern **Input → Process → Output**.

COMMON MISTAKE

Order matters. A variable must be given a value **before** it is used. Using it first causes `NameError: name '...' is not defined`.

اردو میں سمجھیں

سیکوئنس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔ یہ پائتھن کا عام طریقہ ہے۔ کسی ویری ایبل کو استعمال کرنے سے پہلے اس میں ویلیو رکھنا ضروری ہے۔

CHECK YOURSELF

1. Predict the output:

```
x = 4
y = x * 2
x = 10
print(x, y)
```

2. What is the default control structure in Python?

Answers: 1. 10 8 — y was calculated when x was still 4; changing x later does not change y. 2. Sequence.

3.5.2 Selection Statements: if and if-else ★ Golden

LEARN IT

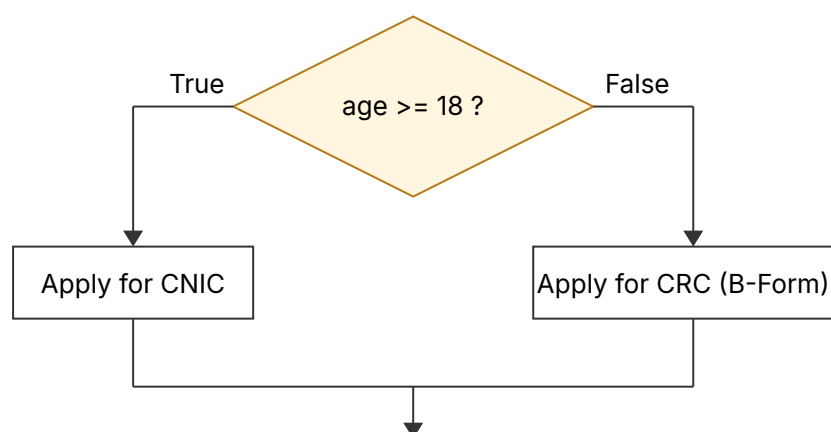
Selection changes the flow of a program based on a **condition** — an expression that is either True or False. It helps a program respond intelligently to different situations.

1. The **if statement** executes a block of code **only if** the condition is True. If the condition is False, the block is completely skipped.

```
if condition:
    statement(s)    # runs only when condition is True
```

2. The **if-else statement** provides **two paths**: if the condition is true, do this; **else** (otherwise), do that. Exactly one of the two blocks always runs.

```
if condition:
    block A         # condition True
else:
    block B         # condition False
```



An if-else statement: one diamond (condition), two paths, then the paths join again.

WORKED EXAMPLE

The if statement.

```
age = int(input("Enter your age: "))
if age ≥ 18:
    print("You are an Adult.")    # runs only if age ≥ 18
print("Program continues...")    # runs every time
```

Run 1 (user types 20):

OUTPUT

```
Enter your age: 20
You are an Adult.
Program continues...
```

Run 2 (user types 15):

OUTPUT

```
Enter your age: 15
Program continues...
```

The last line is not indented, so it is outside the `if` block and runs every time.

The if-else statement.

```
age = int(input("Enter your age: "))
if age ≥ 18:
    print("Apply for CNIC.")
else:
    print("Apply for CRC (B-Form).")
```

Sample run (user types 16):

OUTPUT

```
Enter your age: 16
Apply for CRC (B-Form).
```

Even or odd — a classic if-else program:

```
n = 7
if n % 2 == 0:
    print(n, "is even")
else:
    print(n, "is odd")
```

OUTPUT

```
7 is odd
```

$7 \% 2 = 1$, and $1 == 0$ is False, so the `else` block runs.

REMEMBER IT

Formatting rule: put a **colon (:)** after `if` and `else`, and **indent** the code inside the blocks. `else` never has a condition of its own.

COMMON MISTAKE

Common mistakes: forgetting the colon (`SyntaxError: expected ':'`), writing a condition after `else` (`else age < 18:` is wrong), and not indenting `else` at the same level as its `if`.

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔ if کا بلاک صرف تب چلتا ہے جب شرط درست (True) ہو۔ if-else میں دو راستے ہوتے ہیں: شرط درست ہو تو پہلا بلاک، ورنہ else والا بلاک چلتا ہے۔ if اور else کے بعد کولن (:) اور اندر کے کوڈ کی انڈینٹیشن ضروری ہے۔

EXAM TIP

Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`, the condition, both blocks, and a sample run.

CHECK YOURSELF

1. Predict the output when `marks = 32`:

```
marks = 32
if marks ≥ 33:
    print("Pass")
else:
    print("Fail")
print("Done")
```

2. Find the two errors:

```
n = 5
if n > 0
    print("Positive")
else n ≤ 0:
    print("Not positive")
```

Answers: 1. Fail then Done . 2. Missing colon after `if n > 0`; else must not have a condition — write `else:` .

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

LEARN IT

3. The **if-elif-else statement** handles **many conditions**. (`elif` is short for "else if".) Python checks the conditions **from top to bottom**. As soon as it finds a True condition, it runs that block and **ignores all the rest**. If none is True, the `else` block runs.

4. **Nested if** is an `if` inside another `if`. It is used when a decision needs to pass **several levels of checks**: the inner check happens only if the outer check passed.

WORKED EXAMPLE

Grading with if-elif-else.

```
marks = int(input("Enter marks: "))
if marks ≥ 80:
    print("A-1 Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 60:
    print("B Grade")
else:
    print("Fail or below B")
```

Sample run (user types 75):

OUTPUT

```
Enter marks: 75
A Grade
```

Trace for different marks:

marks	≥ 80?	≥ 70?	≥ 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Loan approval with nested if.

```
age = 20
salary = 35000
if age ≥ 18:
    # passed first check
    if salary ≥ 30000:
        # passed second check
        print("Loan APPROVED!")
    else:
        print("Salary too low for loan.")
else:
    print("Too young for loan.")
```

OUTPUT

```
Loan APPROVED!
```

age	salary	Outer check (age ≥ 18)	Inner check (salary ≥ 30000)	Output
20	35000	True	True	Loan APPROVED!
20	25000	True	False	Salary too low for loan.
16	35000	False	(never checked)	Too young for loan.

When you only need a yes/no answer, two levels can often be joined with `and` — but then you lose the separate messages:

```
age = 20
salary = 25000
if age ≥ 18 and salary ≥ 30000:
    print("Loan APPROVED!")
else:
    print("Loan not approved.")
```

OUTPUT
Loan not approved.

REMEMBER IT

In an `if-elif-else` ladder, put the **strictest condition first** (highest marks first). Think of a ladder: you step off at the first rung that fits.

COMMON MISTAKE

Wrong order gives wrong answers. Here 85 marks gets "B Grade", because `85 ≥ 60` is True first and the other conditions are ignored:

```
marks = 85
if marks ≥ 60:
    print("B Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 80:
    print("A-1 Grade")
```

OUTPUT
B Grade

This is a **logical error**: no crash, just a wrong result. Fix it by checking `≥ 80` first.

اروویں سمجھیں

`if-elif-else` میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔ نیسٹڈ ایف (Nested if) میں ایک `if` دوسرے `if` کے اندر ہوتا ہے، اور اندر والی شرط تب ہی دیکھی جاتی ہے جب باہر والی شرط درست ہو۔

EXAM TIP

Grade-calculator programs (`if-elif-else`) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks carefully (8 spaces for the inner block), and give a sample run. Tracing tables like the ones above earn full marks in "predict the output" questions.

CHECK YOURSELF

1. Predict the output:

```
temp = 30
if temp > 35:
    print("Hot")
elif temp > 25:
    print("Warm")
elif temp > 15:
    print("Pleasant")
else:
    print("Cold")
```

2. Predict the output:

```
x = 15
if x > 10:
    if x > 20:
        print("A")
    else:
        print("B")
else:
    print("C")
```

Answers: 1. Warm (30 > 35 is False; 30 > 25 is True, so the rest are skipped even though 30 > 15 is also True). 2. B (outer check 15 > 10 passes; inner check 15 > 20 fails).

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

LEARN IT

A **repetition statement (loop)** lets a block of code run **again and again**. This greatly reduces the need to write the same code many times. Each single run of the loop body is called an **iteration**.

1. The for loop is used when the **exact number of iterations is known**. It takes each value from a sequence one by one and stores it in the **loop variable** (e.g. `i`).

The `range()` function generates the sequence of numbers for the loop:

Syntax	How it works	Example	Values produced
<code>range(stop)</code>	From 0 up to, but not including , stop.	<code>range(3)</code>	0 1 2
<code>range(start, stop)</code>	From start up to, but not including, stop.	<code>range(2, 5)</code>	2 3 4
<code>range(start, stop, step)</code>	From start towards stop, jumping by step.	<code>range(1, 6, 2)</code>	1 3 5

```
for i in range(3):
    print(i)
```

```
for i in range(1, 6, 2):
    print(i)
```

OUTPUT

```
0
1
2
```

```
for i in range(2, 5):
    print(i)
```

OUTPUT

```
1
3
5
```

```
for i in range(0, 6, 2):
    print("Even:", i)
```

OUTPUT

```
2
3
4
```

OUTPUT

```
Even: 0
Even: 2
Even: 4
```

You can see all the values of a range at once by converting it to a list. A negative step counts **down**; if start equals stop, the range is empty:

```
print(list(range(5)))
print(list(range(2, 5)))
print(list(range(1, 6, 2)))
print(list(range(10, 0, -2)))
print(list(range(5, 5)))
```

OUTPUT

```
[0, 1, 2, 3, 4]
[2, 3, 4]
[1, 3, 5]
[10, 8, 6, 4, 2]
[]
```

WORKED EXAMPLE

Multiplication table of 7:

```
for i in range(1, 6):  
    print(f"7 x {i} = {7 * i}")
```

OUTPUT

```
7 x 1 = 7  
7 x 2 = 14  
7 x 3 = 21  
7 x 4 = 28  
7 x 5 = 35
```

Adding numbers with an accumulator. An **accumulator** is a variable that starts at 0 and collects a running total.

```
total = 0  
for i in range(1, 11):  
    total += i  
print("Sum of 1 to 10 =", total)
```

OUTPUT

```
Sum of 1 to 10 = 55
```

i	1	2	3	4	5	...	10
total after	1	3	6	10	15	...	55

Looping over a string or list. A `for` loop can also take items directly from any sequence:

```
for ch in "Sindh":  
    print(ch, end=" ")  
print()  
for m in [85, 90, 75]:  
    print(m + 5)
```

OUTPUT

```
S i n d h  
90  
95  
80
```

REMEMBER IT

range stops one short: `range(1, 11)` gives 1 to **10**. To include a number `n`, write `range(..., n + 1)`.
Number of values in `range(a, b)` = `b - a`.

COMMON MISTAKE

Writing `range(1, 10)` to print 1 to 10 misses 10 — an "off-by-one" logical error. Also, the loop body must be indented; a `for` line without an indented block gives `IndentationError: expected an indented block after 'for' statement`.

لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔ for لوپ تب استعمال ہوتا ہے جب پہلے سے معلوم ہو کہ کتنی بار دہرانا ہے۔ range(stop) صفر سے شروع ہوتا ہے، start، range(start, stop) سے، اور range(start, stop, step) ہر بار step کے برابر چھلانگ لگاتا ہے۔ آخری عدد (stop) کبھی شامل نہیں ہوتا۔

EXAM TIP

Expect "predict the output" questions on range() with a step, and programs such as "print a table", "sum of first n numbers", "even numbers from 1 to 20". Always state what range produces before tracing.

CHECK YOURSELF

1. Predict the output:

```
for i in range(10, 0, -3):
    print(i, end=" ")
```

2. Predict the output:

```
s = 0
for k in range(2, 11, 2):
    s += k
print(s)
```

3. Find the error:

```
for i in range(3):
    print(i)
```

Answers: 1. 10 7 4 1. 2. 30 (2 + 4 + 6 + 8 + 10). 3. The body is not indented: IndentationError. Indent print(i) by 4 spaces.

3.5.3 The while Loop, break and continue ★ Golden

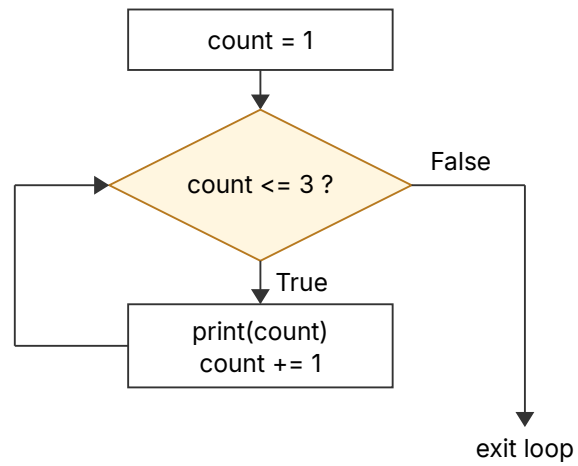
LEARN IT

2. The **while loop** repeats a block of code **as long as a condition remains True**. It is used when the exact number of iterations is **not known in advance** (for example, "keep asking until the password is correct").

A while loop needs three parts: (1) **initialise** a variable before the loop, (2) **test** the condition, (3) **update** the variable inside the loop. If you forget the update, the condition never becomes False and the loop never stops — an **infinite loop**.

Loop control statements change the normal flow of a loop:

- **break** — exits (stops) the loop **entirely and immediately**.
- **continue** — **skips the rest of the current iteration** and jumps to the next one.



A while loop: test the condition, run the body, go back and test again.

```
count = 1
while count ≤ 3:
    print(count)
    count += 1  # VERY IMPORTANT: update the condition!
```

OUTPUT

```
1
2
3
```

WORKED EXAMPLE

Trace of the while loop above:

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends

Sum of digits — we do not know how many digits the number has, so `while` fits well:

```
n = 4072
total = 0
while n > 0:
    digit = n % 10
    total += digit
    n = n // 10
print("Sum of digits =", total)
```

OUTPUT

Sum of digits = 13

Digits taken: 2, 7, 0, 4 → 2 + 7 + 0 + 4 = 13. `n` becomes 407, 40, 4, 0, and then the loop stops.

Password check — repeat until the user gets it right:

```
password = input("Enter password: ")
while password != "sindh123":
    print("Wrong password, try again.")
    password = input("Enter password: ")
print("Welcome!")
```

Sample run (user types `abc`, `sindh`, `sindh123`):

OUTPUT

```
Enter password: abc
Wrong password, try again.
Enter password: sindh
Wrong password, try again.
Enter password: sindh123
Welcome!
```

break and continue side by side:

```
for i in range(5):
    if i == 3:
        break
    print(i)
print("Loop ended")
```

```
for i in range(5):
    if i == 3:
        continue
    print(i)
```

OUTPUT

```
0
1
2
Loop ended
```

At $i = 3$ the loop stops completely, so 3 and 4 are never printed.

OUTPUT

```
0
1
2
4
```

At $i = 3$ only that iteration is skipped; the loop carries on with 4.

while True with break — a loop that runs "forever" until **break** stops it:

```
n = 1
while True:
    if n * n > 50:
        break
    n += 1
print("First number whose square is more than 50:", n)
```

OUTPUT

```
First number whose square is more than 50: 8
```

REMEMBER IT

break = "exit the room"; continue = "skip this turn". For every **while** loop, check the **I-T-U** rule: Initialise, Test, Update.

COMMON MISTAKE

Infinite loop — find the error:

```
count = 1
while count ≤ 5:
    print(count)
```

`count` is never updated, so `count ≤ 5` stays True and the program prints 1 forever. Fix: add `count += 1` inside the loop. (To stop a running program in the terminal, press **Ctrl + C**.)

اردو میں سمجھیں

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔ لوپ کے اندر ویری ایبل کو اپڈیٹ کرنا لازمی ہے ورنہ لوپ کبھی ختم نہیں ہوگا (Infinite Loop)۔ **break** لوپ کو فوراً مکمل طور پر روک دیتا ہے، جبکہ **continue** صرف موجودہ چکر چھوڑ کر اگلے چکر پر چلا جاتا ہے۔

EXAM TIP

"Differentiate **break** and **continue** with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners like seeing the difference in the output (0 1 2 vs 0 1 2 4).

CHECK YOURSELF

1. Predict the output:

```
n = 1
while n < 20:
    n = n * 3
print(n)
```

2. Predict the output:

```
for i in range(1, 8):
    if i % 3 == 0:
        continue
    if i == 7:
        break
    print(i, end=" ")
```

Answers: 1. 27 (n: 1 → 3 → 9 → 27; 27 < 20 is False, so the loop stops). 2. 1 2 4 5 (3 and 6 are skipped by continue; at 7 the loop breaks).

3.5.3 Nested Loops and for vs while ★ Golden

LEARN IT

A **nested loop** is a loop inside another loop. The **inner loop runs completely, from start to finish, for every single iteration of the outer loop**. If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

```
for outer in range(1, 3):          # runs 2 times (1, 2)
    for inner in range(1, 4):      # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

OUTPUT

```
Outer: 1, Inner: 1
Outer: 1, Inner: 2
Outer: 1, Inner: 3
Outer: 2, Inner: 1
Outer: 2, Inner: 2
Outer: 2, Inner: 3
```

WORKED EXAMPLE

A star triangle. The outer loop chooses the row; the inner loop prints that many stars on the same line (`end=""`), and the empty `print()` moves to the next line.

```
for row in range(1, 5):
    for col in range(row):
        print("*", end="")
    print()
```

OUTPUT

```
*
**
***
****
```

Row 1: inner loop runs once → *. Row 2: twice → **. And so on up to row 4.

for loop vs while loop

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
<code>for i in range(1, 4): print(i)</code>	<code>i = 1 / while i ≤ 3: / print(i) / i += 1</code>

REMEMBER IT

Ask yourself: "Do I know **how many times**?" Yes → `for`. "Do I only know **when to stop**?" → `while`.
Nested loops are like a clock: the minute hand (inner loop) goes all the way round for each step of the hour hand (outer loop).

COMMON MISTAKE

Indentation decides which loop a line belongs to. In the triangle program, if `print()` were indented under the inner loop, every star would go on its own line.

اردو میں سمجھیں

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔ جب دہرانے کی تعداد معلوم ہو تو `for` لوپ، اور جب صرف رکنے کی شرط معلوم ہو تو `while` لوپ بہتر ہے۔

EXAM TIP

"Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are typical. For pattern programs, explain what the outer and inner loops each control.

CHECK YOURSELF

1. Predict the output:

```
count = 0
for i in range(3):
    for j in range(2):
        count += 1
print(count)
```

2. Which loop would you use to keep reading numbers until the user enters 0?

Answers: 1. 6 (3×2). 2. A `while` loop, because the number of inputs is not known in advance.

Practical Activity: Shopping Bill Calculator

LEARN IT

This program combines all three control structures: **sequence** (steps in order), **selection** (checking "done", checking the price, checking for a discount) and **repetition** (a `while` loop that keeps adding items). It uses a **flag** — a Boolean variable (`shopping`) that controls when the loop stops. `item.lower()` converts the text to lowercase so that "done", "Done" and "DONE" all work.

```
print("===== SHOPPING BILL =====")
total = 0.0
count = 0
shopping = True

# REPETITION: keep adding items
while shopping:
    item = input("Enter Item Name (or 'done' to stop): ")
    # SELECTION: check if user wants to stop
    if item.lower() == "done":
        shopping = False
    else:
        price = float(input(f"Enter Price of {item}: "))
        # SELECTION: validate price
        if price <= 0:
            print("Invalid price!")
        else:
            total += price
            count += 1

# Summary and discount
if total >= 5000:
    discount = total * 0.20 # 20% off
else:
    discount = 0

final = total - discount
print(f"Total Items: {count}")
print(f"Total Bill: {total}")
print(f"Final to Pay: {final}")
```

WORKED EXAMPLE

Sample run. The user buys Rice for 3000, types a wrong price -50 for Soap, buys Oil for 2500, then types DONE:

OUTPUT

```
===== SHOPPING BILL =====
Enter Item Name (or 'done' to stop): Rice
Enter Price of Rice: 3000
Enter Item Name (or 'done' to stop): Soap
Enter Price of Soap: -50
Invalid price!
Enter Item Name (or 'done' to stop): Oil
Enter Price of Oil: 2500
Enter Item Name (or 'done' to stop): DONE
Total Items: 2
Total Bill: 5500.0
Final to Pay: 4400.0
```

Step	item	price	total	count
1	Rice	3000.0	3000.0	1
2	Soap	-50.0 (invalid, not added)	3000.0	1
3	Oil	2500.0	5500.0	2
4	DONE → shopping = False	—	5500.0	2

After the loop: $5500.0 \geq 5000$, so discount = $5500.0 \times 0.20 = 1100.0$, and final = $5500.0 - 1100.0 = 4400.0$.

REMEMBER IT

Label the parts of any program with comments, as here: # REPETITION, # SELECTION. It helps you, your teacher and the examiner see the structure.

COMMON MISTAKE

`total` and `count` must be set to 0 **before** the loop. If you put `total = 0.0` inside the loop, it resets on every item and the bill is wrong.

اردو میں سمجھیں

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: `while` لوپ سے اشیاء بار بار شامل ہوتی ہیں، `if-else` سے غلط قیمت روکی جاتی ہے اور 5000 یا زیادہ کے بل پر 20 فیصد رعایت دی جاتی ہے۔

CHECK YOURSELF

1. What will "Final to Pay" be if the user buys one item for 4000 and then types done?
2. What is the purpose of the variable `shopping`?

Answers: 1. Final to Pay: 4000.0 (4000 is less than 5000, so no discount). 2. It is a flag that keeps the `while` loop running; setting it to `False` stops the loop.

3.6 Libraries in Python

LEARN IT

A **library** (also called a **module**) is a collection of pre-written code that offers ready-made functions. Instead of writing a square-root algorithm from scratch, you import `math` and call `math.sqrt()`. Libraries save time, reduce bugs and let you stand on other programmers' work.

There are five common ways to import a library. Method 3 (import only what you need) is the best habit: it is clearer and uses less memory than `from math import *`.

Method	Syntax	How you call the function
1. Entire module	<code>import math</code>	<code>math.sqrt(25)</code>
2. Alias (nickname)	<code>import math as m</code>	<code>m.sqrt(16)</code>
3. Specific function	<code>from math import sqrt</code>	<code>sqrt(36)</code>
4. Several functions	<code>from math import sqrt, pi</code>	<code>sqrt(9)</code> , <code>pi</code>
5. Everything (*)	<code>from math import *</code>	<code>sqrt(64)</code> — avoid this

WORKED EXAMPLE

Five imports, one result each:

```
import math
print(math.sqrt(25))
import math as m
print(m.sqrt(16))
from math import sqrt
print(sqrt(36))
from math import sqrt, pi
print(pi)
from math import *
print(sqrt(64))
```

OUTPUT

```
5.0
4.0
6.0
3.141592653589793
8.0
```

COMMON MISTAKE

`from math import *` dumps every name into your program. If you later write a function called `sqrt`, you silently overwrite the library one. Prefer methods 1–4.

اردو میں سمجھیں

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے `import` سے پروگرام میں لایا جاتا ہے۔ سب سے اچھا طریقہ یہ ہے کہ صرف وہ فنکشن درآمد کریں جس کی ضرورت ہو، ستارہ `*` سے سب کچھ نہ کھینچیں۔

CHECK YOURSELF

1. Write the line that lets you call `ceil` without writing `math.` in front.
2. Why is an alias such as `import pandas as pd` useful?

Answers: 1. `from math import ceil`. 2. The library name is long; a short nickname keeps later lines short and is a community convention.

3.6.1 Built-in Libraries

LEARN IT

Built-in libraries come pre-installed with Python. You only import them; you do not run `pip`. Three that every Class XI student must know are `math`, `random` and `datetime`.

math

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

OUTPUT

```
3.141592653589793
5
4
16.0
9.0
```

random and datetime

```
import random
print(random.randint(1, 1))
print(0.0 ≤ random.random() < 1.0)

import datetime
now = datetime.datetime(2026, 9, 27, 11, 59, 40)
print(now.strftime("%d-%m-%Y"))
```

OUTPUT

```
1
True
27-09-2026
```

`randint(a, b)` includes both ends. `random()` is a float in `[0.0, 1.0)`. `strftime` formats a date; `%d-%m-%Y` is day-month-year.

REMEMBER IT

`ceil` always rounds **up** (ceiling); `floor` always rounds **down**. `round()` (built-in, not in `math`) goes to the nearest integer and is a different function.

اردو میں سمجھیں

`math` حساب کے لیے، `random` بے ترتیب عدد کے لیے اور `datetime` تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔ انہیں انسٹال نہیں کرنا پڑتا، صرف `import` کافی ہے۔

CHECK YOURSELF

1. What does `math.ceil(4.0)` return?
2. Write a line that prints a random integer from 1 to 6 (a dice).

Answers: 1. 4 (already an integer). 2. `print(random.randint(1, 6))` after `import random`.

3.6.2 Third-Party Libraries

LEARN IT

Third-party libraries are written by the community. They are **not** bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

WORKED EXAMPLE

Installing and importing Pandas (you will use this in Class XII):

```
pip install pandas
import pandas as pd
```

COMMON MISTAKE

`pip` is a **terminal** command, not a Python statement. Do not type `pip install pandas` inside a `.py` file. In VS Code, open the Terminal panel.

اردو میں سمجھیں

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔ نمپائی، پینڈا، میٹپلاٹ لیب اور ٹینسر فلو اسی قسم کی ہیں۔

CHECK YOURSELF

1. Which library would you choose to draw a bar chart of class marks?
2. Is `math` a third-party library?

Answers: 1. Matplotlib. 2. No — `math` is built-in.

3.7 Debugging Code in Python

LEARN IT

Debugging is the process of identifying, tracing and fixing bugs so a program works correctly. Every program you write will have bugs. The skill is not "never make a mistake"; it is "find the mistake quickly".

Python errors fall into three families:

Type	When it appears	What you notice	Example
Syntax error	Before the program even starts	Python refuses to run	<code>print("Hello" — missing)</code>

Runtime error (exception)	While the program is running	It crashes with a traceback	<code>x = 10 / 0 → ZeroDivisionError</code>
Logical error	Program finishes "successfully"	The output is wrong	<code>area = length + width</code> instead of <code>*</code>

WORKED EXAMPLE

Reading a traceback. When Python crashes it prints a roadmap: the file, the line number, and the error type.

```
x = int("abc")
```

OUTPUT

```
Traceback (most recent call last):
  File "debug_program.py", line 1, in <module>
    x = int("abc")
ValueError: invalid literal for int() with base 10: 'abc'
```

Diagnosis: on **line 1** we tried to convert the letters "abc" into an integer, which raised a `ValueError`.

EXAM TIP

Examiners love "identify the error type" questions. Ask: (1) Does it refuse to start? → syntax. (2) Does it crash mid-run? → runtime. (3) Does it run but give a wrong answer? → logical. Write the name of the error **and** how to fix it.

REMEMBER IT

Fix syntax errors first (the program cannot start). Then run it. Then check the output against a hand-worked example to catch logical errors.

COMMON MISTAKE

A logical error is the most dangerous because Python reports nothing. Always test with values whose answer you already know (for a rectangle 3 by 4, area must be 12, not 7).

اروین سمجھیں

ڈیبگنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔ نحوی غلطی پر پروگرام چلتا ہی نہیں، رن ٹائم پر چلتے چلتے رک جاتا ہے، منطقی غلطی پر چل تو جاتا ہے مگر جواب غلط دیتا ہے — یہ سب سے خطرناک ہے۔

CHECK YOURSELF

1. `print("Hi")` — which error type?
2. A program that should add two numbers multiplies them instead. Which error type?
3. `int("12.5")` raises which named exception?

Answers: 1. Syntax error. 2. Logical error. 3. `ValueError` (runtime).

Chapter 3 Review

Chapter summary

- A program is a set of instructions; programming languages are low-level (close to hardware) or high-level (close to English). Python is a high-level, interpreted, readable language.
- Python is widely used in web development, data science, AI and automation — skills that grow a career in Pakistan and abroad.
- An IDE (such as VS Code plus the Python extension) combines an editor, a terminal, a debugger and syntax highlighting.
- Variables store values; Python types are dynamic. Core types: `int`, `float`, `str`, `bool`, `list`, `dict`.
- `print()` displays output; `input()` always returns a **string** — convert with `int()` or `float()` before arithmetic.
- Operators: arithmetic (`+`, `-`, `*`, `/`, `//`, `%`, `**`), assignment (`=`, `+=`, ...), relational, logical (`and`, `or`, `not`), membership (`in`), bitwise (`&`, `|`, `^`, `~`, `<<`, `>>`).
- Control structures: sequence (top to bottom), selection (`if` / `if-else` / `if-elif-else` / nested), repetition (`for` / `while`).
- `range(stop)` goes 0 ... stop-1; `range(start, stop, step)` jumps by step. `break` leaves a loop; `continue` skips one iteration.
- Use `for` when the count is known; use `while` when you wait for a condition (and always update that condition).
- Libraries extend Python. Built-in: `math`, `random`, `datetime`. Third-party (need `pip`): NumPy, Pandas, Matplotlib, TensorFlow.
- Errors: syntax (grammar, will not run), runtime (crash, traceback), logical (runs, wrong answer).

Key terms

Term	Meaning	اردو
Program	A set of instructions a computer can follow.	پروگرام
High-level language	A language close to English (Python, Java, C++).	اعلیٰ سطحی زبان
Interpreter	Runs a program line by line without a separate compile step.	انٹرپریٹر
IDE	Software that combines editor, terminal and debugger.	انٹیگریٹڈ ڈویلپمنٹ انوائرنمنٹ
Variable	A named box that holds a value.	متغیر
Data type	The kind of value: <code>int</code> , <code>float</code> , <code>str</code> , <code>bool</code> , <code>list</code> , <code>dict</code> .	ڈیٹا کی قسم
Operator	A symbol that performs an operation on operands.	عامل
Operand	A value an operator works on.	معمول

Sequence	Instructions executed from top to bottom.	تسلسل
Selection	Choosing a path with <code>if / else</code> .	انتخاب
Repetition / loop	Repeating a block with <code>for</code> or <code>while</code> .	تکرار / لوپ
Nested	A structure placed inside another of the same kind.	تو در تو
Library / module	A collection of ready-made functions.	لائبریری / ماڈیول
pip	The tool that installs third-party Python libraries.	پپ
Syntax error	A grammar mistake; the program will not start.	نحوی غلطی
Runtime error	A crash while the program is running (exception).	رن ٹائم غلطی
Logical error	The program runs but the result is wrong.	منطقی غلطی
Traceback	Python's report of where a crash happened.	ٹریس بیک
Flag	A Boolean variable that controls a loop or decision.	فلیگ
Integer division	<code>//</code> division that drops the fraction.	صحیح تقسیم

Practice: multiple choice

1. Python is best described as a:

- a. low-level compiled language
- b. high-level interpreted language
- c. machine language
- d. hardware description language

2. `input("Age: ")` always returns a:

- a. int
- b. float
- c. str
- d. bool

3. The value of `17 // 5` is:

- a. 3.4
- b. 2
- c. 3
- d. 3.0

4. The value of `17 % 5` is:

- a. 2
- b. 3
- c. 3.4
- d. 0

5. `2 ** 3` equals:

- a. 6
- b. 8
- c. 9
- d. 5

6. Which operator is bitwise AND?
- a. `and`
 - b. `&&`
 - c. `&`
 - d. `^`
7. `list(range(2, 5))` is:
- a. `[2, 3, 4, 5]`
 - b. `[2, 3, 4]`
 - c. `[2, 5]`
 - d. `[0, 1, 2, 3, 4]`
8. A `while` loop is preferred when:
- a. you know the exact number of repeats
 - b. you must wait for a condition
 - c. you never need a counter
 - d. the body must run exactly once
9. `break` inside a loop:
- a. skips one iteration
 - b. restarts the loop
 - c. leaves the loop immediately
 - d. pauses for input
10. `from math import sqrt` lets you write:
- a. `math.sqrt(9)` only
 - b. `sqrt(9)`
 - c. `import.sqrt(9)`
 - d. `m.sqrt(9)`
11. Which library is built-in?
- a. Pandas
 - b. NumPy
 - c. random
 - d. Matplotlib
12. `print("Hi")` is a:
- a. syntax error
 - b. runtime error
 - c. logical error
 - d. warning only
13. `10 / 0` is a:
- a. syntax error
 - b. runtime error
 - c. logical error
 - d. style error
14. `area = length + width` when you meant to multiply is a:
- a. syntax error
 - b. runtime error
 - c. logical error
 - d. import error
15. `math.ceil(4.3)` is:
- a. 4
 - b. 4.0
 - c. 5
 - d. 4.3

Practice: short questions

1. Differentiate between a compiler and an interpreter.
2. Why must the result of `input()` often be converted with `int()`?
3. State the difference between `/` and `//` with one example each.
4. Write the truth table of the logical operator `and` for two Boolean values.
5. Explain `if-elif-else`. Why is the order of conditions important?
6. Differentiate between a `for` loop and a `while` loop.

7. What do `break` and `continue` do?
8. Name two built-in and two third-party Python libraries and their uses.
9. Differentiate between syntax, runtime and logical errors with one example each.
10. What three pieces of information does a Python traceback give you?

Practice: long / problem questions

1. Write a complete Python program that reads a student's marks (0–100) and prints the grade: 80+ A-1, 70–79 A, 60–69 B, otherwise Fail. Include input conversion and a sample run.
2. Write a program that prints all even numbers from 0 to 10 using `range`, then rewrite it with a `while` loop. Dry-run both.
3. Trace this nested loop and write every line of output:


```
for o in range(1, 3):
    for i in range(1, 4):
        print(o, i)
```
4. A shop gives 20% discount if the bill is 5000 or more. Write a program that keeps reading item prices until the user types 0, prints the total, the discount and the amount to pay. Show a sample run with prices 3000, 2500, 0.
5. Classify each bug and write the corrected line:
 - (a) `print("Total is", total`
 - (b) `avg = total / 0` when there are no items
 - (c) `discount = total + 0.20` when 20% off was intended.
6. Explain how you would use `math`, `random` and `datetime` in a small "lucky student of the day" program for a class of 40. Write the code.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. a | 7. b | 10. b | 13. b |
| 2. c | 5. b | 8. b | 11. c | 14. c |
| 3. c | 6. c | 9. c | 12. a | 15. c |

Short questions — model answers

1. A compiler translates the whole program into machine code before it runs (C, C++). An interpreter translates and runs line by line (Python). Compiled programs often run faster; interpreted ones are easier to test.
2. `input()` always returns a string. Adding strings concatenates them (`"2"+"3"` is `"23"`), so numeric work needs `int()` or `float()`.
3. `/` is true division: `17/5` is `3.4`. `//` is integer (floor) division: `17//5` is `3`.
4. `F and F` $\rightarrow$ `F`; `F and T` $\rightarrow$ `F`; `T and F` $\rightarrow$ `F`; `T and T` $\rightarrow$ `T`. Output is True only when both operands are True.
5. Python tests conditions from the top. The first True block runs and the rest are skipped. If you put `marks  $\geq$  60` above `marks  $\geq$  80`, an 85 would match the weaker condition first and get the wrong grade.
6. `for`: known count, counter updates automatically, good with `range`. `while`: unknown count, you must update the condition yourself or you get an infinite loop.
7. `break` leaves the loop immediately. `continue` skips the rest of this iteration and starts the next one.
8. Built-in: `math` (roots, ceil/floor), `random` (random integers). Third-party: Pandas (tables), Matplotlib (charts). (NumPy and TensorFlow also accepted.)

9. Syntax: missing bracket, will not run. Runtime: $10/0$, crashes. Logical: $\text{area} = l + w$, runs but answer is wrong.
10. The error type (e.g. `ValueError`), the file name, and the line number that caused the crash.

Long questions — model answers

```
1. marks = int(input("Enter marks: "))
if marks ≥ 80:
    print("A-1 Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 60:
    print("B Grade")
else:
    print("Fail or below B")
```

Sample: input 85 → A-1 Grade; input 72 → A Grade; input 40 → Fail or below B.

```
2. for i in range(0, 11, 2):
    print(i)
n = 0
while n ≤ 10:
    print(n)
    n += 2
```

Both print 0, 2, 4, 6, 8, 10.

3. Output (six lines): 1 1 / 1 2 / 1 3 / 2 1 / 2 2 / 2 3. The inner loop runs fully for each value of the outer loop.

```
4. total = 0.0
while True:
    p = float(input("Price (0 to stop): "))
    if p == 0:
        break
    if p > 0:
        total += p
    discount = total * 0.20 if total ≥ 5000 else 0
    print("Total", total)
    print("Discount", discount)
    print("Pay", total - discount)
```

Sample: $3000 + 2500 = 5500$; discount 1100; pay 4400.

5. (a) Syntax — add `)`: `print("Total is", total)`. (b) Runtime `ZeroDivisionError` — only divide when the count of items is not 0. (c) Logical — 20% of total is $\text{total} * 0.20$, not $\text{total} + 0.20$.

```
6. import random, datetime, math
n = random.randint(1, 40)
today = datetime.datetime.now().strftime("%d-%m-%Y")
print("Lucky roll", n, "on", today)
print("Root of roll", math.sqrt(n))
```

`random` picks the student number, `datetime` stamps the date, `math` shows a sample extra calculation.

Self-assessment: I can...

- ☐ classify a language as high-level or low-level and explain why Python is popular.
- ☐ read input, convert types, and print formatted output.
- ☐ write if, if-else, if-elif-else and nested selections.
- ☐ import built-in libraries (math, random, datetime) in more than one style.
- ☐ classify a bug as syntax, runtime or logical and read a traceback.
- ☐ declare variables and identify int, float, str, bool, list and dict values.
- ☐ evaluate arithmetic, relational, logical and bitwise expressions by hand.
- ☐ write for and while loops, including range, break and continue.
- ☐ name pip and at least two third-party libraries.
- ☐ combine sequence, selection and repetition in a small complete program.

Data and Analysis

This chapter teaches you how raw facts become organised information in a database, how tables are linked with keys, how to draw an ER diagram, and how to build tables, forms, queries and charts in Microsoft Access. Banks, NADRA, college libraries and fee offices all run on these ideas.

اس باب میں آپ سیکھیں گے کہ خام حقائق کو ڈیٹا بیس میں کیسے منظم کیا جاتا ہے، ٹیبلز کو کیسے جوڑا جاتا ہے، ای آر خاکہ کیسے بنتا ہے، اور مائیکروسافٹ ایکسیس میں ٹیبل، فارم، کوئری اور چارٹ کیسے بنائے جاتے ہیں۔

What you will learn

- Distinguish data, information and a database, and explain what a DBMS does.
- Choose candidate, primary, alternate and foreign keys and state integrity rules.
- Explain referential integrity, Cascade Update and Cascade Delete with before/after tables.
- Develop the college-library case study from requirements to the final ER schema.
- Create tables in Access (Datasheet and Design view) and pick data types.
- Identify tables, records and fields in a given example.
- Draw ER diagrams (entities, attributes, relationships) and give cardinality 1:1, 1:M, M:N.
- Write a relational schema and resolve an M:N relationship with a junction table.
- State the job of tables, forms, queries and reports.
- Build a form with the Form Wizard and a SELECT query with criteria, AND/OR, totals and charts.

4.1.1 Data, Information and Database

LEARN IT

Data are unprocessed facts, figures or observations. They can be **quantitative** (numbers: 80, 75, 90) or **qualitative** (descriptions: the colours of cars in a parking lot). Data on its own is not very useful — a list of marks does not yet tell you whether the class is strong.

When you organise, process and interpret data, it becomes **information**. Information has meaning and supports a decision. Example: "The class average is 80%, so we can move to the next chapter."

A **database** is an organised collection of **related** data, stored in a structured way so it can be accessed, managed and updated easily. Paper registers and loose files make searching painful; a database is built for search, update and sharing.

Data → Processing → Information → Decision

Aspect	Data	Information
Definition	Raw facts and figures	Processed, meaningful data
Example	85, 90, 75	Average marks = 83.3%
Usefulness	Little, on its own	Useful for decisions

WORKED EXAMPLE

Five CS XI marks: 80, 75, 90, 70, 85. Sum = 400. Average = $400 \div 5 = 80\%$. The numbers are data; "the class average is 80%" is information; a **Student** table holding every student's marks is a database.

REMEMBER IT

Exam line: *Data is raw; information is processed data used for decision-making.* Always add one pair of examples.

COMMON MISTAKE

Do not call a jumble of WhatsApp numbers a database. Related + structured + easy to update is what makes it a database.

اروین سمجھیں

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔ ڈیٹا بیس متعلقہ ڈیٹا کا منظم ذخیرہ ہے جسے آسانی سے تلاش اور تبدیل کیا جا سکتا ہے۔

CHECK YOURSELF

1. Is "Nadia, Class 11, 72 marks" data or information?
2. Why is a stack of unmarked answer sheets not a database?

Answers: 1. Data (raw facts about one student). 2. It is not organised in a structured, searchable, updatable collection of related records.

4.1.2 Database Management System (DBMS)

LEARN IT

A **DBMS** is software that lets you create, manage and interact with databases. It sits between the user and the stored files. You never edit raw data files by hand — you talk to the DBMS (Access, MySQL, Oracle, SQLite) and it handles storage, security and consistency.

Advantages you must be able to list:

- **Less redundancy.** The department name is stored once, not in every student row.
- **Consistency.** Change a phone number once; every screen that uses it shows the new number.
- **Security.** Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- **Integrity.** Rules stop duplicate roll numbers and empty primary keys.

WORKED EXAMPLE

In a paper register, Ali's address is written on the admission page, the fee page and the sports page. If he moves house, two of the three pages stay wrong. In a DBMS, Address lives in one **Student** table. Every form and report reads that single field.

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔ اس سے تکرار کم ہوتی ہے، ڈیٹا ایک جیسا رہتا ہے، صرف مجاز افراد دیکھ سکتے ہیں، اور غلط اندراج رک جاتی ہے۔

CHECK YOURSELF

1. Name two DBMS programs.
2. Which advantage stops two students sharing the same roll number?

Answers: 1. MS Access, MySQL (Oracle, SQLite also). 2. Integrity (unique / primary-key constraint).

4.2 Database Components (Tables, Records, Fields)

LEARN IT

The basic structure is a **table** (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

A **record** (row, tuple) is one complete instance — everything about Ali. A **field** (column, attribute) is one kind of fact — the whole Name column, or the whole Grade column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

The row 101 | Ali | CS | A is one **record**. The Name column is one **field**.

REMEMBER IT

Memory: **row = record = one person. column = field = one question on the admission form.**

COMMON MISTAKE

Do not mix two entities in one table (students and library books together). Each entity gets its own table; keys will link them later.

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔ ایک قطار ایک ریکارڈ (مثلاً علی کی پوری معلومات) ہے، ایک کالم ایک فیلڈ (مثلاً صرف نام) ہے۔

CHECK YOURSELF

1. How many records and how many fields are in the table above?
2. Is "CS" a record or a field value?

Answers: 1. 3 records, 4 fields. 2. A value stored in the Course field.

4.3.1 Keys and Integrity Constraints ★ Golden

LEARN IT

A **key** is a field (or a set of fields) that uniquely identifies records and links tables. Integrity constraints are the rules that keep those keys honest.

Candidate key. Any field (or combination) that *could* uniquely identify a row. A table may have several. In an Employee table: EmpID, EmpLicense# and EmpPassport are all candidate keys.

Primary key (PK). The one candidate you actually choose. Rules: unique, never NULL, only one PK per table (it may be a **composite** of two columns). Example: EmpID.

Alternate key. The leftover candidate keys. After EmpID is chosen, EmpLicense# and EmpPassport become alternate keys — still unique, just not “the” identifier.

Foreign key (FK). A field in this table that refers to the PK of another table. It *may* contain duplicates (many employees in one department) and, in some designs, NULL.

WORKED EXAMPLE

Classroom analogy. Roll Number, B-Form number and phone number can all identify a student (candidate keys). The school picks Roll Number as the primary key. B-Form and phone become alternate keys. In a **Result** table, RollNo is a foreign key pointing back to Student.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

EXAM TIP

Board questions: “Differentiate primary and foreign key” (table of 4 points) and “Why can a PK not be NULL?” Because a NULL would mean “this row has no identity”, so the DBMS could not tell rows apart.

COMMON MISTAKE

A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL). Roll numbers are issued by the school and stay unique.

اردو میں سمجھیں

کینڈیڈٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔ پرائمری کی منتخب شدہ شناخت ہے — منفرد اور خالی نہیں۔ جو بچ جائیں وہ آئرنیٹ کیز ہیں۔ فارن کی دوسرے ٹیبل کی پرائمری کی طرف اشارہ کرتی ہے اور ٹیبلز کو جوڑتی ہے۔

CHECK YOURSELF

1. Can a table have two primary keys?
2. Can a foreign key have duplicate values?

Answers: 1. No — only one PK (it may be composite). 2. Yes — many child rows may point to the same parent.

4.3 Relational Database Management System (RDBMS)

LEARN IT

An **RDBMS** stores data in several tables and uses primary and foreign keys to keep relationships. Banks, universities and NADRA use RDBMS software. Extra strengths on top of a plain DBMS:

- Reduces redundancy by splitting data into related tables.
- Supports many users at once, with permissions.
- Is queried with **SQL** (Structured Query Language) — a standard language of SELECT, INSERT, UPDATE, DELETE.

WORKED EXAMPLE

Instead of one giant spreadsheet of students-and-departments, an RDBMS has Department(DeptID, DeptName) and Student(RollNo, Name, DeptID FK). CS is stored once. SQL: SELECT Name FROM Student WHERE DeptID = 1;

اردو میں سمجھیں

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیڑ سے ان کے تعلقات برقرار رکھتا ہے۔ ایس کیو ایل سے سوال پوچھے جاتے ہیں۔ اس سے تکرار کم اور کئی صارفین محفوظ طریقے سے کام کر سکتے ہیں۔

CHECK YOURSELF

1. What language do you use to query an RDBMS?
2. Why split Department out of the Student table?

Answers: 1. SQL. 2. To store the department name once (less redundancy) and update it in one place.

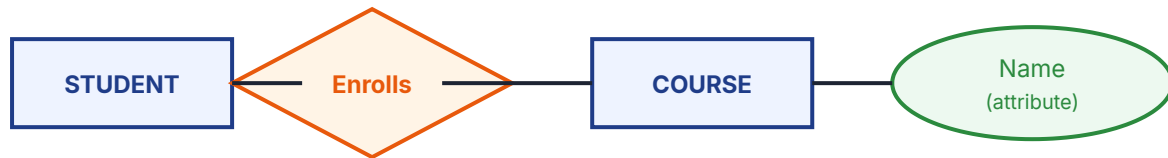
4.4 Entity Relationship Model (ER-Model) ★ Golden

LEARN IT

The **ER model** is the architect's blueprint: you draw what will be stored and how it connects *before* you create tables in Access. Three building blocks:

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class

Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has
--------------	--------------------------------	---------	-----------------------



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Cardinality (types of relationship)

Type	Meaning	Example
One-to-One (1:1)	One record $\leftrightarrow$ exactly one record. Rare; often for security.	One student has one ID card.
One-to-Many (1:M)	One record in A $\leftrightarrow$ many in B. The most common type.	One teacher teaches many students.
Many-to-Many (M:N)	Many in A $\leftrightarrow$ many in B.	Many students enrol in many classes.

COMMON MISTAKE

The M:N rule. A relational database cannot store M:N directly — it duplicates data. Break it into two 1:M relationships with a **junction table** (also called a bridge or associative table). Example: Enrollment(StudentID, ClassID, Date).

EXAM TIP

Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two foreign keys.

اردو میں سمجھیں

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔ مستطیل ہستی، بیضوی صفت، لوزی تعلق۔ تعلقات 1:1، M:1 یا M:N ہوتے ہیں۔ M:N کو دو M:1 میں توڑنے کے لیے جنکشن ٹیبل بنانا پڑتا ہے۔

CHECK YOURSELF

1. What symbol is used for a relationship?
2. A hospital: one patient, many appointments. Cardinality?

Answers: 1. Diamond. 2. 1:M (Patient 1 — has — M Appointment).

4.5 Referential Integrity ★ Golden

LEARN IT

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it. It stops **orphan records**.

If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student. You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

Parent — STUDENT

ID	Name
101	Ali
102	Sara

Child — RESULT

ID	Grade
101	A
999	F — rejected

Cascade operations

Cascade Update. Change the PK in the parent, and every matching FK in the child changes automatically. Ali's ID 101 → 201 updates Enrollments too. Nothing is left pointing at a vanished 101.

Cascade Delete. Delete the parent, and related child rows are deleted too. If Sara (102) leaves school, her Enrolment and Result rows disappear. That prevents "ghost" marks for a student who is gone.

WORKED EXAMPLE

Before Cascade Update

Student.ID	Enrol.StudentID
101	101

After changing 101 → 201 with Cascade Update ON

Student.ID	Enrol.StudentID
201	201

If Cascade Update were OFF, Access would refuse the change, or (in a sloppy file) leave Enrol stuck on 101 — an orphan.

COMMON MISTAKE

Cascade Delete is powerful and dangerous. Deleting one parent can wipe years of child history. In a real school, you more often **archive** a student (a Status field) than delete them.

EXAM TIP

Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔ کیسکیڈ اپ ڈیٹ والدین کی کی بدلنے پر بچوں کو خود بدل دیتی ہے؛ کیسکیڈ ڈیلیٹ والدین مٹانے پر متعلقہ بچے ریکارڈ بھی مٹا دیتی ہے۔

CHECK YOURSELF

1. What is an orphan record?
2. Cascade Delete: parent Teacher is removed. What happens to that teacher's Class rows?

Answers: 1. A child row whose FK value does not exist in the parent. 2. They are deleted automatically (if Cascade Delete is on).

4.6 Relational Schema Development

LEARN IT

A **relational schema** is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate. You write it *before* clicking around in Access.

It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships. Benefits: less redundancy, integrity planned in advance, easier to grow later.

WORKED EXAMPLE

Written schema for a tiny college:

```
Department (DeptID PK, DeptName)
Student     (RollNo PK, Name, Class, DeptID FK → Department)
Course      (CourseID PK, Title, Credit, DeptID FK → Department)
```

REMEMBER IT

Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

ریلیشنل اسکیم ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔ سافٹ ویئر کھولنے سے پہلے یہ بنانا چاہیے۔

CHECK YOURSELF

1. Name two things a schema must show.
2. Give one reason to write a schema before opening Access.

Answers: 1. Tables/fields and keys/relationships. 2. It reduces redundancy and mistakes (any valid benefit).

4.7 Case Study — Library Management System

LEARN IT

A college library needs a database. Walk the seven steps examiners expect.

Step 1 — Requirements. A book has ID, title, author, publisher. A member (student/teacher) has ID, name, contact. A member can borrow many books; a book can be borrowed by different members over time. Record Issue Date and Return Date.

Step 2 — Entities. Book, Member. (Borrowing will need a third entity in step 6.)

Step 3 — Attributes. Book: BookID (PK), Title, Author, Publisher. Member: MemberID (PK), Name, ContactInfo.

Step 4 — Relationship. Member *borrow*s Book.

Step 5 — Cardinality. One member borrows many books over time; one book is borrowed by many members over time → **M:N**.

Step 6 — Resolve M:N. Add junction entity **BorrowRecord**: BorrowID (PK), MemberID (FK), BookID (FK), IssueDate, ReturnDate. Now Member 1—M BorrowRecord M—1 Book (two 1:M links).

Step 7 — Final schema.

```
Member      (MemberID PK, Name, ContactInfo)
Book        (BookID PK, Title, Author, Publisher)
BorrowRecord (BorrowID PK, MemberID FK, BookID FK, IssueDate, ReturnDate)
```



Final library ER: two 1:M relationships through the BorrowRecord junction table.

WORKED EXAMPLE

Sample rows after Sara borrows “Khushwant” on 5 Oct 2026:

BorrowID	MemberID	BookID	IssueDate	ReturnDate
5001	102	B-17	2026-10-05	

The same book can later be borrowed by Ali as BorrowID 5002 — no duplicated book title, no duplicated member address.

اردو میں سمجھیں

لائبریری میں ممبر اور کتاب کا تعلق بہت-سے-بہت-سے ہے، اس لیے بورو ریکارڈ نامی پل ٹیبل بناتے ہیں جس میں دونوں کی فارن کیز اور اجراء/واپسی کی تاریخ ہوتی ہے۔

CHECK YOURSELF

1. Why is BorrowRecord needed?
2. Where do IssueDate and ReturnDate belong, and why not on Book?

Answers: 1. To break M:N into two 1:M links. 2. On BorrowRecord — they belong to a *transaction*, not to the book itself (a book is issued many times).

4.8 Database Objects (Tables, Forms, Queries, Reports)

LEARN IT

Four objects you will meet in Access. Learn the one-line jobs cold:

Object	Job	You ...
Table	Stores the data (foundation; every other object depends on it)	keep the actual records here
Form	Friendly screen for enter / edit / view (text boxes, drop-downs)	type data in; fewer mistakes than a raw grid
Query	Ask a question: filter, sort, calculate	e.g. "students above 80%"
Report	Formatted output for printing or sharing	read only — never type into a report

REMEMBER IT

Store → Enter → Search → Print. Tables, Forms, Queries, Reports.

COMMON MISTAKE

A query does not copy the table into a second storage place. It is a saved question. Change the table, re-run the query, and the answer updates.

اروین سمجھیں

ٹیبل ڈیٹا رکھتی ہے، فارم سے اندراج ہوتا ہے، کوئری سوال پوچھ کر چھاننتی ہے، رپورٹ چھپائی کے لیے سنواری ہوئی پیشکش ہے — اس میں ڈیٹا بھرا نہیں جاتا۔

CHECK YOURSELF

1. Which object would you use for a printed fee slip?
2. Which object actually holds the fee amount?

Answers: 1. Report. 2. Table.

4.9 Creation of Tables in Microsoft Access

LEARN IT

Plan first (4.9.1): purpose, field list, data type of each field, which field is the PK.

Datasheet View (4.9.2) — fast, spreadsheet-like, good for beginners, weak on rules.

1. Create tab → Table.
2. A blank grid appears.
3. Double-click Field1, Field2, ... to rename.
4. Type data; Access guesses the data type.

Limitation: you cannot easily say "Marks cannot exceed 100". Professionals use Design View.

Design View (4.9.3) — full control before any data is typed.

1. Create tab → Table Design.
2. Type each Field Name (StudentID, Name, ...).
3. Pick the Data Type from the dropdown.
4. Right-click the unique field → Primary Key (key icon).
5. Save the table.

Data type	Use it for
Short Text	Names, addresses, phone numbers (you do not add phone numbers). Up to 255 characters.
Number	Values you calculate with: Age, Marks, Quantity.
Date/Time	Date of birth, admission date, appointment time.
Currency	Fees, salary.
AutoNumber	Access fills a unique sequential number — excellent PK.

COMMON MISTAKE

Phone numbers look numeric but are Short Text. Leading zeros (0333...) would vanish in a Number field, and you never sum two phone numbers.

اردو میں سمجھیں

پہلے ٹیبل کا منصوبہ بنائیں۔ ڈیٹا شیٹ ویو تیز لیکن کمزور ہے؛ ڈیزائن ویو میں فیلڈ، ڈیٹا ٹائپ اور پرائمری کی خود منتخب کرتے ہیں۔ فون نمبر شارٹ ٹیکسٹ رکھیں۔

CHECK YOURSELF

1. Best data type for StudentID if you want Access to generate it?
2. Best data type for FeeAmount?

Answers: 1. AutoNumber. 2. Currency.

4.10 Designing Forms for Data Manipulation

LEARN IT

A form is the screen people actually type into. It cuts errors and can show validation messages. Every form **must** have a data source — a table or a query. A form with no source has nowhere to save.

Form Wizard (the exam method):

1. Create tab → Form Wizard.
2. Select the table (e.g. Student).
3. Move fields with > (one) or >> (all).
4. Choose layout. **Columnar** = one record at a time, best for data entry.
5. Name it ("Student Registration") → Finish.

WORKED EXAMPLE

A Student Registration form shows Student ID (AutoNumber, locked), Full Name, Class, Department as labelled boxes, plus Save. The clerk never sees the raw datasheet and cannot accidentally overwrite the next row.

اروین سمجھیں

فارم اندراج کی سادہ اسکرین ہے۔ ہر فارم کسی ٹیبل یا کوئری سے جڑا ہوتا ہے۔ فارم وزرڈ سے ٹیبل چنائیں، فیلڈز منتخب کریں، کالم نما لے آؤں بہتر ہے، نام دے کر ختم کریں۔

CHECK YOURSELF

1. Can a form exist without a table or query?
2. Which layout is best for entering one student at a time?

Answers: 1. No — it needs a data source. 2. Columnar.

4.11 Creating Queries in MS Access ★ Golden

LEARN IT

A query retrieves, filters and analyses data **without changing the original tables**. A simple SELECT query:

1. Create tab → Query Design.
2. Show Table → add Student → Close.
3. Double-click the fields you want; they drop into the grid.
4. Red Run (!) button executes it.

Criteria restrict rows. Type a value in the Criteria row under a field.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"

AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Wildcards in Access: * = any characters, ? = one character. Like "A*" finds Ali, Ahmed, Ayesha.

WORKED EXAMPLE

Equivalent SQL (what Access writes behind the grid):

```
SELECT StudentName, Department
FROM Student
WHERE Department = "Computer Science";

SELECT StudentName, Department
FROM Student
WHERE StudentName = "Ali" AND Department = "Computer Science";

SELECT StudentName, Department
FROM Student
WHERE Department = "IT" OR Department = "Computer Science";
```

EXAM TIP

Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

COMMON MISTAKE

Text criteria need quotes: "IT". Numbers do not: ≥ 80 . Mixing these is a common cause of empty results.

اردو میں سمجھیں

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔ ایک شرط Criteria قطار میں، AND ایک ہی قطار کی دو کالموں میں، OR الگ قطار (or) میں۔ وائلڈ کارڈ * کئی حروف، ? ایک حرف۔

CHECK YOURSELF

1. Criteria for marks at least 80?
2. You want CS or IT. Same row or two rows?

Answers: 1. ≥ 80 under Marks. 2. Two rows (OR).

4.11.4 Data Summarization and Statistical Analysis

LEARN IT

Grouping combines rows that share a value (all CS students together) and then an **aggregate** calculates a number for the group. In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
----------	--------------------

COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

WORKED EXAMPLE

Average marks per department: add Department and Marks; Totals Σ; Department → Group By; Marks → Avg; Run.

Department	Avg of Marks
Computer Science	82.5
IT	76.0

Check: CS marks 80, 85 → average $(80+85)/2 = 82.5$. IT 70, 82 → 76.0.

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

اردو میں سمجھیں

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔ ایکسیس میں Σ بٹن Total قطار کھولتا ہے۔

CHECK YOURSELF

1. Which Total setting goes under Department?
2. Which function counts how many books each member has borrowed?

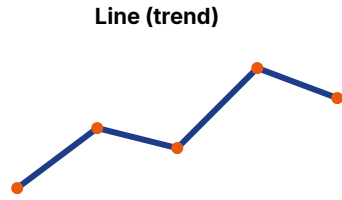
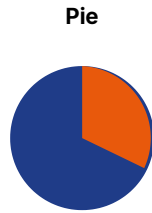
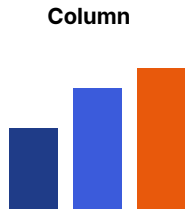
Answers: 1. Group By. 2. COUNT (of BorrowID or BookID) grouped by MemberID.

4.11.6 Data Visualization in MS Access (Charts)

LEARN IT

Charts turn numbers into a picture so a pattern jumps out. Pick the chart that matches the question:

Chart	Shape	Best for
Bar (horizontal)	Bars sideways	Categories with long names (English, Mathematics, ...)
Column (vertical)	Bars standing	Side-by-side comparison (2024 / 2025 / 2026 admissions)
Pie	Slices of a circle	Parts of one whole (percent of CS vs IT vs Eng). Few slices.
Line	Dots joined	A trend over time (visitors per month)



Column = compare categories. Pie = share of a total. Line = change over time.

EXAM TIP

"Which chart?" — line for time, pie for composition, bar/column for comparing categories. Mention one reason.

اروین سمجھیں

بار لمبے ناموں والی اقسام، کالم سادہ موازنہ، پائی حصوں کا تناسب، لائن وقت کے ساتھ رجحان۔ چارٹ کوئری یا ٹیبل کے اعداد کو تصویر بناتا ہے۔

CHECK YOURSELF

1. Website visitors each month — which chart?
2. Share of boys vs girls in one class — which chart?

Answers: 1. Line. 2. Pie (two parts of a whole).

Chapter 4 Review

Chapter summary

- Data = raw facts; information = processed data used for decisions; a database stores related data in a structured, updatable way.
- A DBMS is the software layer that provides less redundancy, consistency, security and integrity.
- Table = rows (records) and columns (fields). One table, one entity.
- Candidate keys could identify a row; the primary key is the one you choose (unique, not NULL); leftover candidates are alternate keys; a foreign key points at another table's PK.
- An RDBMS keeps several related tables and is queried with SQL.
- ER symbols: rectangle entity, oval attribute, diamond relationship. Cardinality 1:1, 1:M, M:N. M:N needs a junction table.
- Referential integrity forbids orphan FKs. Cascade Update copies a changed PK into children; Cascade Delete removes children with the parent.
- Write a relational schema before building. Library case: Member and Book are M:N, resolved by BorrowRecord.
- Objects: tables store, forms enter, queries search, reports print.
- Access Design View sets field names, data types and the PK. Phone numbers are Short Text; money is Currency; auto IDs are AutoNumber.
- Forms need a table/query source; Wizard + columnar layout for data entry.
- Query criteria: one value; AND on the same row; OR on two rows. Totals Σ for Group By / Count / Sum / Avg / Max / Min.
- Bar/column compare categories, pie shows parts of a whole, line shows a trend over time.

Key terms

Term	Meaning	اردو
Data	Raw facts and figures.	ڈیٹا / خام حقائق
Information	Processed data that supports a decision.	معلومات
Database	Organised collection of related data.	ڈیٹا بیس
DBMS	Software that creates and manages databases.	ڈیٹا بیس مینجمنٹ سسٹم
Table	Rows and columns about one entity.	ٹیبل
Record	One row — one instance.	ریکارڈ
Field	One column — one attribute.	فیلڈ
Primary key	Chosen unique, non-empty identifier.	پرائمری کی
Foreign key	Field that refers to another table's PK.	فارن کی

Candidate key	Any field(s) that could be the PK.	کینڈیڈیٹ کی
Alternate key	A candidate not chosen as PK.	آلٹرنیٹ کی
RDBMS	DBMS that stores related tables linked by keys.	ریلیشنل ڈی بی ایم ایس
SQL	Language to query relational databases.	ساختہ استفساری زبان
Entity	Object we store data about (ER rectangle).	ہستی
Attribute	Property of an entity (ER oval).	صفت
Relationship	Link between entities (ER diamond).	تعلق
Cardinality	1:1, 1:M or M:N.	عددییت
Junction table	Bridge table that splits an M:N into two 1:M.	جنکشن ٹیبل
Referential integrity	Every FK matches a live PK (no orphans).	حوالہ جاتی درستگی
Cascade Update	Parent PK change copied to child FKs.	کیسکیڈ اپ ڈیٹ
Cascade Delete	Deleting the parent deletes related children.	کیسکیڈ ڈیلیٹ
Schema	Written blueprint of tables, keys, relationships.	اسکیما / نقشہ
Form	Screen for entering data.	فارم
Query	Saved question that filters/calculates data.	کوئری
Report	Formatted printable output.	رپورٹ
AutoNumber	Access-generated unique integer, often the PK.	آٹو نمبر

Practice: multiple choice

- Processed data used for a decision is called:
 - a field
 - information
 - a query
 - redundancy
- A complete row about one student is a:
 - field
 - report
 - record
 - schema
- A primary key cannot contain:
 - numbers
 - text
 - NULL
 - an AutoNumber

4. EmpLicense# after EmpID is chosen as PK is an:
- a. foreign key
 - b. alternate key
 - c. orphan
 - d. report
5. A foreign key:
- a. must be unique in the child table
 - b. may contain duplicates
 - c. cannot exist in an RDBMS
 - d. is always AutoNumber
6. The ER symbol for a relationship is a:
- a. rectangle
 - b. oval
 - c. diamond
 - d. circle
7. The most common cardinality is:
- a. 1:1
 - b. 1:M
 - c. M:N stored as-is
 - d. 0:0
8. M:N is stored in an RDBMS by:
- a. duplicating both tables
 - b. a junction table
 - c. deleting one entity
 - d. using a pie chart
9. A child row whose StudentID is not in Student is:
- a. a candidate key
 - b. an orphan record
 - c. a form
 - d. a primary key
10. Changing parent ID 101 → 201 and seeing the child follow is:
- a. Cascade Delete
 - b. Cascade Update
 - c. a SELECT query
 - d. a pie chart
11. Which Access object is for printing a report card?
- a. Table
 - b. Form
 - c. Query
 - d. Report
12. Phone numbers should be stored as:
- a. Number
 - b. Currency
 - c. Short Text
 - d. Date/Time
13. AND criteria go:
- a. on two different rows of the grid
 - b. on the same Criteria row
 - c. in a report header
 - d. only in SQL Server
14. To average marks per department you set Department to:
- a. Sum
 - b. Where
 - c. Group By
 - d. Expression
15. The best chart for admissions in 2024, 2025, 2026 is a:
- a. pie chart
 - b. column chart
 - c. junction table
 - d. foreign key

Practice: short questions

1. Differentiate data and information with one example each.
2. List four advantages of a DBMS.
3. Define record and field with reference to a Student table.
4. Differentiate primary key and foreign key (four points).
5. Why can a relational database not store M:N directly?
6. What is referential integrity? Give one rejected-insert example.
7. Differentiate Cascade Update and Cascade Delete.
8. State the four Access objects and the job of each in one line.
9. Why are phone numbers Short Text, not Number?
10. How do you express OR in Query Design?

Practice: long / problem questions

1. Draw a complete ER diagram for Student–Course enrolment (M:N). Resolve it with a junction entity. Write the relational schema with keys marked.
2. A Result table uses StudentID as a foreign key. Explain referential integrity, then show before/after tables for (a) Cascade Update of ID 101 → 201 and (b) Cascade Delete of student 102.
3. Work through the library case study: requirements, entities, attributes, relationship, cardinality, junction table, final schema. Include two sample BorrowRecord rows.
4. Write step-by-step how to create a Student table in Access Design View (fields RollNo Number PK, Name Short Text, Marks Number, Fee Currency, AdmDate Date/Time) and then a columnar form with the Wizard.
5. Write the Query Design settings and the SQL to list CS students with Marks ≥ 80 , sorted by Name. Then write a totals query that shows COUNT of students and AVG(Marks) for each Department.
6. A principal wants to see (i) fee collected per class, (ii) the share of each class in total enrolment, (iii) enrolment over five years. Which chart for each, and why?

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. b | 7. b | 10. b | 13. b |
| 2. c | 5. b | 8. b | 11. d | 14. c |
| 3. c | 6. c | 9. b | 12. c | 15. b |

Short questions — model answers

1. Data = raw facts (80, 75, 90). Information = processed meaning ("average is 81.7%, the class is ready").
2. Less redundancy, consistency, security (authorised access), integrity (valid, unique keys). (Sharing / backup also accepted.)
3. Record = one student's full row (101, Ali, CS, A). Field = one column such as Name.
4. PK uniquely identifies a row, cannot be NULL, one per table, lives in the parent. FK refers to a PK, may duplicate, may be NULL, lives in the child, creates the relationship.
5. It would duplicate attributes of both entities. Split into two 1:M links with a junction table that holds both FKs plus extra attributes (dates).

6. Every FK must match an existing parent PK. Inserting Result.StudentID = 999 is rejected if 999 is not in Student — that would have been an orphan.
7. Cascade Update copies a changed parent PK into child FKs. Cascade Delete removes child rows when the parent is deleted.
8. Table stores; form enters; query searches/filters; report prints.
9. They are identifiers, not quantities; leading zeros would vanish; you never add two numbers of a phone.
10. Put one value on the Criteria row and the other on the or: row of the same field (or of different fields if that is the question).

Long questions — model answers

1. Entities STUDENT and COURSE, relationship Enrolls (M:N). Junction ENROLLMENT(EnrollID PK, StudentID FK, CourseID FK, Date). Schema: Student(StudentID PK, Name, Class); Course(CourseID PK, Title); Enrollment as above. Diagram: 1 — M — M — 1 through the junction.
2. (a) Parent 101 Ali → 201 Ali; child Result rows that said 101 now say 201. (b) Delete parent 102 Sara; all Result and Enrolment rows with 102 disappear. Integrity definition: FK must match a live PK; this stops orphans and keeps reports true.
3. Follow steps 1–7 in the chapter. Sample: (5001, 102, B-17, 2026-10-05, NULL); (5002, 101, B-17, 2026-10-12, 2026-10-20) — same book, two members, no duplicated titles.
4. Create → Table Design; enter the five fields and types; right-click RollNo → Primary Key; Save as Student. Create → Form Wizard → Student → >> all fields → Columnar → name Student Registration → Finish.
5. Grid: Name, Department, Marks. Criteria under Department: "CS" (or "Computer Science"); Criteria under Marks: ≥ 80 (same row = AND); Sort under Name: Ascending. SQL: `SELECT Name, Department, Marks FROM Student WHERE Department="CS" AND Marks  $\geq$  80 ORDER BY Name`; Totals query: Department Group By, StudentID Count, Marks Avg.
6. (i) Column or bar — compare categories (classes). (ii) Pie — each class is a part of one whole enrolment. (iii) Line — a trend over years.

Self-assessment: I can...

- | | |
|--|---|
| ☐ explain data vs information vs database vs DBMS. | ☐ point at records and fields in a table. |
| ☐ choose PK / FK / candidate / alternate keys correctly. | ☐ draw ER symbols and label 1:1, 1:M, M:N. |
| ☐ resolve M:N with a junction table. | ☐ explain referential integrity and both cascade operations with examples. |
| ☐ write a relational schema including the library case study. | ☐ match tables, forms, queries and reports to a task. |
| ☐ create a table in Design View and pick data types. | ☐ build a Wizard form and a criteria query (AND, OR, totals). |
| ☐ choose bar, column, pie or line for a given question. | |

Application and Impacts of Computing

This chapter shows you how computing technologies such as Artificial Intelligence, the Internet of Things and Data Analytics are changing life in Pakistan — in cities, hospitals, farms and classrooms. You will also learn how to judge information sources, weigh the good and bad effects of computing, and see how assistive technologies give everyone an equal chance.

اس باب میں آپ سیکھیں گے کہ مصنوعی ذہانت (AI)، انٹرنیٹ آف تھنگز (IoT) اور ڈیٹا اینالیٹکس ہماری روزمرہ زندگی کو کیسے بدل رہے ہیں۔ آپ معلومات کے ذرائع کو پرکھنا، کمپیوٹنگ کے مثبت اور منفی اثرات سمجھنا، اور معذور افراد کی مدد کرنے والی ٹیکنالوجی کے بارے میں بھی جانیں گے۔

What you will learn

- Explain what Artificial Intelligence (AI) is and give everyday examples of it.
- Name and explain the six components of an IoT system, with examples of each.
- Compare IoT, AI and Data Analytics by definition, purpose, type of work and output.
- Discuss the uses, benefits and challenges of AI in education in Pakistan.
- Analyse positive and negative impacts of computing in five fields, and practise responsible use.
- Describe how IoT, Data Analytics and AI work together in a smart system.
- List the five components of Data Analytics in the correct order.
- Describe uses of IoT in smart cities, industry, healthcare, education, homes and farming.
- Classify information sources as primary, secondary or tertiary, and judge when a source is unreliable.
- Explain the four categories of assistive technologies, their importance, and related careers.

5.1 Introduction to Computing Impacts

LEARN IT

Computing means using computers and computer-based devices to store, process and share information. Today computing is an important part of your daily life. You use computers, mobile phones, the internet and smart devices almost everywhere — at home, in college and on the road.

In Pakistan, computing is becoming **vital** (very necessary) in:

- **Schools and colleges** — online admissions, e-books, online results of the Sindh boards.
- **Hospitals** — computerised patient records, digital X-rays, online appointment systems.
- **Banks** — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- **Farms** — weather forecasts on mobile phones, soil sensors, tube-well control.
- **Industries** — automatic machines in textile mills in Karachi, Hyderabad and Faisalabad.

An **impact** is the effect something has on people and society. Computing has **positive impacts** (speed, convenience, new jobs) and **negative impacts** (privacy risks, addiction, job loss). In this chapter you will study three key technologies first — AI, IoT and Data Analytics — and then their uses and impacts.

WORKED EXAMPLE

A day in Sara's life in Karachi — spot the computing:

Step 1. 7:00 am — Her phone unlocks by recognising her face (AI).

Step 2. 7:30 am — She books a ride on a ride-hailing app; GPS shows the car's location (internet + location data).

Step 3. 9:00 am — College attendance is marked by a biometric (fingerprint) machine (IoT).

Step 4. 2:00 pm — She pays her fee through a mobile banking app (computing in business).

Step 5. 8:00 pm — YouTube suggests a physics lecture based on what she watched yesterday (AI).

Conclusion: in a single day, Sara uses computing in at least five different fields without even thinking about it.

اردو میں سمجھیں

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔ پاکستان میں اسکول، اسپتال، بینک، کھیت اور صنعتیں سب کمپیوٹر اور انٹرنیٹ پر انحصار کرنے لگے ہیں۔ کمپیوٹنگ کے فائدے بھی ہیں اور نقصانات بھی، جنہیں ہم اس باب میں سمجھیں گے۔

CHECK YOURSELF

1. Name four places in Pakistan where computing has become vital.
2. What is meant by the "impact" of computing?

Answers: 1. Any four of: schools, hospitals, banks, farms, industries. 2. The effect computing has on people and society — it can be positive or negative.

5.1.1 Artificial Intelligence (AI) ★ Golden

LEARN IT

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

Understand the name in two parts:

- **Artificial** — it is *created by humans*; it is not natural.
- **Intelligence** — it can perform tasks that *require thinking*.

AI systems **use data to make decisions**. The more good data an AI system sees, the better it becomes at recognising patterns (repeating features) and predicting what to do.

Examples from the source:

1. **Face recognition** on a mobile phone to unlock the screen.
2. **YouTube suggesting videos** based on your interests.
3. **Chatbots** answering questions on a website (for example, a bank's help chat).

More Pakistani examples: Google Maps predicting traffic on Shahrah-e-Faisal, Urdu voice typing on your phone, banks flagging a suspicious card transaction, and AI tools that translate English into Urdu.

AI in smart systems

AI rarely works alone. It often acts as the **"brain"** of a system, working together with the **Internet of Things (IoT)** and **Data Analytics**:

- **IoT devices gather data** — they are the "eyes" (sensors).
- **Analytics sorts the data** — it organises and studies it.
- **AI makes the final decision** — it is the "brain".
- An **action** follows — something happens in the real world.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

WORKED EXAMPLE

Smart air-conditioner in a hot Hyderabad room — trace the system:

Step 1 (IoT – eyes). A temperature sensor reads the room temperature: 36 °C.

Step 2 (Analytics). The readings of the last 10 minutes are organised: 34, 35, 35, 36 °C — the room is getting hotter.

Step 3 (AI – brain). The AI compares this with the comfortable limit (say 26 °C) and with the fact that a person is in the room (motion sensor). Decision: "Turn the AC on."

Step 4 (Action). A switch turns the AC on automatically.

No human pressed a button — the system sensed, thought and acted.

REMEMBER IT

Eyes – Sort – Brain – Do. IoT is the *eyes*, Analytics *sorts*, AI is the *brain*, and the actuator *does* the action. Think of a cricket fielder: his eyes see the ball (IoT), he judges speed and direction (analytics), decides to dive (AI), and his body moves (action).

COMMON MISTAKE

Mistake: "AI means robots." **Fix:** Most AI is just *software* — YouTube suggestions, spam filters and face unlock have no robot body. A robot may *use* AI, but AI itself is the thinking software.

اردو میں سمجھیں

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔ AI ڈیٹا کی مدد سے فیصلے کرتی ہے، جیسے چہرہ پہچان کر فون کھولنا یا یوٹیوب پر ویڈیوز تجویز کرنا۔ ایک اسمارٹ سسٹم میں IoT آنکھوں کا کام کرتا ہے اور AI دماغ کا۔

EXAM TIP

Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks).

To score full marks: (1) write the two-part meaning of the name — *artificial* = *made by humans*, *intelligence* = *can think/decide*; (2) say "AI uses data to make decisions"; (3) give at least two examples; (4) for the "together" question, draw the flow IoT → Analytics → AI → Action and explain it with the room-AC example.

CHECK YOURSELF

1. Why is AI called "artificial"?
2. In a smart system, which technology is the "brain" and which is the "eyes"?
3. Is a spam filter in Gmail an example of AI? Why?

Answers: 1. Because it is created by humans. 2. AI is the brain; IoT is the eyes. 3. Yes — it uses data (past spam emails) to decide whether a new email is spam, a task that needs "thinking".

5.1.2 Internet of Things (IoT) & Components ★ Golden

LEARN IT

The **Internet of Things (IoT)** is a system in which **physical devices are connected to the Internet** and can **collect and share data** to perform tasks **automatically**.

The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so they can "talk" to each other and to us.

Source example: A **smartwatch** collects your heart-rate data and sends it to your mobile phone, where it is displayed.

Components of IoT systems

IoT relies on **six foundational components** to work properly.

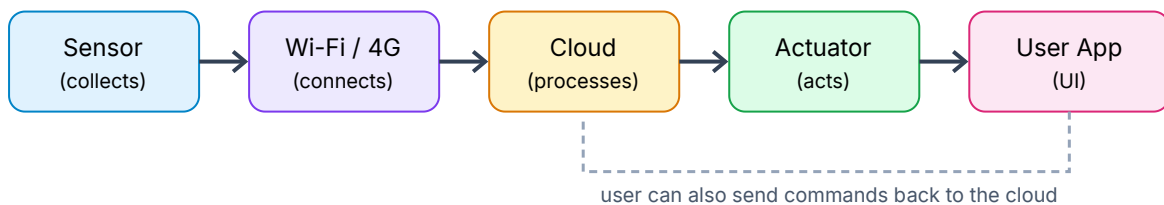
#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information. Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC

5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands

Understanding each component

- **Sensors** turn something in the real world (heat, movement, a fingerprint, smoke, wetness of soil) into data. A smoke sensor in a Karachi shopping mall "smells" smoke and reports it.
- **Actuators** do the opposite: they take a digital command and make something *move* or *change*. The water valve that opens, the motor that turns, the door that slides open at a bank entrance.
- **Connectivity** is the "road" for data. Short distances use Bluetooth (watch to phone); a house uses Wi-Fi; low-power sensor networks use **ZigBee**; devices far away in fields use mobile networks (4G/5G from Jazz, Zong, Telenor/Ufone).
- **Data processing** is where raw numbers ("36") become meaning ("too hot — turn on AC"). Its four steps are *collect* → *organise* → *analyse* → *decide*.
- **Cloud computing** stores and processes the huge amount of data that thousands of sensors produce every second — far more than a single small device can hold.
- **User Interface** is how *you* see the data and give orders — an app on your phone, a touch screen on a smart fridge, or saying "Hey Google, turn off the lights".

How the components connect



The six IoT components working as one system (data processing happens inside the cloud).

WORKED EXAMPLE

Smart fire alarm in a Karachi factory — identify all six components:

Step 1 — Sensor: A smoke sensor detects smoke near the machines.

Step 2 — Connectivity: It sends the reading over the factory Wi-Fi.

Step 3 — Cloud computing: The reading is stored on an online server along with the history of all sensors.

Step 4 — Data processing: The software checks: "Is the smoke level above the safe limit?" Yes → decide "Fire risk!"

Step 5 — Actuator: Water valves of the sprinklers open and the alarm bell switch turns on.

Step 6 — User interface: The manager's mobile app shows a red alert: "Smoke detected – Hall 3".

REMEMBER IT

Sensor senses, Actuator acts. Both words give you the answer: **sensor** = see/sense (input), **actuator** = act (output). Remember the six components with "**Some Ants Carry Data Carefully Up**" — Sensors, Actuators, Connectivity, Data processing, Cloud, UI.

COMMON MISTAKE

Mistake: Calling a fan controller or water valve a "sensor". **Fix:** Ask "does it *measure* something or *do* something?" A valve opens and closes — it *does*, so it is an **actuator**. A soil moisture sensor only *measures*, so it is a **sensor**.

اردو میں سمجھیں

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے ڈیٹا جمع اور شیئر کرتے ہیں۔ اس کے چھ بنیادی اجزاء ہیں: سینسرز (معلومات جمع کرتے ہیں)، ایکچویٹرز (جسمانی کام انجام دیتے ہیں جیسے موٹر چلانا یا دروازہ کھولنا)، کنیکٹیویٹی (وائی فائی، بلوٹوتھ)، ڈیٹا پروسیسنگ، کلاؤڈ کمپیوٹنگ، اور یوزر انٹرفیس (موبائل ایپ)۔

EXAM TIP

"Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — *physical devices, connected to the internet, collect and share data automatically*. Then list all **six** components, each with a one-line meaning and one example. A labelled flow diagram (Sensor → Wi-Fi → Cloud → Actuator → App) earns extra credit.

Short questions often ask: "Differentiate between a sensor and an actuator" — sensor collects data (input, eyes and ears), actuator performs physical action (output, converts digital signals into movement).

CHECK YOURSELF

1. Classify as sensor or actuator: (a) fingerprint scanner, (b) automatic door opener, (c) motion detector, (d) electric motor.
2. Name four types of connectivity used in IoT.
3. Why is cloud computing needed in IoT?

Answers: 1. (a) sensor, (b) actuator, (c) sensor, (d) actuator. 2. Wi-Fi, Bluetooth, ZigBee, 4G/5G. 3. Because IoT devices produce massive amounts of data; the cloud stores and processes it over the internet instead of on small local hard drives.

5.1.3 Data Analytics & Components

LEARN IT

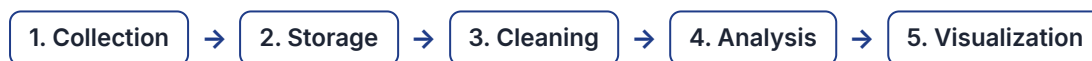
Data Analytics is the process of **collecting, organising, cleaning, studying and presenting data** to find useful information. It helps people make better decisions.

Data means raw facts and figures (for example, "72, 45, 88"). **Information** is data that has been processed so it has meaning ("Ali scored 72%; he passed"). Data Analytics is the bridge from data to information.

Source example: A college collects exam marks, calculates percentages, and identifies which students need extra help. This entire process is Data Analytics.

Components of Data Analytics

Data Analytics comprises **five key components**, which happen in order like a pipeline:



1. **Data Collection** — Gathering raw information from multiple sources: websites, surveys, sensors, hospital records, bank transactions.
2. **Data Storage** — Storing data safely and properly after collection. It can be stored in files, databases, spreadsheets or the cloud.
3. **Data Cleaning** — Identifying and correcting errors in data. Raw data may contain:
 - **Duplication** of records (the same student entered twice),
 - **Incorrect values** (typos, e.g. marks "870" instead of "87"),
 - **Missing** files or records (a student's marks not entered),
 - **Improper formats** (dates written as 05/10/2026 in one place and "5 Oct 26" in another).
4. **Data Analysis** — Studying the clean data to identify patterns, trends and relevant information for problem-solving (e.g. finding out which subjects students failed the most).
5. **Data Visualization** — Creating visual representations of data (plots, charts, graphs) to provide valuable insights at a glance (e.g. a pie chart showing pass vs fail percentage).

WORKED EXAMPLE

A college in Sukkur analyses Physics test results (out of 50) for 6 students.

Step 1 — Collection: Marks are gathered from the teacher's register.

Roll No	Name	Marks (raw)
1	Ayesha	42
2	Bilal	18
3	Hina	35
3	Hina	35
4	Kashif	250
5	Sana	(missing)

Step 2 — Storage: The marks are typed into a spreadsheet saved on Google Drive (cloud).

Step 3 — Cleaning: Hina's record is a *duplicate* → delete one copy. Kashif's 250 is an *incorrect value* (more than 50) → the register shows 25, so correct it. Sana's marks are *missing* → ask the teacher; she was absent, so mark her "absent".

Step 4 — Analysis: Clean marks: 42, 18, 35, 25. Pass mark is 17 (33% of 50 = 16.5, rounded up). All 4 passed. Percentages: Ayesha 84%, Bilal 36%, Hina 70%, Kashif 50%. Average = $(42 + 18 + 35 + 25) \div 4 = 120 \div 4 = 30$ marks = 60%. Bilal is closest to failing → needs extra help.

Step 5 — Visualization: A bar chart shows each student's percentage; Bilal's short bar makes the problem visible at a glance.

REMEMBER IT

"Cows Sit Calmly And Vote" — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar (store), pick out stones (clean), cook (analyse), serve beautifully on a plate (visualise).

COMMON MISTAKE

Mistake: Skipping cleaning and going straight to analysis. **Fix:** Dirty data gives wrong answers. In the example, Kashif's wrong "250" would have made the average $(42 + 18 + 35 + 250) \div 4 = 86.25$ — more than 50, which is impossible. Always clean first.

ارو میں سمجھیں

ڈیٹا اینالٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔ اس کے پانچ اجزاء ہیں: ڈیٹا جمع کرنا، محفوظ کرنا، صاف کرنا (غلطیاں اور دہرائے گئے ریکارڈ ہٹانا)، تجزیہ کرنا، اور چارٹ یا گراف کی صورت میں دکھانا۔

CHECK YOURSELF

1. List the five components of Data Analytics in order.
2. Name four kinds of errors removed during data cleaning.
3. A pie chart of pass/fail percentage belongs to which component?

Answers: 1. Collection, Storage, Cleaning, Analysis, Visualization. 2. Duplicate records, incorrect values (typos), missing records, improper formats. 3. Data Visualization.

Comparison: IoT vs AI vs Data Analytics

LEARN IT

These three technologies are often used together, so students mix them up. The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

WORKED EXAMPLE

One problem, three technologies — a K-Electric smart meter:

IoT: The smart electricity meter measures units used every 15 minutes and sends them over the mobile network. Output: raw readings.

Data Analytics: The company cleans and studies a month of readings and draws a graph showing that most electricity is used between 7 pm and 11 pm. Output: report/graph.

AI: Software learns each area's pattern, predicts tomorrow's demand, and decides how to plan load management. Output: smart decision.

REMEMBER IT

Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

COMMON MISTAKE

Mistake: Writing that IoT "makes decisions". **Fix:** In the comparison table IoT's main purpose is *data collection and automation*; the *intelligent decision* belongs to AI.

اردو میں سمجھیں

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔ AI ذہین سافٹ ویئر ہے جو فیصلے اور پیش گوئیاں کرتا ہے۔ ڈیٹا اینالیٹکس ڈیٹا کا جائزہ لے کر رپورٹس اور گراف بناتا ہے۔ تینوں مل کر ایک اسمارٹ نظام بناتے ہیں۔

CHECK YOURSELF

1. Which of the three works mainly with hardware?
2. What is the output of Data Analytics?

Answers: 1. IoT. 2. Reports, graphs and insights.

5.2 Uses of IoT in Different Areas

LEARN IT

IoT **improves efficiency** (doing more with less effort), **saves time**, and **enhances the overall quality of life**. It plays a significant role in developing modern **infrastructure** (the basic systems of a country — roads, power, water, hospitals). The source lists six areas.

1. Smart Cities

- **Traffic management:** Sensors adjust traffic signals automatically to reduce congestion (crowding). Example: signals on a busy Karachi road stay green longer for the direction with more cars.
- **Street lighting:** Lights turn ON/OFF based on daylight or movement, saving electricity.
- **Parking:** Sensors guide drivers to empty parking spaces, e.g. in a large shopping mall car park.
- **Waste management:** Smart bins send alerts when full, so trucks collect only full bins instead of driving to every bin.

2. Industrial Automation

- Sensors check machine **temperature and vibration** to prevent breakdowns (called predictive maintenance).
- Automated machines perform repeated tasks precisely.
- Quality-control sensors check products during manufacturing — for example, in the **textile industries of Sindh**, cameras and sensors spot faults in cloth as it is woven.

3. Healthcare (Smart Health)

- **Smartwatches** continuously measure heart rate and vital signs (pulse, oxygen level).
- **Remote monitoring devices** send patient data directly to doctors — vital for **rural areas of Pakistan**, where a specialist may be hundreds of kilometres away (e.g. a heart patient in Tharparkar monitored by a doctor in Karachi).
- **Emergency alert devices** inform hospitals instantly, for example when an elderly person falls.

4. Education (Smart Learning)

- **Smart classrooms:** digital boards and connected projectors.
- **Attendance systems:** biometric automated attendance for students and teachers (used in many Sindh government schools and colleges).
- **Resource management:** automated classroom lighting and AC control — off when the room is empty.

5. Home Automation

- **Smart lighting:** controlled remotely via mobile apps.
- **Security:** real-time monitoring via Wi-Fi cameras — you can watch your house gate on your phone.
- **Voice control:** using assistants (like Alexa or Google Assistant) to control TVs and appliances.

6. Smart Farming

Smart farming is **crucial for agriculture in Pakistan** (especially Sindh and Punjab) to **save water** and **improve yields** (the amount of crop produced).

- **Soil moisture sensors:** check if crops actually need water.
- **Automated irrigation:** turns water pumps ON/OFF based on soil data.
- **Drones:** monitor crop health from above, spotting disease or dry patches.

Smart farming process:



WORKED EXAMPLE

A cotton farmer near Nawabshah uses smart irrigation:

Step 1. A soil moisture sensor in the field reads 18% moisture.

Step 2. It sends the reading over the 4G network to a cloud service.

Step 3. The cloud AI checks: cotton needs at least 25% moisture, and the weather forecast shows no rain → decision "Needs water: YES".

Step 4. The actuator (pump switch) turns the tube-well ON.

Step 5. When moisture reaches 30%, the AI decides "enough" and the pump turns OFF.

Result: water is used only when needed — important because Sindh faces water shortages — and the farmer sees everything on his phone app.

REMEMBER IT

Remember the six areas with "**CHIEF H**": **C**ities, **H**ealth, **I**ndustry, **E**ducation, **F**arming, **H**omes.

COMMON MISTAKE

Mistake: Giving only the name of the area ("smart cities") in an exam answer. **Fix:** Always add *what* IoT does there — e.g. "smart bins send alerts when full" — and, if possible, a Pakistani example.

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔ اسمارٹ شہروں میں ٹریفک سگنل اور اسٹریٹ لائٹس خود کار ہوتی ہیں، اسپتالوں میں مریضوں کا ڈیٹا براہ راست ڈاکٹر تک پہنچتا ہے، اور کھیتوں میں مٹی کی نمی کے سینسر بتاتے ہیں کہ فصل کو پانی چاہیے یا نہیں، جس سے سندھ اور پنجاب میں پانی کی بچت ہوتی ہے۔

CHECK YOURSELF

1. How do smart bins help waste management?
2. Why is remote health monitoring especially useful in Pakistan?
3. In smart farming, which device is the actuator?

Answers: 1. They send alerts when full, so only full bins are emptied. 2. Rural areas lack doctors; devices send patient data directly to doctors far away. 3. The water pump (its switch).

5.2.1 Scope of AI in Education in Pakistan

LEARN IT

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving **teaching, learning** and **administration** (managing the school: admissions, records, timetables).

Practical uses:

- **Learning Management Systems (LMS)** — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI features can recommend lessons to each student.
- **Automated grading and digital testing** — MCQ tests marked instantly by computer.
- **AI-based career counselling** — software that studies a student's marks and interests and suggests suitable fields.

Benefits:

- Helps students learn **at their own speed**.
- **Identifies students who need extra help**.
- Supports students with disabilities (**Assistive AI**), e.g. text-to-speech for a visually impaired student.

Major challenges in Pakistan:

- Limited **internet access in rural areas**.
- Lack of **computers and digital devices** for all students.
- Need for proper **teacher training** to use AI tools.
- **Data privacy and security** issues — students' personal data must be protected.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices
AI career counselling	Supports students with disabilities	Teacher training needed

WORKED EXAMPLE**Scenario — Fatima in Larkana and an AI tutor app:**

Step 1. Fatima takes a 10-question online maths quiz on the college LMS. It is graded instantly (automated grading).

Step 2. The AI notices she got all 3 trigonometry questions wrong.

Step 3. It recommends two short videos and five easy practice questions on trigonometry — she learns at her own speed.

Step 4. Her teacher's dashboard flags "Fatima – needs help in trigonometry" (identifies students who need help).

Challenge: At home Fatima's village has weak 3G signals, so the videos buffer — showing the rural internet challenge.

COMMON MISTAKE

Mistake: Writing that AI will replace teachers. **Fix:** The source says AI *improves* teaching and needs *teacher training* — teachers use AI as a tool; they are not replaced.

اردو میں سمجھیں

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔ اس سے طلبہ اپنی رفتار سے سیکھ سکتے ہیں اور کمزور طلبہ کی نشاندہی ہوتی ہے۔ لیکن دیہی علاقوں میں انٹرنیٹ کی کمی، کمپیوٹرز کی قلت، اساتذہ کی تربیت اور ڈیٹا کی حفاظت بڑے چیلنج ہیں۔

CHECK YOURSELF

1. Give three practical uses of AI in education.
2. State any two challenges of AI in education in Pakistan.

Answers: 1. LMS, automated grading and digital testing, AI-based career counselling. 2. Any two: limited rural internet, lack of devices, need for teacher training, data privacy and security.

5.2.2 Career Connection in IoT and AI

LEARN IT

Learning computing skills opens doors to many employers in Pakistan: **software houses** (e.g. in Karachi, Lahore and Islamabad technology parks), **telecom companies** (Jazz, Telenor), **banks**, and **government IT departments** (e.g. NADRA, provincial IT boards).

- **IoT Engineer:** designs smart systems and connected devices (e.g. a smart irrigation kit).
- **Data Analyst:** analyses the data generated by IoT systems and produces reports.

Key skills needed:

- **Programming** — Python, C++.
- **Networking** — Wi-Fi, protocols (rules for how devices communicate).
- **Electronics** — sensors and circuits.
- **Problem-solving.**

REMEMBER IT

Skills = "**PNEP**": **P**rogramming, **N**etworking, **E**lectronics, **P**roblem-solving. You start building the first one this year when you learn Python!

ارو میں سمجھیں

کمپیوٹنگ کی مہارتیں سیکھ کر آپ سافٹ ویئر ہاؤسز، ٹیلی کام کمپنیوں (جیسے جاز، ٹیلی نار)، بینکوں اور سرکاری آئی ٹی محکموں میں کام کر سکتے ہیں۔ IoT انجینئر اسمارٹ آلات بناتا ہے اور ڈیٹا اینالسٹ ان سے پیدا ہونے والے ڈیٹا کا تجزیہ کرتا ہے۔ اس کے لیے پروگرامنگ، نیٹ ورکنگ، الیکٹرانکس اور مسئلہ حل کرنے کی صلاحیت ضروری ہے۔

CHECK YOURSELF

1. What does an IoT Engineer do?
2. Name the four key skills for IoT/AI careers.

Answers: 1. Designs smart systems and connected devices. 2. Programming (Python, C++), networking, electronics, problem-solving.

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

LEARN IT

Information sources are ways to find knowledge and data. The internet has vast information, but **not all of it is accurate**. Sources are divided into three main categories.

1. Primary sources give **original, firsthand information** straight from people, events or official records. They are **highly authentic** (genuine).

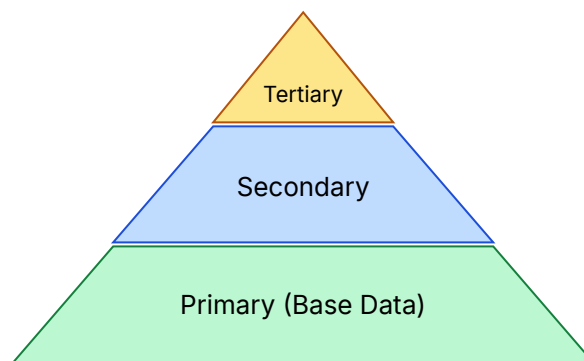
Examples: interviews, photographs, diaries, NADRA official records.

2. Secondary sources contain information that has **already been collected and explained by someone else**. They **summarise or analyse primary data**.

Examples: textbooks, news reports, research papers, Pakistan Bureau of Statistics reports.

3. Tertiary sources provide **highly summarised and organised information** derived from both primary and secondary sources. They are used for **quick reference**.

Examples: encyclopedias, Wikipedia, dictionaries, handbooks.



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

Many examples, sorted

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life

When do sources become unreliable?

In the digital age you must check facts carefully. A source is **unreliable** (cannot be trusted) if:

Source type	Reasons for unreliability
Primary	Information is incomplete, data is recorded incorrectly, or false information was deliberately provided.
Secondary	The information is outdated, lacks proper references/citations, or has not been verified by an expert.
Tertiary	Comes from an unknown/unverified website, or the information is oversimplified and misses key facts.

REMEMBER IT

Rule of thumb (Author – Date – References): If an online article does not state the **author's name**, the **date of publication**, or list its **references**, it should not be trusted for academic work. Remember "**ADR**" — like checking a bus has a driver, a route and a ticket before you board.

WORKED EXAMPLE

Classify these — worked solutions:

(a) Your grandfather's handwritten diary from 1947. Written by a person who lived through the event → firsthand → **Primary**.

(b) A Dawn newspaper article analysing the budget. A journalist collected facts and explained them → **Secondary**.

(c) An Oxford English–Urdu dictionary. Organised, summarised reference built from many sources → **Tertiary**.

(d) A Bureau of Statistics report on literacy in Sindh. It analyses census (primary) data → **Secondary**.

(e) A photo you took of a road accident. Original record of the event → **Primary**.

Method: Ask (1) "Was the creator *there* / is it the original record?" → Primary. (2) "Is someone *explaining or analysing* original data?" → Secondary. (3) "Is it a *quick-reference summary* of many sources?" → Tertiary.

CHECK YOURSELF

Exercise — classify these as Primary (P), Secondary (S) or Tertiary (T):

1. A video interview with a cricketer after a match.
2. Your Computer Science textbook.
3. Wikipedia page on "Mohenjo-daro".
4. A birth certificate issued by the union council.
5. A research paper reviewing ten studies on smog in Lahore.
6. A handbook of physics formulas.
7. Original exam answer sheets of students.
8. A TV news report summarising a court decision.

Answers: 1. P 2. S 3. T 4. P 5. S 6. T 7. P 8. S

COMMON MISTAKE

Mistake: Thinking "primary" means "most important" or "tertiary" means "useless". **Fix:** The names describe *how far the information is from the original event*. Primary = straight from the source; tertiary = summarised summary. Each is useful for a different job. Also, Wikipedia is **tertiary** — not primary — because it summarises other sources.

اروین سمجھیں

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔ پرائمری ذرائع براہ راست اور اصل معلومات دیتے ہیں (جیسے انٹرویو، تصاویر، نادرا کا ریکارڈ)۔ سیکنڈری ذرائع وہ ہیں جن میں کسی اور نے پرائمری معلومات کی وضاحت یا تجزیہ کیا ہو (جیسے درسی کتابیں، خبریں)۔ ٹریشیری ذرائع خلاصہ شدہ معلومات دیتے ہیں جیسے انسائیکلوپیڈیا، وکی پیڈیا اور لغت۔ جس آن لائن مضمون میں مصنف کا نام، تاریخ یا حوالہ جات نہ ہوں اس پر بھروسہ نہ کریں۔

EXAM TIP

Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short), and "When does a source become unreliable?"

Write the definition key phrase for each — *firsthand/original*, *collected and explained by someone else*, *highly summarised for quick reference* — plus two examples each. Draw the pyramid with primary at the base. For unreliability, give one reason for each type and quote the author–date–references rule.

5.4 Impacts of Computing in Various Fields

LEARN IT

Computing has drastically changed society. It makes life **faster and more connected**, but it also brings challenges like **privacy risks** and **technology addiction**. For every field, think of both sides.

Field	Positive impact (+)	Negative impact (–)
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Education	Online classes, e-books and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain).
Communication	Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai).	People interact less in person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Smart and responsible use

- **Protect personal information** and use **strong passwords** (long, with letters, numbers and symbols; never share your OTP or CNIC details with callers).
- **Do not share fake news** or unverified WhatsApp forwards.
- **Balance screen time** with physical sports and activities.

WORKED EXAMPLE

Scenario analysis — a bank in Karachi moves to mobile banking:

Step 1 — Identify the field: Business (banking).

Step 2 — Positive: Customers transfer money in seconds from home; no queue at the branch; the bank processes more transactions with fewer staff (speed and efficiency).

Step 3 — Negative: Some cashier jobs are no longer needed (job loss); hackers may attempt cyber-attacks or phishing SMS to steal accounts.

Step 4 — Responsible use: Customers should use strong passwords, never share OTPs, and report suspicious messages.

REMEMBER IT

For the five fields, remember "**E-C-B-H-A**" — **E**ducation, **C**ommunication, **B**usiness, **H**ealthcare, **A**griculture — and always write one + and one – for each.

COMMON MISTAKE

Mistake: Writing only positive impacts in a "discuss the impacts" question. **Fix:** "Impact" means *both* sides. Examiners expect a balanced answer: advantages, disadvantages, and how to use technology responsibly.

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔ تعلیم، مواصلات، کاروبار، صحت اور زراعت میں اس نے کام تیز اور آسان بنائے ہیں، لیکن اس کے ساتھ اسکرین کی لت، سماجی تنہائی، ملازمتوں کا خاتمہ، ہیکنگ اور پرائیویسی کے خطرات بھی پیدا ہوئے ہیں۔ ذمہ دارانہ استعمال کے لیے مضبوط پاس ورڈ رکھیں، جعلی خبریں نہ پھیلائیں، اور کھیل کود کے ساتھ اسکرین ٹائم میں توازن رکھیں۔

CHECK YOURSELF

1. Give one positive and one negative impact of computing in healthcare.
2. Why can't many farmers in Pakistan benefit from smart agriculture tools?
3. State three rules of smart and responsible use.

Answers: 1. + Digital records and telemedicine help treat remote patients; – patient data privacy is at risk if servers are hacked. 2. Poor internet access in rural areas. 3. Protect personal information/use strong passwords; do not share fake news; balance screen time with physical activity.

5.5 Assistive Technologies ★ Golden

LEARN IT

Assistive technologies are **devices, software and systems** that help people with **disabilities or functional limitations** (difficulty in seeing, hearing, moving, speaking, remembering, etc.). They allow more people to **participate equally** in school, work and society.

There are **four categories**:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Mobility · Sensory · Communication · Cognitive

- A **screen reader** is software that reads aloud the text on a screen, so a blind person can use a phone or computer. **VoiceOver** is Apple's screen reader; Android has TalkBack.
- A **Braille display** is a device with small pins that rise and fall to form Braille dots, so a blind user can read screen text with their fingers.
- A **speech-generating device** speaks out words that a person types or selects on a screen.
- A **smart wheelchair** can be controlled by a joystick, voice or head movement and may avoid obstacles with sensors.

WORKED EXAMPLE

Scenario — choose the right assistive technology:

- (a) Ahmed, a student in Hyderabad, is blind and wants to read his e-textbook. → **Sensory aid**: a screen reader or Braille display.
- (b) Mrs Rashida cannot climb the stairs of her house after a knee operation. → **Mobility aid**: stair lift (or smart wheelchair).
- (c) Zain cannot speak clearly due to a medical condition but can type. → **Communication aid**: text-to-speech app or speech-generating device.
- (d) Nani often forgets to take her medicine. → **Cognitive aid**: digital reminder app.
- (e) Usman has dyslexia (difficulty reading). → **Cognitive/learning aid**: audiobooks and spell-checkers.

Method: identify the *limitation* (moving, seeing/hearing, speaking, memory/reading) → choose the matching category.

REMEMBER IT

Four categories = "**My Small Cat Cries**": **Mobility**, **Sensory**, **Communication**, **Cognitive**. Link each to a body function: *legs* (mobility), *eyes/ears* (sensory), *mouth* (communication), *mind* (cognitive).

COMMON MISTAKE

Mistake: Putting text-to-speech and screen readers in the same category every time. **Fix**: In the source, the **screen reader** is a *sensory* aid (supports vision), while a **text-to-speech app** is listed as a *communication* aid (helps a person with speech difficulty express themselves). Decide by *whose problem* it solves.

اروین سمجھیں

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور معاشرے میں برابر حصہ لے سکیں۔ ان کی چار اقسام ہیں: نقل و حرکت میں مدد (اسمارٹ ویبل چیئر)، حسی مدد (سماعتی آلہ، اسکرین ریڈر، بریل ڈسپلے)، گفتگو میں مدد (اسپیچ جنریٹنگ ڈیوائس) اور ذہنی و تعلیمی مدد (آڈیو بکس، اسپیل چیکر، یاد دہانی ایپس)۔

EXAM TIP

Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ____" (MCQ).

For full marks write the definition (devices, software and systems; help people with disabilities; equal participation), then all **four** categories each with purpose + two examples. Finish with one line on importance (equal access, independence, inclusion).

CHECK YOURSELF

1. Which category does a hearing aid belong to?
2. Give two examples of cognitive/learning aids.
3. Define assistive technologies in one sentence.

Answers: 1. Sensory aids. 2. Any two: audiobooks, spell-checkers, digital reminder apps. 3. Devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society.

5.5.1 Importance of Assistive Technologies

LEARN IT

Assistive technologies are important for three main reasons:

1. **Equal access to education:** Screen readers allow blind students to read digital textbooks equally with their peers (classmates).
2. **Independence in daily life:** Mobility aids reduce the need for constant human assistance — a person in a smart wheelchair can go to the market without someone pushing.
3. **Social inclusion:** They break down barriers, allowing individuals to work, study and socialise confidently.

Equal Access → Independence → Inclusion

WORKED EXAMPLE

How one tool changes a life: A visually impaired student in Karachi uses a screen reader on her laptop. **Equal access** — she reads the same e-books as her class. **Independence** — she no longer needs someone to read to her. **Inclusion** — she joins the class WhatsApp group, takes online tests and later gets a job, contributing to society.

REMEMBER IT

Remember "**All**": **A**ccess, **I**ndependence, **I**nclusion.

ارو میں سمجھیں

معاون ٹیکنالوجیز کی اہمیت تین باتوں میں ہے: تعلیم تک برابر رسائی (ناہینا طلبہ اسکرین ریڈر سے کتابیں پڑھ سکتے ہیں)، روزمرہ زندگی میں خود مختاری (کسی کی مسلسل مدد کی ضرورت کم ہوتی ہے)، اور سماجی شمولیت (لوگ اعتماد کے ساتھ پڑھ، کام اور میل جول کر سکتے ہیں)۔

CHECK YOURSELF

1. State three reasons why assistive technologies are important.
2. How do mobility aids support independence?

Answers: 1. Equal access to education, independence in daily life, social inclusion. 2. They reduce the need for constant human assistance.

5.5.2 Career Connection in Assistive Tech

LEARN IT

Designing assistive technology is a **highly rewarding career path** — you earn a living while improving people's lives.

Career	What they do
Assistive Software Developer	Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
Biomedical Engineer	Designs artificial limbs and hearing aids.
UX/UI Accessibility Expert	Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them).

UX means *user experience* (how easy and pleasant a product is to use); **UI** means *user interface* (the screens and buttons you interact with). **Accessibility** means designing so that *everyone*, including people with disabilities, can use it.

اردو میں سمجھیں

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔ اسسٹو سافٹ ویئر ڈویلپر اسپیشل ٹو ٹیکسٹ جیسے ٹولز بناتا ہے، ہائیو میڈیکل انجینئر مصنوعی اعضاء اور سماعتی آلات تیار کرتا ہے، اور UX/UI ایکسیبلٹی ماہر یقینی بناتا ہے کہ ویب سائٹس نابینا افراد کے لیے بھی آسانی سے استعمال ہو سکیں۔

CHECK YOURSELF

1. Who designs artificial limbs and hearing aids?
2. What does a UX/UI accessibility expert ensure?

Answers: 1. A biomedical engineer. 2. That websites are easy to navigate for visually impaired users.

Chapter 5 Review

Chapter summary

- Computing is vital in Pakistan's schools, hospitals, banks, farms and industries; it has both positive and negative impacts.
- **AI** is "artificial" (made by humans) and "intelligent" (does tasks needing thinking); it uses data to learn, recognise patterns and make decisions.
- In smart systems, IoT is the "eyes" that gather data, Analytics sorts it, and AI is the "brain" that decides the action.
- **IoT** is a system of physical devices connected to the internet that collect and share data to perform tasks automatically.
- The six IoT components are sensors (collect), actuators (act), connectivity, data processing, cloud computing and user interface.
- **Data Analytics** is a pipeline: collection → storage → cleaning → analysis → visualization.
- Data cleaning removes duplicates, incorrect values, missing records and improper formats.
- IoT outputs raw data (hardware); AI outputs smart decisions (software); Data Analytics outputs reports, graphs and insights (data).
- IoT is used in smart cities, industrial automation, healthcare, education, home automation and smart farming.
- Smart farming (soil sensor → cloud AI → water pump) saves water and improves yields in Sindh and Punjab.
- AI in Pakistani education: LMS, automated grading, career counselling; challenges are rural internet, lack of devices, teacher training and data privacy.
- IoT/AI careers (IoT engineer, data analyst) need programming, networking, electronics and problem-solving.
- Information sources: primary (firsthand/original), secondary (collected and explained by others), tertiary (summarised for quick reference).
- Do not trust an online article that lacks an author, a publication date or references.
- Computing helps education, communication, business, healthcare and agriculture but risks screen addiction, isolation, job loss, cyber-attacks and privacy loss.
- **Assistive technologies** — mobility, sensory, communication and cognitive aids — give equal access, independence and social inclusion to people with disabilities.

Key terms

Term	Meaning	اردو
Computing	Using computers and digital devices to store, process and share information.	کمپیوٹنگ
Artificial Intelligence (AI)	Human-made technology that lets machines think, learn and decide using data.	مصنوعی ذہانت

Internet of Things (IoT)	Physical devices connected to the internet that collect and share data to work automatically.	انٹرنیٹ آف تھنگز، اشیاء کا انٹرنیٹ
Sensor	Device that collects information from the physical world.	حساس آلہ (سینسر)
Actuator	Device that performs a physical action in response to a command.	عمل کار آلہ (ایکچویٹر)
Connectivity	The link (Wi-Fi, Bluetooth, ZigBee, 4G/5G) that lets devices send and receive data.	رابطہ کاری
Data processing	Changing raw data into useful information.	ڈیٹا کی پروسیسنگ
Cloud computing	Storing data and using services over the internet instead of local drives.	کلاؤڈ کمپیوٹنگ
User interface (UI)	The part that lets a user see information and give commands to a device.	صارف انٹرفیس
Data Analytics	Collecting, storing, cleaning, analysing and presenting data to find useful information.	ڈیٹا کا تجزیہ
Data cleaning	Finding and correcting errors such as duplicates and missing values.	ڈیٹا کی صفائی
Data visualization	Showing data as charts, plots or graphs.	ڈیٹا کی بصری پیشکش
Smart city	City using IoT for traffic, lighting, parking and waste.	اسمارٹ شہر
Smart farming	Using sensors, automated irrigation and drones in agriculture.	اسمارٹ کاشتکاری
Learning Management System (LMS)	Online platform for lessons, assignments and tests.	آن لائن تعلیمی نظام
Telemedicine	Treating patients remotely using computing and communication.	دور دراز علاج
Primary source	Original, firsthand information.	بنیادی ذریعہ
Secondary source	Information collected and explained by someone else.	ثانوی ذریعہ
Tertiary source	Highly summarised information for quick reference.	ثالثی ذریعہ
Reliability	How far a source can be trusted.	قابل اعتماد ہونا
Privacy	Keeping personal information safe from others.	رازداری

Cyber-attack	An attempt by hackers to damage or steal from computer systems.	سائبر حملہ
Assistive technology	Devices, software and systems that help people with disabilities.	معاون ٹیکنالوجی
Screen reader	Software that reads screen text aloud for blind users.	اسکرین ریڈر
Braille display	Device with raised pins showing text as Braille dots.	بریل ڈسپلے
Accessibility	Designing so everyone, including people with disabilities, can use a product.	رسائی پذیری

Practice: multiple choice

- AI is called "artificial" because it:
 - runs only on electricity
 - is created by humans
 - is always wrong
 - works without data
- Which of the following is an actuator?
 - Temperature sensor
 - Motion sensor
 - Water valve
 - Wi-Fi router
- ZigBee is an example of which IoT component?
 - Connectivity
 - Sensor
 - User interface
 - Actuator
- Google Drive and iCloud are examples of:
 - Sensors
 - Data cleaning
 - Actuators
 - Cloud computing
- Removing duplicate student records belongs to which component of Data Analytics?
 - Data collection
 - Data storage
 - Data cleaning
 - Data visualization
- A pie chart showing pass vs fail percentage is an example of:
 - Data collection
 - Data storage
 - Data cleaning
 - Data visualization
- According to the comparison table, the output of IoT is:
 - Raw data from the environment
 - Smart decisions
 - Reports and graphs
 - Printed documents
- In a smart system, AI acts as the:
 - eyes
 - brain
 - hands
 - wires

9. NADRA's official records are a:
- | | |
|--------------------|----------------------|
| a. tertiary source | b. secondary source |
| c. primary source | d. unreliable source |
10. Wikipedia is classified as a:
- | | |
|---------------------|---------------------|
| a. primary source | b. tertiary source |
| c. secondary source | d. firsthand record |
11. Smart bins that send alerts when full are a use of IoT in:
- | | |
|--------------------|-----------------|
| a. smart farming | b. healthcare |
| c. home automation | d. smart cities |
12. A system in which a soil sensor tells the cloud AI to switch on a water pump is used in:
- | | |
|---------------------|--------------------------|
| a. smart farming | b. industrial automation |
| c. smart classrooms | d. traffic management |
13. A screen reader such as VoiceOver is a:
- | | |
|-----------------|----------------------|
| a. mobility aid | b. communication aid |
| c. sensory aid | d. cognitive aid |
14. A speech-generating device is a:
- | | |
|-----------------|----------------------|
| a. mobility aid | b. cognitive aid |
| c. sensory aid | d. communication aid |
15. An online article should NOT be trusted for academic work if it lacks:
- | | |
|-----------------------|--|
| a. colourful pictures | b. the author's name, publication date or references |
| c. a long title | d. advertisements |

Practice: short questions

1. Define Artificial Intelligence and give two examples from daily life.
2. Define the Internet of Things (IoT) with one example.
3. Differentiate between a sensor and an actuator with two examples of each.
4. What is the role of cloud computing in an IoT system?
5. List the five components of Data Analytics. What kinds of errors are removed in data cleaning?
6. Give two uses of IoT in smart cities and two in healthcare.
7. State two benefits and two challenges of AI in education in Pakistan.
8. Differentiate between primary and secondary sources with two examples each.
9. When does a secondary source become unreliable?
10. Give one positive and one negative impact of computing on (a) communication and (b) business.

Practice: long / problem questions

1. Define IoT. Explain its six components with an example of each. Draw a diagram showing how they connect.
2. Compare IoT, AI and Data Analytics in a table (definition, purpose, type of work, output). Then explain, with an example, how the three work together in a smart system.

3. A college in Mirpurkhas wants to find out why many students are failing Mathematics. Explain how it would use each of the five components of Data Analytics to solve this problem.
4. Explain the three types of information sources with examples. Classify the following: (i) a school textbook, (ii) a photograph of an event, (iii) an encyclopedia, (iv) an interview, (v) a Bureau of Statistics report, (vi) a dictionary. When does each type become unreliable?
5. What are assistive technologies? Explain their four categories with examples, state their importance, and name two related careers.
6. "Computing has both positive and negative impacts on society." Discuss with reference to education, healthcare and agriculture, and suggest three rules for smart and responsible use.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. d | 7. a | 10. b | 13. c |
| 2. c | 5. c | 8. b | 11. d | 14. d |
| 3. a | 6. d | 9. c | 12. a | 15. b |

Short questions — model answers

1. AI is human-made ("artificial") technology that can perform tasks requiring thinking ("intelligence"); it uses data to make decisions. Examples: face recognition to unlock a phone; YouTube suggesting videos; chatbots answering questions.
2. IoT is a system in which physical devices are connected to the internet and can collect and share data to perform tasks automatically. Example: a smartwatch collects heart-rate data and sends it to a mobile phone for display.
3. A sensor collects information from the physical world (input; the "eyes and ears"), e.g. temperature sensor, motion sensor. An actuator performs a physical action in response to a command, converting digital signals into movement (output), e.g. electric motor, automatic door opener.
4. Cloud computing stores IoT data and provides processing services over the internet instead of local hard drives, so it can handle the massive amounts of data produced by many devices. Examples: Google Drive, iCloud, online LMS.
5. Collection, storage, cleaning, analysis, visualization. Cleaning removes duplicated records, incorrect values (typos), missing files/records, and improper formats.
6. Smart cities: sensors adjust traffic signals to reduce congestion; smart bins send alerts when full (also street lighting, parking). Healthcare: smartwatches measure heart rate and vitals; remote monitoring devices send patient data to doctors (also emergency alerts).
7. Benefits: students learn at their own speed; AI identifies students who need extra help (also supports students with disabilities). Challenges: limited internet access in rural areas; lack of computers/devices (also teacher training, data privacy).
8. Primary sources give original, firsthand information from people, events or official records (e.g. interviews, NADRA records). Secondary sources contain information already collected and explained by someone else, summarising or analysing primary data (e.g. textbooks, news reports).
9. When the information is outdated, lacks proper references/citations, or has not been verified by an expert.
10. (a) Communication: + emails and video calls give instant, global communication at almost no cost; – less in-person interaction causes social isolation. (b) Business: + online shopping, digital banking and automation improve speed and efficiency; – manual jobs are lost to machines and banks face cyber-attacks.

Long questions — model answers

- Definition:** IoT is a system in which physical devices are connected to the internet and collect and share data to perform tasks automatically. **Components:** (1) Sensors — collect information from the physical world, e.g. temperature, motion, fingerprint, smoke, soil moisture sensors. (2) Actuators — perform physical actions in response to commands, converting digital signals to movement, e.g. motor, switch, door opener, water valve, fan controller. (3) Connectivity — lets devices send/receive data, e.g. Wi-Fi, Bluetooth, ZigBee, 4G/5G. (4) Data processing — changes raw data into useful information through collecting, organising, analysing and deciding, e.g. checking if a room is too hot to switch on the AC. (5) Cloud computing — stores data and provides services over the internet, handles massive data, e.g. Google Drive, iCloud, LMS. (6) User interface — lets users see information and give commands, e.g. mobile apps, touch screens, voice commands. **Diagram:** Sensor (collects) → Wi-Fi → Cloud (processes) → Actuator (acts) → User App (UI).
- Table:** IoT — network of physical devices connected to the internet; purpose: data collection and automation; works with hardware; output: raw data from environment. AI — technology enabling machines to think and decide; purpose: intelligent decisions and predictions; works with smart software; output: smart decisions or actions. Data Analytics — process of examining data to find patterns; purpose: extracting meaningful insights; works with data (numbers, records); output: reports, graphs, insights. **Working together:** IoT is the eyes that gather data, Analytics sorts it, AI is the brain that decides. Example: a temperature sensor reads 36 °C (IoT) → recent readings are organised showing the room is heating up (Analytics) → AI decides the room is too hot (AI) → a switch turns on the AC (action).
- (1) **Collection:** gather marks of all Maths tests, attendance and survey responses from students. (2) **Storage:** keep them safely in a spreadsheet or database, e.g. on the cloud. (3) **Cleaning:** remove duplicate records, correct typing errors (e.g. 120 out of 100), fill or mark missing records, and make formats (dates, roll numbers) consistent. (4) **Analysis:** study the clean data to find patterns — e.g. which chapters most students got wrong, whether low attendance links to failing. (5) **Visualization:** present results in bar charts or a pie chart of pass vs fail so the principal sees the problem at a glance and plans extra classes for weak chapters.
- Primary:** original, firsthand, highly authentic (interviews, photographs, diaries, NADRA records). **Secondary:** collected and explained by someone else, summarises/analyses primary data (textbooks, news reports, research papers, Bureau of Statistics reports). **Tertiary:** highly summarised and organised from primary and secondary sources for quick reference (encyclopedias, Wikipedia, dictionaries, handbooks). **Classification:** (i) secondary, (ii) primary, (iii) tertiary, (iv) primary, (v) secondary, (vi) tertiary. **Unreliability:** primary — incomplete, recorded incorrectly, or deliberately false; secondary — outdated, no references/citations, not verified by an expert; tertiary — unknown/unverified website, or oversimplified and missing key facts. Rule: do not trust an article without author, date or references.
- Definition:** devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society. **Categories:** (1) Mobility aids — help people move; smart wheelchairs, stair lifts. (2) Sensory aids — support vision and hearing; hearing aids, screen readers (VoiceOver), Braille displays. (3) Communication aids — help people with speech difficulties express themselves; speech-generating devices, text-to-speech apps. (4) Cognitive/learning aids — help with memory, reading, understanding; audiobooks, spell-checkers, reminder apps. **Importance:** equal access to education (screen readers let blind students read digital textbooks), independence in daily life (mobility aids reduce need for constant help), social inclusion (people work, study and socialise confidently). **Careers:** any two of assistive software developer (speech-to-text tools), biomedical engineer (artificial limbs, hearing aids), UX/UI accessibility expert (websites usable by visually impaired users).
- Introduction:** computing makes life faster and more connected but brings privacy risks and addiction. **Education:** + online classes, e-books and learning apps make study flexible and accessible; – excessive screen time reduces attention span and harms physical health. **Healthcare:** + digital patient records and telemedicine let doctors treat remote patients accurately (e.g. rural Sindh); – patient data privacy is at risk if hospital servers are hacked. **Agriculture:** + smart irrigation and weather forecasting greatly increase yields; – poor rural internet means many farmers cannot use these tools. **Responsible use:** protect

personal information and use strong passwords; do not share fake news or unverified forwards; balance screen time with sports and physical activity. **Conclusion:** benefits are large if technology is used wisely.

Self-assessment: I can...

- ☐ define AI and explain why it is called "artificial intelligence".
- ☐ define IoT and name its six components with examples.
- ☐ list the five components of Data Analytics in order and describe data cleaning.
- ☐ describe IoT uses in smart cities, industry, healthcare, education, homes and farms.
- ☐ classify any source as primary, secondary or tertiary and judge its reliability.
- ☐ explain the four categories of assistive technologies and their importance.
- ☐ explain how IoT, Data Analytics and AI work together (eyes → sort → brain → action).
- ☐ tell the difference between a sensor and an actuator.
- ☐ compare IoT, AI and Data Analytics in a table.
- ☐ state uses, benefits and challenges of AI in education in Pakistan.
- ☐ discuss positive and negative impacts of computing in five fields.
- ☐ name careers linked to IoT, AI and assistive technology and the skills they need.

Digital Literacy

This chapter teaches you how to find, collect, organise and present information using digital tools, safely and responsibly. You will learn about types of data, ways to collect it, and how to turn it into a useful report, presentation or infographic. At the end you will follow a complete research project from the first question to the final product.

اس باب میں آپ سیکھیں گے کہ ڈیجیٹل آلات کی مدد سے معلومات کو محفوظ اور ذمہ دارانہ طریقے سے کیسے تلاش کیا جائے، جمع کیا جائے، ترتیب دیا جائے اور پیش کیا جائے۔ آخر میں آپ ایک مکمل تحقیقی منصوبہ (Digital Inquiry Project) شروع سے آخر تک دیکھیں گے۔

What you will learn

- Define digital literacy and explain why it matters in the Information Age.
- Describe five data-collection strategies: interviews, surveys, prototypes, observation and simulations.
- Compare primary and secondary data by source, examples, purpose and control.
- Use spreadsheet formulas such as `=SUM()`, `=AVERAGE()` and `=COUNTIF()` to analyse data.
- Use advanced search operators (" " , `filetype:` , `site:`) and evaluate the reliability of a source.
- Distinguish between qualitative and quantitative data and give examples of each.
- Write clear, unbiased survey questions and recognise badly written ones.
- List the eight steps for designing a data-collection approach, including ethics and consent.
- Choose a suitable digital tool (spreadsheet, presentation, infographic, report, dashboard) to present findings.
- Carry out all 12 steps of a digital inquiry project and produce a final digital artefact.

6.1 Introduction to Digital Literacy

LEARN IT

Digital literacy is the ability to understand and use digital devices, the internet, and online tools in a **safe**, **effective**, and **responsible** manner.

Look at the three key words:

- **Safe** – you protect yourself and others online (for example, you do not share passwords or private details).
- **Effective** – you get the job done well (for example, you find the right information quickly).
- **Responsible** – you use information honestly and fairly (for example, you do not copy someone else's work or spread false news).

In our increasingly digital world, digital literacy is essential for **everyone**, not only for computer experts. It enables you to:

- **find** information,
- **communicate** with others,
- **produce** digital content, and
- **solve problems** through technology.

Did you know? Digital literacy is not only *using devices*; it is *using digital information safely and wisely*. A digitally literate student can:

Locate information → Organise it properly → Turn it into something useful → Share it responsibly

WORKED EXAMPLE

Is this student digitally literate?

Ali can type fast and plays games on his phone all day. When his teacher asks him to find the population of Karachi, he copies the first number he sees on a random website and does not check where it came from.

Step 1 – Can he use devices? Yes. He has basic skills.

Step 2 – Can he locate and judge information? No. He did not check whether the website was reliable.

Step 3 – Conclusion: Ali has *computer skills* but is not yet fully *digitally literate*. A digitally literate student would compare two or three trusted sources (for example, the Pakistan Bureau of Statistics website) before using the number.

REMEMBER IT

Remember the four verbs of a digitally literate student: **Locate – Organise – Create – Share** ("LOCS": think of a *lock* that keeps your digital life safe).

COMMON MISTAKE

Mistake: "Digital literacy means knowing how to use a computer." **Fix:** Using a computer is only the start. Digital literacy also means finding, judging, creating and sharing information safely and responsibly.

ڈیجیٹل خواندگی (Digital Literacy) سے مراد ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنا اور استعمال کرنا ہے۔ ایک ڈیجیٹل خواندہ طالب علم معلومات تلاش کر سکتا ہے، انہیں ترتیب دے سکتا ہے، ان سے کوئی مفید چیز بنا سکتا ہے اور انہیں ذمہ داری سے دوسروں تک پہنچا سکتا ہے۔

CHECK YOURSELF

1. Write the definition of digital literacy in one sentence.
2. Name the four things digital literacy enables you to do.

Answers: 1. The ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. 2. Find information, communicate, produce digital content, and solve problems through technology.

6.2 Digital Literacy in the Information Age

LEARN IT

The **Information Age** is the period we live in now, where vast amounts of data are **produced**, **shared** and **accessed instantly** through the internet, mobile devices and digital platforms (websites, apps, social media).

Simply **having access** to technology is **not enough**. Owning a smartphone does not make you good at using information. You must develop the skills to **understand**, **manage** and **use** information wisely.

Digital literacy goes beyond basic computer skills. It involves three kinds of ability:

Ability	What it means	Example
Technical	Knowing how to operate devices and software.	Creating a Google Form; using a spreadsheet.
Cognitive (thinking)	Thinking critically about information.	Checking whether a news story is true before believing it.
Social	Communicating and behaving well with others online.	Being polite in a class WhatsApp group; respecting privacy.

Together these abilities help you function successfully in a **digital society**.

WORKED EXAMPLE

Applying the three abilities. A video claims "Drinking hot water cures dengue." What does a digitally literate student do?

Technical: Uses a search engine to search `dengue treatment site:.gov`.

Cognitive: Notices the video has no doctor or source, and that official health websites do not support the claim. Concludes it is false.

Social: Does not forward the video, and politely tells the family group that it is not reliable.

REMEMBER IT

Three abilities = "**Tech, Think, Talk**": Technical, Cognitive (think), Social (talk with others).

COMMON MISTAKE

Mistake: Thinking that "more information" automatically means "better information". **Fix:** In the Information Age the problem is not finding *some* information; it is choosing the *right, reliable* information from too much.

اروین سمجھیں

معلوماتی دور (Information Age) میں انٹرنیٹ اور موبائل کے ذریعے بے پناہ معلومات فوراً تیار اور شیئر ہوتی ہیں۔ صرف ٹیکنالوجی تک رسائی کافی نہیں؛ معلومات کو سمجھنے، سنبھالنے اور دانشمندی سے استعمال کرنے کی مہارت ضروری ہے۔ ڈیجیٹل خواندگی میں تکنیکی، ذہنی (Cognitive) اور سماجی صلاحیتیں شامل ہیں۔

CHECK YOURSELF

1. What is the Information Age?
2. Name the three types of ability that digital literacy involves.

Answers: 1. A period in which vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms. 2. Technical, cognitive and social abilities.

6.3 Types of Data ★ Golden

LEARN IT

Data means facts, numbers, words or observations that we collect. The data gathered through a survey, an investigation or a classroom project can be described in two ways: **qualitatively** or **quantitatively**. Knowing the difference is critical, because each type is analysed in a different way.

6.3.1 Qualitative Data

Qualitative data describes what people **think, feel, notice or experience**. It is **non-numerical** and **descriptive**: it is made of words, not numbers. The word comes from *quality* (what something is like).

Examples: students' opinions about online classes; descriptions of study habits; observation notes ("The students looked bored after 20 minutes").

6.3.2 Quantitative Data

Quantitative data deals with **numbers**. The word comes from *quantity* (how much or how many). It is useful when you need to **count, measure, compare, calculate averages**, or **display results in charts**.

Examples: number of students using the internet daily; marks obtained in a test; time spent online in hours.

6.3.3 Comparing Qualitative and Quantitative Data

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

LEARN IT

Going a little further (background). Quantitative data is sometimes split into two kinds:

- **Discrete** data can only take separate, countable values, e.g. number of students in a class (35, 36 – never 35.5).
- **Continuous** data can take any value in a range and is *measured*, e.g. height (162.4 cm) or time online (2.75 hours).

Also note: a question with fixed categories (e.g. "Which activity do you use your phone for most? Lectures / Notes / Social media / Games") gives **categories** that you can **count**. In this chapter's case study such answers are treated as **quantitative (categories)**, because you count how many students chose each option.

WORKED EXAMPLE

Classify each piece of data.

Data	Type	Reason
"I like online classes because I can replay lectures."	Qualitative	An opinion with a reason, in words.
Ayesha scored 78 out of 100.	Quantitative	A score – a number you can compare.
28 out of 40 students use the internet daily.	Quantitative	A count (and gives 70%).
Observer notes: "Students chat during the last period."	Qualitative	A description of behaviour.
Bilal spends 3.5 hours online per day.	Quantitative	A measurement of time.

Quick test: ask "Can I calculate an average of this?" If yes, it is quantitative. If the answer is words, it is qualitative.

REMEMBER IT

QUALitative = **QUAL**ity = words and feelings. **QUANT**itative = **QUANT**ity = numbers and counts. Qualitative answers **why/how**; quantitative answers **how many/how much**.

COMMON MISTAKE

Mistake: Thinking that any data containing a number is quantitative. A phone number or a roll number is a *label*, not a quantity – averaging roll numbers is meaningless. **Fix:** Data is quantitative only if the number measures or counts something.

اردو میں سمجھیں

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کلاسز کے بارے میں رائے۔ مقداری ڈیٹا (Quantitative Data) اعداد پر مشتمل ہوتا ہے، جیسے ٹیسٹ کے نمبر یا آن لائن گزاریے گئے گھنٹے۔ کیفیتی ڈیٹا "کیوں اور کیسے" سمجھنے کے لیے، جبکہ مقداری ڈیٹا موازنہ کرنے اور چارٹ بنانے کے لیے بہترین ہے۔

EXAM TIP

Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a **table** with four rows: **nature, examples, best used for**, plus a one-line definition of each. Always give **two examples** of each type. The source's table (Nature / Examples / Best used for) is exactly what examiners expect.

CHECK YOURSELF

1. Is "time spent online in hours" qualitative or quantitative?
2. Which type of data is best for understanding *why* something happens?
3. Classify: "The canteen food is too oily."

Answers: 1. Quantitative (measurable number). 2. Qualitative. 3. Qualitative (an opinion in words).

6.4 Data-Collection Strategies ★ Golden

LEARN IT

Digital literacy is not limited to finding information that **already exists** on the internet. You should also learn how to gather **original** (new) information yourself. A **data-collection strategy** is a planned method for gathering this information.

The five common strategies are:

Interviews

• Surveys

• Prototypes

• Observation

• Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

REMEMBER IT

Mnemonic for the five strategies: "**I S P O S**" – "**I** **S**aw **P**eople **O**n **S**creens": Interviews, Surveys, Prototypes, Observation, Simulations.

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔
ڈیٹا جمع کرنے کے پانچ عام طریقے یہ ہیں: انٹرویو، سروے، پروٹوٹائپ، مشاہدہ (Observation) اور سمیولیشن (Simulation)۔

EXAM TIP

"List and explain the data-collection strategies" is a frequent long question. Name all **five**, give a one-sentence definition and **one example** for each, and say what type of data each gives. Detailed notes on each follow in the next two topics.

CHECK YOURSELF

1. Name the five data-collection strategies.
2. What does "original information" mean?

Answers: 1. Interviews, surveys, prototypes, observation, simulations. 2. New information that you gather yourself, rather than information that already exists.

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

6.4.1 Interviews

LEARN IT

An **interview** is a **planned conversation** used to gather information **directly from a person**.

- It may be **one-to-one** (you and one person) or in a **small group**.
- Questions are **planned in advance**, so you do not forget anything important.
- Answers are usually **qualitative** (opinions, reasons, experiences).

Strength: you get deep, detailed answers and can ask follow-up questions ("Why do you say that?").

Weakness: it takes a lot of time, so you can only interview a few people.

6.4.2 Surveys / Questionnaires

LEARN IT

A **survey** (or **questionnaire**) is useful when the **same set of questions** needs to be asked to **many people**.

- It may be on **paper** or created with **digital tools** such as **Google Forms** or **Microsoft Forms**.
- **Note:** online survey tools **automatically collect responses** and **display basic charts**, which makes data analysis much easier.

Types of survey and interview questions

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)

Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? ____	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Good questions vs bad questions

A good question is **clear, short, unbiased** (it does not push the person towards one answer) and asks about **one thing at a time**.

Problem	Bad question	Better question
Leading / biased – suggests the "right" answer	Don't you agree that phones waste students' time?	How do phones affect your study time? (Helps a lot / Helps a little / No effect / Harms a little / Harms a lot)
Double-barrelled – two questions in one	Do you use your phone for lectures and games?	Split into two: "Do you use your phone for lectures?" and "Do you use your phone for games?"
Vague – unclear words	Do you use your phone a lot?	How many hours do you use your phone on a normal school day?
Overlapping options	1–3 hrs / 3–5 hrs / 5+ hrs (where does exactly 3 go?)	Less than 3 hrs / 3 to less than 5 hrs / 5 hrs or more
Sensitive / too personal	What is your father's monthly income?	Leave it out unless it is truly needed for the study.

WORKED EXAMPLE

Building a short Google Form, step by step. Research question: "How do Class XI students travel to college?"

Step 1: Open Google Forms → Blank form → Title: "Travel to College Survey". Add a description: "This survey is anonymous and for a class project. Do you agree to take part? (Yes/No)" – this is your **consent** question.

Step 2: Q1 (multiple choice): "How do you usually travel to college?" Walk / Bus / Rickshaw / Motorbike / Car / Other.

Step 3: Q2 (number): "How many minutes does your journey take?"

Step 4: Q3 (open-ended): "What is the biggest problem you face while travelling?"

Step 5: Click *Send* and share the link in the class group. Responses appear in the *Responses* tab with automatic pie/bar charts, and can be opened in Google Sheets.

Result: Q1 and Q2 give quantitative data; Q3 gives qualitative data.

REMEMBER IT

Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house (survey).

COMMON MISTAKE

Mistake: Writing only open-ended questions in a survey for 100 people. You will get 100 paragraphs that are very hard to analyse. **Fix:** Use mostly closed or number questions in a large survey, and only one or two open-ended questions.

اروین سمجھیں

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔ سروے میں ایک ہی سوالات بہت سے لوگوں سے پوچھے جاتے ہیں، کاغذ پر یا گوگل فارمز (Google Forms) جیسے ڈیجیٹل ٹولز کے ذریعے، جو جوابات خودکار طور پر جمع کر کے چارٹ بھی دکھاتے ہیں۔ اچھا سوال واضح، مختصر، غیر جانبدار اور ایک وقت میں ایک ہی بات کے بارے میں ہوتا ہے۔

EXAM TIP

You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the problem (leading, double-barrelled, vague, overlapping) and give the corrected version.

CHECK YOURSELF

1. Name two digital tools for creating surveys.
2. What is wrong with: "Do you enjoy maths and physics?"
3. Are interview answers usually qualitative or quantitative?

Answers: 1. Google Forms and Microsoft Forms. 2. It is double-barrelled (two subjects in one question); ask about each subject separately. 3. Usually qualitative.

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

6.4.3 Prototypes

LEARN IT

A **prototype** is a **simple early version** of a product, model or solution. It allows an idea to be **tested before** time and effort are spent on the final version. Users **try** the prototype and give **feedback**, and this feedback becomes useful **data for improvement**.

Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five classmates to try "finding the timetable". You note where they get confused.

6.4.4 Observation

LEARN IT

In **observation**, the researcher **watches** what people do or what happens in a situation and **records the results carefully. No direct questions are asked.**

The data may be **qualitative** (written notes, e.g. "students push in the queue") or **quantitative** (counts, e.g. "42 students bought samosas in 10 minutes").

Why use it? People sometimes say one thing and do another. Observation shows what really happens.

6.4.5 Simulations

LEARN IT

A **simulation** is a **computer-based model of a real situation. Variables** (the things that can change, like speed, temperature or number of people) can be changed to see what happens.

Simulations are highly useful when **real experiments are costly, risky or difficult** to perform.

Examples: a flight simulator for training pilots (real crashes are dangerous); a traffic simulation to test what happens if a new signal is added (changing real roads is costly); a weather model to predict a cyclone (you cannot experiment with real weather).

WORKED EXAMPLE

Choosing the right method. Which method would you use?

(a) "How many cars cross an intersection at 8 AM?" → Observation. Stand at the corner from 8:00 to 8:15 and tally each car. Asking drivers would be impossible and unnecessary; you just count.

(b) "How do students feel about the new cafeteria food?" → Survey or interview. Feelings are inside people's heads, so you must ask them. Use a survey for many students, or interviews for deeper reasons.

(c) "Will a new college website menu be easy to use?" → Prototype. Build a rough version and let users try it.

(d) "What happens to traffic if the bridge is closed for repairs?" → Simulation. Closing a real bridge to test is costly and disruptive.

REMEMBER IT

Watch it → Observation. **Ask** it → Survey/Interview. **Try** it → Prototype. **Model** it → Simulation.

COMMON MISTAKE

Mistake: Saying that observation involves asking people questions. **Fix:** In observation, **no direct questions** are asked – the researcher only watches and records. Also, a prototype is *not* the finished product; it is an early test version.

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔ مشاہدے (Observation) میں محقق کوئی سوال نہیں پوچھتا بلکہ صرف دیکھ کر نتائج درج کرتا ہے۔ سمیولیشن کسی حقیقی صورتحال کا کمپیوٹر ماڈل ہے، جو اس وقت مفید ہے جب حقیقی تجربہ مہنگا، خطرناک یا مشکل ہو۔

EXAM TIP

Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method **and** justify it using its key feature (e.g. "Simulation, because a real experiment would be risky").

CHECK YOURSELF

1. Why would an engineer use a simulation instead of a real test?
2. Can observation produce quantitative data? Give an example.
3. What becomes the "data" when using a prototype?

Answers: 1. Because the real experiment may be costly, risky or difficult. 2. Yes – for example, counting the number of cars passing a point. 3. The feedback users give after trying the prototype.

6.5 Primary and Secondary Data ★ Golden

LEARN IT

Whenever you use data, it is important to know **where it came from**. The source affects how much **control** the researcher had over collecting it, and how **confidently** the results can be used.

6.5.1 Primary Data

Primary data is collected **first-hand** by the researcher for a **particular purpose**. It is gathered **directly** from people, events or situations, rather than taken from an already published source.

Sources: interviews, surveys, observations, experiments.

6.5.2 Secondary Data

Secondary data has **already been collected by someone else** and is being used again for a **new purpose**. It is taken from existing archives (stored collections of information).

Sources: books, research articles, official reports, web statistics, online databases.

Primary Data

Fresh, direct collection
(You do the work)

Surveys · Interviews · Observation

Secondary Data

Existing archives
(Someone else did it)

Books · Reports · Databases

Primary data is collected by you; secondary data was collected by others.

6.5.3 Difference Between Primary and Secondary Data

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

WORKED EXAMPLE

Primary or secondary? Hira is researching "Internet use among students in Hyderabad".

(a) She sends a Google Form to 50 students in her college. → **Primary** (she collected it herself, for this study).

(b) She uses a PTA (Pakistan Telecommunication Authority) report on the number of broadband users in Pakistan. → **Secondary** (PTA collected it for its own purpose; Hira reuses it).

(c) She counts how many students use phones in the library during one hour. → **Primary** (her own observation).

(d) She quotes a statistic from a research article. → **Secondary**.

Why use both? Her primary data tells her about *her own* college; the secondary data lets her **compare** it with the national picture.

REMEMBER IT

Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

COMMON MISTAKE

Mistake: Thinking data from a website is always primary because "I found it myself". **Fix:** What matters is *who collected* the data. If someone else collected it and you only found it, it is secondary.

اروین سمجھیں

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔ ثانوی ڈیٹا (Secondary Data) پہلے ہی کسی اور نے جمع کیا ہوتا ہے اور اسے نئے مقصد کے لیے دوبارہ استعمال کیا جاتا ہے، جیسے کتابیں، رپورٹس اور آن لائن ڈیٹا بیس۔ بنیادی ڈیٹا پر محقق کا کنٹرول زیادہ اور ثانوی ڈیٹا پر کم ہوتا ہے۔

EXAM TIP

"Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: **Source, Examples, Purpose, Control**. Learn it exactly. Add one example from each side to secure full marks.

CHECK YOURSELF

1. Which type of data gives the researcher high control?
2. Is a government census report primary or secondary data for your project?
3. Give two sources of primary data.

Answers: 1. Primary data. 2. Secondary (someone else collected it). 3. Any two: interviews, surveys, observations, experiments.

6.6 Designing a Data-Collection Approach

LEARN IT

Before collecting data, a researcher must have a **plan**. A plan makes sure that the information gathered is **accurate**, **relevant** (related to the question) and **useful**. Follow these eight steps in order:

1. **Define the research question clearly.** What exactly do you want to find out?
2. **Decide the type of data needed:** quantitative, qualitative, or both.
3. **Select suitable methods** (survey, interview, observation).
4. **Choose the sample or target group** – the people who will give you data.
5. **Prepare the tools** such as forms, questions and observation sheets.
6. **Collect the data** carefully and accurately.
7. **Store the data digitally** in a spreadsheet and **keep a backup**.
8. **Follow ethical practice and respect privacy. Consent is mandatory** (you must get permission).



WORKED EXAMPLE

Applying the eight steps. Topic: water wastage in your college.

1. **Question:** "How much water is wasted at the college taps during break time, and why?"
2. **Data type:** Both – quantitative (number of taps left running, minutes) and qualitative (reasons students give).
3. **Methods:** Observation (count running taps) + a short survey (ask about habits).
4. **Sample:** All 12 taps in the college; 30 students from different classes for the survey.
5. **Tools:** An observation sheet with columns *Tap no.* / *Time* / *Left running?* (Y/N) / *Minutes*; a Google Form with 5 questions.
6. **Collect:** Observe during break for five days; send the form link.
7. **Store:** Enter the observation counts into Google Sheets; download a copy to a USB drive as backup.
8. **Ethics:** Ask the principal's permission; tell students the survey is anonymous and voluntary; do not record names.

REMEMBER IT

Remember the steps with: "**Q**uiet **T**eachers **M**ake **S**tudents **T**hink, **C**heck, **S**ave **E**thically" → Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

COMMON MISTAKE

Mistake: Starting to collect data before writing a clear research question. You then collect lots of data that does not answer anything. **Fix:** Always write the question first. Another mistake: keeping data in only one place – if the file is lost, the project is lost. Always keep a backup.

اردو میں سمجھیں

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں، فارم اور سوالات تیار کریں، احتیاط سے ڈیٹا جمع کریں، اسے اسپریڈشیٹ میں محفوظ کر کے بیک اپ رکھیں، اور اخلاقیات کا خیال رکھیں۔ اجازت (Consent) لینا لازمی ہے۔

CHECK YOURSELF

1. What is the first step in designing a data-collection approach?
2. Which step says "consent is mandatory"?
3. Where should collected data be stored?

Answers: 1. Define the research question clearly. 2. Step 8 – follow ethical practice and respect privacy. 3. Digitally, in a spreadsheet, with a backup copy.

6.7 Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

LEARN IT

Collecting data is only **one part** of the work. You must also **present** your findings so that others can understand them **quickly and easily**. Digital tools make this much easier. The four main tools are spreadsheets, presentations, infographics and reports.

6.7.1 Spreadsheets

LEARN IT

Spreadsheets such as **MS Excel**, **Google Sheets** and **Apple Numbers** arrange information in **rows** (horizontal, numbered 1, 2, 3...) and **columns** (vertical, lettered A, B, C...). The box where a row and column meet is a **cell**, named by its column and row, e.g. **B2**. With a spreadsheet you can:

- **Enter data** systematically (one record per row).
- **Calculate** totals, averages and percentages.
- **Sort** (arrange in order) and **filter** (show only some rows) records.
- **Create charts and graphs** automatically.

A **formula** always starts with = . A **range** such as B2:B4 means "cells B2 to B4".

Formula

What it does

=SUM(B2:B4)	Adds all values in B2 to B4.
=AVERAGE(B2:B4)	Mean: sum ÷ number of values.
=MAX(B2:B4) / =MIN(B2:B4)	Largest / smallest value.
=MEDIAN(B2:B4)	Middle value when sorted.
=COUNT(B2:B4)	Counts cells containing numbers.
=COUNTA(C2:C4)	Counts cells that are not empty (text or numbers).
=COUNTIF(C2:C4,"Yes")	Counts cells that match a condition.

WORKED EXAMPLE

Analysing the case-study spreadsheet. The data from the 6.8 case study (Step 8) is entered like this:

	A	B	C	D	E
1	Student	Hours	Study Use	Main Activity	Distracted?
2	1	3	Yes	Lectures	Sometimes
3	2	5	No	Social Media	Yes
4	3	2	Yes	Notes	No

Step 1 – Total hours: =SUM(B2:B4) → 3 + 5 + 2 = **10**.

Step 2 – Average hours: =AVERAGE(B2:B4) → 10 ÷ 3 = **3.33** hours (to 2 decimal places).

Step 3 – Most and least: =MAX(B2:B4) → **5**; =MIN(B2:B4) → **2**.

Step 4 – Median: sorted values 2, 3, 5 → middle = **3**. =MEDIAN(B2:B4) gives 3.

Step 5 – How many use the phone for study? =COUNTIF(C2:C4,"Yes") → **2**.

Step 6 – Percentage: =COUNTIF(C2:C4,"Yes")/COUNTA(C2:C4) → 2 ÷ 3 = 0.667; format the cell as % → **66.7%**.

Step 7 – How many said "Yes" to distracted? =COUNTIF(E2:E4,"Yes") → **1** (only "Yes" matches; "Sometimes" does not).

Step 8 – Sort: select the data → Data → Sort by column B (largest to smallest) → order becomes Student 2 (5), Student 1 (3), Student 3 (2).

6.7.2 Presentations

LEARN IT

Presentation software such as **PowerPoint**, **Google Slides** and **Canva** helps you explain findings to an **audience** in an organised way. Good practice:

- Begin with a clear **title** and the **research question**.
- Use **short bullet points** instead of long paragraphs.
- Include **tables, charts and key findings**.
- Keep the design **simple and readable** (large font, few colours).

Spreadsheets and presentations work together: you make a chart from the spreadsheet data and insert it into the slides.

Spreadsheet data



Chart / Graph



Insert



Slides

REMEMBER IT

Presentation rule of thumb: **one idea per slide**, and no more than about **six short bullet points**. If you are reading long sentences off the slide, the slide has too much text.

COMMON MISTAKE

Mistake: Typing a result by hand (e.g. writing "3.33" in a cell) instead of using a formula. If the data changes, the typed result becomes wrong. **Fix:** Always use formulas like `=AVERAGE(B2:B4)` so results update automatically. Also remember every formula must begin with `=`.

ارو میں سمجھیں

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔ اسپریڈشیٹ (جیسے ایکسل اور گوگل شیٹس) میں ڈیٹا قطاروں اور کالموں میں ترتیب دیا جاتا ہے، اور `=SUM()` اور `=AVERAGE()` جیسے فارمولوں سے مجموعہ، اوسط اور فیصد نکالے جاتے ہیں۔ پریزنٹیشن سافٹ ویئر (جیسے پاور پوائنٹ) میں مختصر نکات، چارٹس اور سادہ ڈیزائن کے ساتھ نتائج سامعین کے سامنے پیش کیے جاتے ہیں۔

CHECK YOURSELF

1. Write the formula to find the average of cells D2 to D20.
2. Name three spreadsheet programs.
3. Give two rules for a good presentation.

Answers: 1. `=AVERAGE(D2:D20)` . 2. MS Excel, Google Sheets, Apple Numbers. 3. Any two: start with a clear title and research question; use short bullet points; include tables/charts/key findings; keep design simple and readable.

6.7.3 – 6.7.4 Infographics and Reports

6.7.3 Infographics

LEARN IT

An **infographic** (information + graphic) is a **visual way** of presenting information, facts or data so that people can understand the main idea **quickly and easily**. Tools such as **Canva**, **Power BI** or **Tableau** can be used. A good infographic should:

- **Simplify** complex concepts.
- **Highlight** the most important findings and results.
- Use **clear visuals**, **limited text** and **accurate** information.

6.7.4 Reports

LEARN IT

Reports are **structured documents** used to present collected data, findings, analysis and conclusions. Tools like **MS Word** or **Google Docs** are standard. A good report should:

- Start with a clear **title**, **introduction** and **purpose**.
- Present the collected data using **tables or charts**.
- Explain findings in **simple language**.
- Include **conclusions and recommendations** based on the data.

Optional: you may use AI tools carefully to improve wording, but **always review for accuracy**. AI can make mistakes or invent facts, and you are responsible for what you submit.

LEARN IT

Visual rule of thumb: a good chart or infographic can explain complex data **faster** than a long paragraph. However, visuals must always be **accurate** and based strictly on the **real data collected** – never exaggerated.

WORKED EXAMPLE

Which tool should you choose?

Situation	Best tool	Why
You must calculate the average marks of 40 students.	Spreadsheet	Formulas calculate instantly.
You will explain results to the class in assembly.	Presentation	Designed for a live audience.
You want a one-page poster for the notice board.	Infographic	Main result understood at a glance.
The principal wants the full study with method and recommendations.	Report	Detailed, structured document.

REMEMBER IT

Match tool to job: **Spreadsheet = Calculate, Presentation = Tell, Infographic = Show quickly, Report = Explain fully.**

COMMON MISTAKE

Mistake: Making a chart look more dramatic than the data – for example, starting the bar-chart axis at 18 instead of 0 so that 20 looks twice as big as 19. **Fix:** Start bar-chart axes at 0, label axes, and show real numbers only.

اردو میں سمجھیں

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے، اس کے لیے کینوا (Canva) اور پاور بی آئی (Power BI) جیسے ٹولز استعمال ہوتے ہیں۔ رپورٹ ایک منظم دستاویز ہے جس میں عنوان، تعارف، ڈیٹا، نتائج، اختتامیہ اور سفارشات شامل ہوتی ہیں۔ تصاویر اور چارٹس ہمیشہ اصل ڈیٹا کے مطابق اور درست ہونے چاہئیں، مبالغہ آمیز نہیں۔

CHECK YOURSELF

1. Name two tools for making infographics.
2. What must you always do if you use AI to improve your report's wording?
3. What should a report end with?

Answers: 1. Any two: Canva, Power BI, Tableau. 2. Review it carefully for accuracy. 3. Conclusions and recommendations based on the data.

6.8 Case Study: Digital Inquiry Project ★ Golden

LEARN IT

This case study shows how to apply **all** the ideas of this chapter to solve a real-world problem from start to finish. An **inquiry** is an investigation to find the answer to a question.

Research question: "How does daily mobile phone use affect the study habits of students?"

Scenario: Students notice classmates using phones during free periods and while studying. Some say phones **help** them learn; others say phones **waste time**. The class must investigate whether phone use helps or harms study habits.

The digital inquiry workflow

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

The project is carried out in **12 steps** that fit into this five-stage workflow:

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data

Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

REMEMBER IT

Workflow in five words: **Ask → Search → Collect → Analyse → Create**. Every research project you ever do will follow this pattern.

اروین سمجھیں

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟" تحقیق پانچ مراحل میں ہوتی ہے: سوال، تلاش، ڈیٹا جمع کرنا، تجزیہ اور آخری پروڈکٹ بنانا۔ یہ کام بارہ قدموں میں مکمل کیا جاتا ہے۔

EXAM TIP

Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps with one or two lines each, and finish with the artefact. Learn the five-stage workflow as your skeleton.

CHECK YOURSELF

1. What is the research question in the case study?
2. Name the five stages of the digital inquiry workflow.

Answers: 1. "How does daily mobile phone use affect the study habits of students?" 2. Question, Search, Collect, Analyse, Create.

6.8 Steps 1 – 3: Advanced Search and Methodology ★ Golden

The **methodology** is the plan of methods a researcher uses to answer the question.

Step 1: Perform advanced searches

LEARN IT

Students first search for **background information** using search engines (such as Google or Bing). They use **advanced search operators** – special symbols or words typed into the search box – to **narrow down** results and find **reliable secondary data**.

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Other useful operators (background): - (minus) excludes a word, e.g. phone addiction -game; OR finds either word, e.g. smartphone OR mobile; site:.gov.pk searches only Pakistani government websites.

LEARN IT

Source evaluation checklist. Before trusting a source, ask:

- | | |
|---|--|
| ☐ Is the author/organisation clearly mentioned? | ☐ Is the information recent ? |
| ☐ Is the website reliable (.edu, .gov, research-based)? | ☐ Does the source provide facts/references , or is it biased (one-sided, trying to sell something)? |

WORKED EXAMPLE

Building a search query step by step. You want official PDF reports about screen time of students on Pakistani government websites.

Step 1 – Plain search: screen time students → millions of results, many blogs and adverts.

Step 2 – Exact phrase: "screen time" students → results must contain the exact phrase.

Step 3 – Only PDFs: "screen time" students filetype:pdf → mostly reports and papers.

Step 4 – Only government sites: "screen time" students filetype:pdf site:.gov.pk → a small number of highly relevant, reliable results.

Step 5 – Evaluate: open a result and check the checklist: author named? year recent? references given?

Step 2: Identify data types

LEARN IT

Students decide what kind of **primary data** they will collect:

- **Quantitative:** number of hours used daily; percentage who feel distracted.
- **Qualitative:** opinions on why they get distracted; suggestions for better use.

Step 3: Select data-collection methods

LEARN IT

Because they need to ask the **same questions to many students**, a **survey/questionnaire** (via Google Forms or on paper) is the **most efficient** method. Interviews would take too long for so many people, and observation could not reveal opinions.

REMEMBER IT

Three operators, three jobs: " " = **exact**, **filetype:** = **format**, **site:** = **where**. Note: there is **no space** after the colon (`site:.edu` , not `site: .edu`).

COMMON MISTAKE

Mistake: Assuming the first search result is the most reliable. Search results can be adverts or popular blogs. **Fix:** Always apply the source evaluation checklist. A website ending in .edu or .gov is usually more reliable, but you should still check the date and author.

اردو میں سمجھیں

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوئیشن مارکس ("") بالکل وہی جملہ تلاش کرتے ہیں، filetype:pdf صرف پی ڈی ایف فائلیں اور site:.edu صرف تعلیمی ویب سائٹس دکھاتا ہے۔ ہر ذریعے کو مصنف، تاریخ، قابل اعتماد ویب سائٹ اور حوالوں کی بنیاد پر پرکھا جاتا ہے۔ پھر ڈیٹا کی اقسام طے کی جاتی ہیں اور بہت سے طلبہ سے سوال کرنے کے لیے سروے کو بہترین طریقہ چنا جاتا ہے۔

EXAM TIP

"Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the four-point source evaluation checklist.

CHECK YOURSELF

1. Write a query that finds the exact phrase "online learning" only on educational websites.
2. Why was a survey chosen in Step 3?
3. Give one quantitative and one qualitative data item from Step 2.

Answers: 1. "online learning" site:.edu . 2. Because the same questions had to be asked to many students, and a survey is the most efficient way to do that. 3. Quantitative: hours used daily (or % who feel distracted). Qualitative: opinions on why they get distracted (or suggestions for better use).

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection

★ Golden

Step 4: Prepare the survey

LEARN IT

Design **clear, unbiased** questions. Each question is linked to the type of data it produces:

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Step 5: Choose the sample

LEARN IT

The **sample** is the group of people who provide the data. It is usually impossible to ask everyone (the whole **population**), so a smaller group is chosen. The students decide to survey **40 Class XI students from different sections** to get a **representative** mix of habits. "Representative" means the sample looks like the whole group – it includes different kinds of students, not just your friends.

Step 6: Collect primary data (ethically)

LEARN IT

Ethics means doing what is morally right. When conducting the survey, students must follow strict ethical guidelines:

- Ask **permission** before collecting responses (**consent**).
- Avoid collecting **sensitive personal information**.
- Do **not record names** unless absolutely necessary (**anonymity**).
- Keep the responses **private**.
- Use the data **only for educational purposes**.

WORKED EXAMPLE

Why "different sections" matters. Suppose the group surveyed only the 40 students of the Pre-Engineering section who sit in the front rows.

Problem: these students might study more than average, so the results would show less phone distraction than is true for the whole class.

Fix: pick students from every section (e.g. 10 each from Pre-Medical, Pre-Engineering, Computer Science and Commerce). Now the sample better represents all Class XI students.

Ethical opening text for the form: "This survey is for a class project on phone use and study habits. It is anonymous – do not write your name. Your answers will be kept private and used only for this project. Do you agree to take part? ☐ Yes ☐ No"

REMEMBER IT

Ethics in five words: **Consent, No-Sensitive, Anonymous, Private, Educational** – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

COMMON MISTAKE

Mistake: Thinking a bigger sample is always enough. A sample of 200 students all from one section is still biased. **Fix:** A sample must be **representative**, not just large.

اردو میں سمجھیں

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔ پانچویں قدم میں نمونہ (Sample) چنا جاتا ہے: مختلف سیکشنز کے چالیس طلبہ تاکہ نتائج پوری جماعت کی نمائندگی کریں۔ چھٹے قدم میں اخلاقی اصولوں کے ساتھ ڈیٹا جمع کیا جاتا ہے: اجازت لیں، حساس معلومات نہ لیں، نام نہ لکھیں (گمنامی)، جوابات کو راز میں رکھیں اور صرف تعلیمی مقصد کے لیے استعمال کریں۔

EXAM TIP

Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words **consent** and **anonymity** – examiners look for them. You may also be asked to classify the four survey questions by data type.

CHECK YOURSELF

1. What is a sample?
2. What does anonymity mean in a survey?
3. What type of data does "Suggest one way to use phones responsibly" give?

Answers: 1. The group of people who provide the data. 2. Not recording the names of participants (unless absolutely necessary). 3. Qualitative.

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation

★ Golden

Step 7: Use secondary data

LEARN IT

Students include a reliable online article stating that "**excessive screen time lowers concentration.**" They will **compare** this secondary data with their own primary survey results. If both agree, the conclusion is stronger. If they disagree, the students must think about why (for example, a different age group or country).

Step 8: Organise data in a spreadsheet

LEARN IT

Responses are entered into **MS Excel** or **Google Sheets** to **sort**, **calculate totals** and **find averages**. Each row is one student; each column is one question.

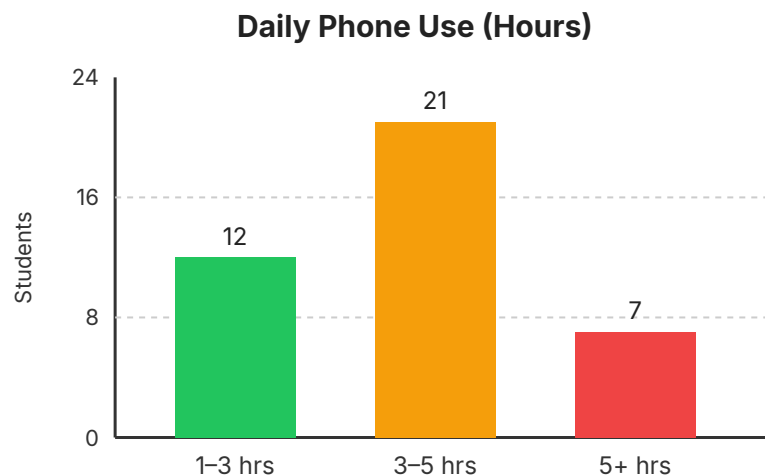
Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

(The full sheet has 40 rows, one per student. The formulas for this table are worked out in topic 6.7.1.)

Step 9: Create charts and graphs

LEARN IT

The spreadsheet data is converted into **visuals** to make it easy to digest. In the source, the "Daily Phone Use (Hours)" bar chart groups the 40 students into three bars: 1–3 hrs, 3–5 hrs and 5+ hrs, with the 3–5 hrs bar the tallest.



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

WORKED EXAMPLE

Making the chart in Google Sheets.

Step 1 – Count each group. If the hours are in B2:B41, use `=COUNTIF(B2:B41,"<3")` for under 3 hrs, `=COUNTIFS(B2:B41,"≥3",B2:B41,"<5")` for 3 to under 5 hrs, and `=COUNTIF(B2:B41,"≥5")` for 5 hrs or more. (This gives non-overlapping groups.)

Step 2 – Check the total. $12 + 21 + 7 = 40$ ✓ – every student is counted exactly once.

Step 3 – Percentages. $12 \div 40 = 30\%$; $21 \div 40 = 52.5\%$; $7 \div 40 = 17.5\%$. Total = 100% ✓.

Step 4 – Insert chart. Select the group labels and counts → Insert → Chart → choose *Column chart*. Add the title "Daily Phone Use (Hours)" and label the axes.

Step 5 – Choose the right chart. Bar/column chart to compare groups; pie chart to show parts of a whole (e.g. % in each group); line chart to show change over time.

REMEMBER IT

Chart chooser: **Bar = Compare**, **Pie = Parts of a whole**, **Line = Change over time**.

COMMON MISTAKE

Mistake: Using overlapping groups like "1–3" and "3–5": a student who uses the phone for exactly 3 hours could be counted twice or put in the wrong bar. **Fix:** Define groups clearly, e.g. "under 3", "3 to under 5", "5 or more", and check that the group counts add up to the sample size.

اردو میں سمجھیں

ساتویں قدم میں ایک قابلِ اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے موازنہ کرنے کے لیے لیا جاتا ہے۔ آٹھویں قدم میں جوابات ایکسل یا گوگل شیٹس میں درج کر کے ترتیب، مجموعہ اور اوسط نکالی جاتی ہے۔ نویں قدم میں اسپریڈشیٹ کے ڈیٹا سے چارٹ اور گراف بنائے جاتے ہیں تاکہ نتائج آسانی سے سمجھ آئیں۔

EXAM TIP

You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full formula with `=`, the correct range, and the result.

CHECK YOURSELF

1. What is the secondary data used in Step 7?
2. Which formula counts how many students answered "Yes" in column C, rows 2 to 41?
3. Which type of chart is best for comparing the three hour-groups?

Answers: 1. A reliable online article stating "excessive screen time lowers concentration". 2. `=COUNTIF(C2:C41, "Yes")`. 3. A bar (column) chart.

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations

★ Golden

Step 10: Analyse the results

LEARN IT

To **analyse** means to study the data carefully to find **patterns** and **trends**. By looking at the spreadsheet and charts, the group identifies:

- **Most students** (the majority) use phones for **3–5 hours daily**.
- **Half** use it for **educational purposes** (lectures/notes).
- **A clear trend:** students who use phones mostly for **entertainment** report much **higher distraction** levels than those who use them for study.

Step 11: Write the conclusion

LEARN IT

A **conclusion** answers the research question using the evidence:

"The research shows mobile phones have both positive and negative effects. They provide access to learning materials. However, excessive use of social media and gaming reduces concentration. Therefore, phones should not be banned, but managed wisely."

Step 12: Recommendations

LEARN IT

Recommendations are practical suggestions based on the data. The students recommend:

- Turn off **social media notifications** while studying.
- Set **daily screen-time limits**.
- Take **physical breaks** from screens to protect health.

WORKED EXAMPLE

From numbers to a finding. Using the estimated chart counts (12, 21, 7 out of 40):

Step 1: $21 \div 40 = 52.5\%$. Since $52.5\% > 50\%$, the statement "most students use phones for 3–5 hours" is supported.

Step 2: "Half use it for educational purposes" means about 20 of the 40 students chose Lectures or Notes as their main activity ($20 \div 40 = 50\%$). Formula:

`=COUNTIF(D2:D41, "Lectures")+COUNTIF(D2:D41, "Notes")` .

Step 3 – Compare groups: filter the sheet to show only Social Media/Games users and count how many said "Yes" to distracted; then do the same for Lectures/Notes users. If the entertainment group has a much higher share of "Yes", that is the trend.

Step 4 – Link to secondary data: the article said excessive screen time lowers concentration; the survey agrees, so the conclusion is stronger.

REMEMBER IT

Conclusion formula: "**The data shows ... However ... Therefore ...**" – state the main finding, mention the other side, then give the overall answer.

COMMON MISTAKE

Mistake: Writing a conclusion that is not based on the data, e.g. "Phones should be banned" when the data showed phones also help learning. **Fix:** Every sentence in the conclusion and recommendations must be supported by your findings. Also, a trend between two things (entertainment use and distraction) shows a *link*; be careful before claiming one definitely *causes* the other.

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف اسے تعلیم کے لیے استعمال کرتے ہیں، اور تفریح کے لیے استعمال کرنے والے زیادہ توجہ بٹنے کی شکایت کرتے ہیں۔ گیارہویں قدم میں نتیجہ لکھا جاتا ہے کہ فون پر پابندی نہیں بلکہ اس کا دانشمندانہ استعمال ہونا چاہیے۔ بارہویں قدم میں سفارشات دی جاتی ہیں: پڑھائی کے دوران نوٹیفیکیشن بند کریں، اسکرین ٹائم کی حد مقرر کریں اور وقفے لیں۔

EXAM TIP

You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows from a finding.

CHECK YOURSELF

1. According to Step 10, which group reported higher distraction?
2. Should phones be banned according to the conclusion?
3. Give two of the students' recommendations.

Answers: 1. Students who use phones mostly for entertainment. 2. No – they should not be banned but managed wisely. 3. Any two: turn off social media notifications while studying; set daily screen-time limits; take physical breaks from screens.

6.8 The Final Digital Artefact ★ Golden

LEARN IT

A **digital artefact** is a final product created using digital technologies. Students must create **one final artefact** that answers the original research question. It is **not just a decorated file**; it must present **evidence** and **conclusions**.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

A **dashboard** is an interactive screen of charts; the viewer can click filters (e.g. "show only Commerce students") and the charts update.

Standard artefact structure

1. Title
2. Research Question
3. Introduction

4. Method of Data Collection
5. Primary Data Findings
6. Secondary Data Summary
7. Charts and Graphs
8. Conclusion
9. Recommendations
10. References

WORKED EXAMPLE

Planning a 10-slide presentation artefact for the phone study (one slide per structure item):

Slide 1 Title: "Phones and Study Habits – Class XI Survey". **Slide 2** Research question. **Slide 3** Introduction: why the class investigated this. **Slide 4** Method: Google Form survey, 40 students, different sections, anonymous with consent. **Slide 5** Primary findings: 52.5% use phones 3–5 hours; half use for study. **Slide 6** Secondary data: article on screen time and concentration. **Slide 7** The bar chart of daily phone use. **Slide 8** Conclusion: "manage wisely, don't ban". **Slide 9** Three recommendations. **Slide 10** References: the article's title, author, website and date.

REMEMBER IT

The structure follows the inquiry itself: **What** (title, question, intro) → **How** (method) → **Found** (primary, secondary, charts) → **So what** (conclusion, recommendations) → **Credit** (references).

COMMON MISTAKE

Mistake: Spending all your time on colours and animations and forgetting evidence. **Fix:** An artefact is judged on whether it answers the research question with evidence. Also never skip **References** – listing your secondary sources is honest and avoids plagiarism.

اروین سمجھیں

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش کرتی ہے؛ یہ صرف سچی ہوئی فائل نہیں۔ اس کی چار صورتیں ہیں: رپورٹ، پریزنٹیشن، انفوگرافک اور ڈیش بورڈ۔ معیاری ساخت میں عنوان، تحقیقی سوال، تعارف، طریقہ کار، بنیادی و ثانوی ڈیٹا، چارٹس، نتیجہ، سفارشات اور حوالہ جات شامل ہیں۔

EXAM TIP

Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must present **evidence and conclusions**, not just decoration.

CHECK YOURSELF

1. Which artefact lets viewers explore data with interactive filters?
2. What is the last item in the standard artefact structure?
3. Name two tools for making an infographic artefact.

Answers: 1. A dashboard (Power BI, Tableau). 2. References. 3. Canva and Adobe Express.

Careers in Digital Research

LEARN IT

The skills in this chapter lead to real jobs:

Career	What they do	Chapter skill used
Market Research Analyst	Designs surveys to collect primary data to guide business decisions.	Surveys, primary data
UX (User Experience) Researcher	Gathers qualitative and quantitative data to improve app and website designs.	Prototypes, interviews, observation
Data Journalist	Uses advanced search skills to research stories and presents them via infographics.	Advanced search, infographics
Digital Marketing Specialist	Collects audience data and presents results through dashboards and reports.	Dashboards, reports

WORKED EXAMPLE

A day for a UX researcher. A Karachi food-delivery app wants to redesign its checkout. The UX researcher (1) builds a clickable **prototype**, (2) **observes** 8 users trying it and counts how many seconds each takes (quantitative), (3) **interviews** them about what confused them (qualitative), and (4) presents a short **report** with recommendations to the design team.

REMEMBER IT

Link each career to one tool: Market Researcher → **Survey**; UX Researcher → **Prototype**; Data Journalist → **Search + Infographic**; Digital Marketer → **Dashboard**.

ارو میں سمجھیں

اس باب کی مہارتوں سے کئی کیریئر بنتے ہیں: مارکیٹ ریسرچ اینالسٹ کاروباری فیصلوں کے لیے سروے بناتا ہے، یو ایکس (UX) ریسرچر ایپس اور ویب سائٹس بہتر بنانے کے لیے ڈیٹا جمع کرتا ہے، ڈیٹا جرنلسٹ جدید سرچ اور انفوگرافکس استعمال کرتا ہے، اور ڈیجیٹل مارکیٹنگ اسپیشلسٹ ڈیش بورڈز اور رپورٹس کے ذریعے نتائج پیش کرتا ہے۔

CHECK YOURSELF

1. Which career designs surveys to guide business decisions?
2. What does UX stand for?

Answers: 1. Market Research Analyst. 2. User Experience.

Chapter 6 Review

Chapter summary

- **Digital literacy** is the ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly – not just using devices, but finding, evaluating and creating information.
- In the **Information Age**, access to technology is not enough; digital literacy needs technical, cognitive and social abilities.
- **Qualitative data** is descriptive and non-numerical (opinions, reasons, behaviour) and explains why/how.
- **Quantitative data** is numerical, countable or measurable (hours, scores, percentages) and is used for comparing and charting.
- The five **data-collection strategies** are interviews, surveys, prototypes, observation and simulations.
- **Interviews** are planned conversations giving mostly qualitative data; **surveys** ask the same questions to many people, on paper or with Google/Microsoft Forms.
- Good survey questions are clear, short, unbiased and ask about one thing; avoid leading, double-barrelled, vague and overlapping questions.
- A **prototype** is an early version tested for feedback; **observation** records what happens without asking questions; a **simulation** is a computer model used when real experiments are costly, risky or difficult.
- **Primary data** is collected first-hand (high control); **secondary data** was collected by others and reused (limited control).
- Designing a data-collection approach has eight steps: question, data type, method, sample, tools, collect, store with backup, and ethics (consent is mandatory).
- **Spreadsheets** organise and calculate (e.g. `=SUM()` , `=AVERAGE()` , `=COUNTIF()`); **presentations** deliver to an audience; **infographics** give quick visual summaries; **reports** give detailed structure.
- Visuals must always be accurate and based strictly on real data, never exaggerated.
- **Advanced search operators** such as " " , `filetype:pdf` and `site:.edu` help find reliable sources efficiently; sources must be evaluated for author, date, reliability and bias.
- **Ethics**: get consent, avoid sensitive information, keep anonymity, keep responses private, and use data only for educational purposes.
- The **digital inquiry project** follows Question → Search → Collect → Analyse → Create in 12 steps.
- A **digital artefact** (report, presentation, infographic or dashboard) answers the research question using evidence and visuals, following a ten-part structure ending in references.

Key terms

Term	Meaning	اردو
Digital literacy	Ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly.	ڈیجیٹل خواندگی
Information Age	Period in which vast amounts of data are produced, shared and accessed instantly.	معلوماتی دور

Data	Facts, numbers, words or observations collected for study.	مواد / ڈیٹا
Qualitative data	Descriptive, non-numerical data about thoughts, feelings and experiences.	کیفیتی ڈیٹا
Quantitative data	Numerical data that can be counted or measured.	مقداری ڈیٹا
Interview	Planned conversation to gather information directly from a person or small group.	انٹرویو / بالمشافہ گفتگو
Survey / Questionnaire	Same set of questions asked to many people.	سروے / سوالنامہ
Prototype	Simple early version of a product used to test an idea and get feedback.	ابتدائی نمونہ
Observation	Watching and recording what happens without asking questions.	مشاہدہ
Simulation	Computer-based model of a real situation in which variables can be changed.	نقلی ماڈل (سمیولیشن)
Variable	A factor that can be changed or measured.	متغیر
Primary data	Data collected first-hand by the researcher for a specific purpose.	بنیادی ڈیٹا
Secondary data	Data already collected by someone else and reused for a new purpose.	ثانوی ڈیٹا
Research question	The clear question an investigation tries to answer.	تحقیقی سوال
Sample	The group of people who provide the data.	نمونہ
Representative sample	A sample that reflects the mix of the whole group.	نمائندہ نمونہ
Consent	Permission given by a person before their data is collected.	رضامندی / اجازت
Anonymity	Not recording people's names so they cannot be identified.	گمنامی
Ethics	Rules of right and fair behaviour when collecting and using data.	اخلاقیات
Spreadsheet	Software that arranges data in rows and columns and performs calculations.	اسپریڈشیٹ / حسابی جدول
Formula	A spreadsheet instruction starting with = that calculates a result.	فارمولا

Infographic	A visual presentation of information so the main idea is understood quickly.	تصویری معلومات (انفوگرافک)
Report	Structured document presenting data, findings, analysis and conclusions.	رپورٹ
Dashboard	Interactive screen of charts and filters for exploring data.	ڈیش بورڈ
Advanced search operator	Special symbol or word (e.g. " ", filetype:, site:) that narrows search results.	جدید تلاش آپریٹر
Methodology	The plan of methods used to answer a research question.	طریقہ کار
Digital artefact	Final product made with digital technology that answers the research question with evidence.	ڈیجیٹل آرٹیفیکٹ / ڈیجیٹل تخلیق
Recommendation	A practical suggestion based on the findings.	سفارش

Practice: multiple choice

- Digital literacy is best described as the ability to:
 - type quickly on a keyboard
 - understand and use digital devices, the internet and online tools safely, effectively and responsibly
 - repair computer hardware
 - write computer programs
- Which of the following is qualitative data?
 - Marks obtained in a test
 - Number of hours spent online
 - Students' opinions about online classes
 - Percentage of students using the internet
- Which method is most suitable when the same questions must be asked to many people?
 - Survey
 - Interview
 - Prototype
 - Simulation
- Simulations are especially useful when:
 - you want people's opinions
 - the data already exists in a report
 - you want to watch people without asking questions
 - real experiments are costly, risky or difficult
- To find out how many cars cross an intersection at 8 AM, the best method is:
 - Interview
 - Observation
 - Survey
 - Prototype
- Which of these is primary data?
 - Responses to a Google Form you created for your project
 - An official government report
 - A published research article
 - Web statistics from a website

7. A researcher has limited control over:
- a. their own interviews
 - b. their own survey questions
 - c. their own observations
 - d. secondary data
8. The first step in designing a data-collection approach is to:
- a. collect the data
 - b. choose the sample
 - c. define the research question clearly
 - d. store the data with a backup
9. Which spreadsheet formula finds the mean of cells B2 to B6?
- a. =AVERAGE(B2:B6)
 - b. =SUM(B2:B6)
 - c. =COUNT(B2:B6)
 - d. =MAX(B2:B6)
10. The search query `digital distraction site:.edu` will:
- a. find only PDF files
 - b. search only educational websites
 - c. find the exact phrase "digital distraction" only
 - d. exclude educational websites
11. The operator `filetype:pdf` is used to:
- a. search for an exact phrase
 - b. search only government websites
 - c. convert a web page into PDF
 - d. find only PDF documents
12. "Do not record names unless absolutely necessary" describes the principle of:
- a. consent
 - b. accuracy
 - c. anonymity
 - d. sampling
13. In the case study, how many Class XI students were surveyed?
- a. 20
 - b. 40
 - c. 50
 - d. 100
14. Which tools are suggested for a dashboard artefact?
- a. Power BI, Tableau
 - b. MS Word, Google Docs
 - c. PowerPoint, Google Slides
 - d. Google Forms, Microsoft Forms
15. A prototype is:
- a. the final finished product
 - b. a published report
 - c. a type of survey question
 - d. a simple early version of a product used to test an idea

Practice: short questions

1. Define digital literacy. Name the three kinds of ability it involves.
2. Why is simply having access to technology not enough in the Information Age?
3. Differentiate between qualitative and quantitative data, giving two examples of each.
4. How is an interview different from a survey?
5. What is a prototype, and how does it produce useful data?
6. When is a simulation used? Give one example.
7. Compare primary and secondary data in terms of purpose and control.
8. List any four ethical guidelines to follow when collecting primary data.
9. Explain three advanced search operators with an example of each.

10. What is a digital artefact? Name its four options with one suitable tool each.

Practice: long / problem questions

1. Explain the eight steps of designing a data-collection approach. Apply each step to the research question: "How much water is wasted at the college taps during break time, and why?"
2. **Design a survey for** finding out how Class XI students in your college travel to college and how long the journey takes. Write a consent statement and at least six questions (mixing closed, number, rating and open-ended types), state the data type of each question, describe your sample, and explain how you will keep the data ethical.
3. Ten students recorded their daily phone use (hours) in cells B2 to B11: 2, 4, 3, 5, 6, 4, 3, 4, 1, 4. (a) Write the spreadsheet formulas and give the results for the total, mean, median, mode, maximum and minimum. (b) Write a formula to count students using the phone for more than 4 hours, and find the percentage. (c) Group the data into "1–2 hrs", "3–4 hrs" and "5+ hrs" and name a suitable chart.
4. Describe all 12 steps of the Digital Inquiry Project on the question "How does daily mobile phone use affect the study habits of students?", including the final digital artefact and its structure.
5. **Design a survey for** investigating "How do Class XI students use AI tools for homework?" Then explain which secondary sources you would search for (with advanced search queries), and which artefact you would create to present your findings to the principal, with reasons.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. d | 7. d | 10. b | 13. b |
| 2. c | 5. b | 8. c | 11. d | 14. a |
| 3. a | 6. a | 9. a | 12. c | 15. d |

Short questions — model answers

1. Digital literacy is the ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. It involves technical, cognitive and social abilities.
2. In the Information Age vast amounts of data are produced and shared instantly. Owning a device does not mean you can judge or use this information well. Users must develop skills to understand, manage and use information wisely, and to tell reliable information from unreliable.
3. Qualitative data is descriptive and non-numerical; it explains why and how (e.g. students' opinions about online classes; observation notes). Quantitative data is numerical, countable or measurable; it is used for comparing and making charts (e.g. marks in a test; hours spent online).
4. An interview is a planned conversation with one person or a small group, giving mostly qualitative, in-depth answers. A survey asks the same set of questions to many people (on paper or via Google/Microsoft Forms) and is efficient for large groups; online surveys also collect responses and make charts automatically.
5. A prototype is a simple early version of a product, model or solution. Users try it and give feedback; this feedback is data used to improve the product before time and effort are spent on the final version.
6. A simulation is used when real experiments are costly, risky or difficult. Example: a flight simulator to train pilots, or a traffic simulation to test the effect of closing a road.
7. Purpose: primary data is prepared for a specific study; secondary data is reused for a new purpose. Control: the researcher has high control over primary data (questions and process) and limited control over secondary data because it already exists.

8. Any four: ask permission (consent); avoid sensitive personal information; do not record names unless necessary (anonymity); keep responses private; use data only for educational purposes.
9. Quotes " " find an exact phrase, e.g. "study habits" students. filetype: finds a file format, e.g. "academic performance" filetype:pdf gives only PDFs. site: limits to a type of website, e.g. digital distraction site:.edu searches only educational sites.
10. A digital artefact is a final product created with digital technology that answers the research question with evidence and conclusions. Options: digital report (MS Word/Google Docs), presentation (PowerPoint/Canva/Slides), infographic (Canva/Adobe Express), dashboard (Power BI/Tableau).

Long questions — model answers

1. **Eight steps applied to water wastage.** (1) **Define the question:** "How much water is wasted at the college taps during break time, and why?" (2) **Data type:** both – quantitative (number of taps left running, minutes running) and qualitative (students' reasons). (3) **Methods:** observation to count running taps; a short survey to learn habits and reasons. (4) **Sample:** all 12 taps observed over 5 break times; 30 students from different classes surveyed. (5) **Tools:** an observation sheet (Tap no., Time, Left running Y/N, Minutes) and a Google Form. (6) **Collect** carefully, at the same time each day, recording accurately. (7) **Store** data in Google Sheets and keep a backup copy (download or USB). (8) **Ethics:** get the principal's permission; the survey is voluntary and anonymous with a consent question; data used only for the project.
2. **Survey on travel to college (model).** *Consent:* "This anonymous survey is for a Class XI project on travel to college. Do not write your name. Answers will be kept private and used only for this project. Do you agree to take part? Yes / No." *Questions:* Q1 How do you usually travel to college? Walk / Bus / Rickshaw / Motorbike / Car / Other – quantitative (categories). Q2 How many minutes does your journey usually take? ____ – quantitative. Q3 How many days a week are you late because of travel? 0 / 1 / 2 / 3 or more – quantitative. Q4 "My journey to college is safe." 1 (strongly disagree) to 5 (strongly agree) – quantitative (rating). Q5 Do you travel alone? Yes / No – quantitative (count). Q6 What is the biggest problem you face while travelling? – qualitative. Q7 Suggest one way the college could make travel easier. – qualitative. Questions are unbiased, ask one thing each, and options do not overlap. *Sample:* about 40 Class XI students, roughly 10 from each section, to be representative. *Ethics:* consent, no sensitive information (no home address), no names (anonymity), private storage, educational use only. Data is collected through Google Forms and exported to Google Sheets for analysis.
3. **Spreadsheet problem.** (a) $\text{=SUM(B2:B11)} \rightarrow 2+4+3+5+6+4+3+4+1+4 = 36$. $\text{=AVERAGE(B2:B11)} \rightarrow 36 \div 10 = 3.6$ hours. Sorted: 1, 2, 3, 3, 4, 4, 4, 4, 5, 6; $\text{=MEDIAN(B2:B11)} \rightarrow (4 + 4) \div 2 = 4$. $\text{=MODE(B2:B11)} \rightarrow 4$ (appears 4 times). $\text{=MAX(B2:B11)} \rightarrow 6$; $\text{=MIN(B2:B11)} \rightarrow 1$. (b) $\text{=COUNTIF(B2:B11, ">4")}$ $\rightarrow$ values 5 and 6 $\rightarrow 2$ students; percentage $\text{=COUNTIF(B2:B11, ">4")}/\text{COUNT(B2:B11)} \rightarrow 2 \div 10 = 20\%$. (c) 1–2 hrs: 2 students (2, 1); 3–4 hrs: 6 students (4, 3, 4, 3, 4, 4); 5+ hrs: 2 students (5, 6). Check: $2 + 6 + 2 = 10 \checkmark$. A **bar/column chart** is suitable for comparing the groups (or a pie chart to show 20%, 60%, 20%).
4. **Twelve steps.** (1) Advanced searches with operators such as "mobile phone use" "study habits" students, "academic performance" filetype:pdf, digital distraction site:.edu; evaluate sources (author, recent, reliable site, facts/bias). (2) Identify data types: quantitative (hours daily, % distracted) and qualitative (reasons, suggestions). (3) Select method: survey via Google Forms/paper, because the same questions go to many students. (4) Prepare the survey: hours daily (quantitative), main activity (quantitative categories), how the phone helps (qualitative), one responsible-use suggestion (qualitative). (5) Sample: 40 Class XI students from different sections. (6) Collect ethically: consent, no sensitive data, anonymity, privacy, educational use only. (7) Secondary data: an article saying excessive screen time lowers concentration, to compare with results. (8) Organise in Excel/Google Sheets (Student, Hours, Study Use, Main Activity, Distracted?), sort, total, average. (9) Charts: bar chart of daily phone use (1–3, 3–5, 5+ hrs). (10) Analyse: majority use 3–5 hours; half use for education; entertainment users report more distraction. (11) Conclusion: phones have positive and negative effects; should be managed wisely, not banned. (12) Recommendations: turn off social media notifications, set screen-time limits, take physical breaks. **Artefact:** one of report, presentation, infographic or dashboard, structured as Title, Research Question, Introduction, Method, Primary Findings, Secondary Summary, Charts, Conclusion, Recommendations, References.

5. AI tools for homework (model). *Survey:* consent statement (anonymous, voluntary, educational use only). Q1 Have you used an AI tool (e.g. ChatGPT, Gemini) for homework? Yes / No – quantitative. Q2 How many times a week? 0 / 1–2 / 3–5 / 6 or more – quantitative (non-overlapping categories). Q3 What do you use it for most? Explaining topics / Checking answers / Writing assignments / Translation / Other – quantitative (categories). Q4 "AI tools help me understand my subjects better." 1–5 rating – quantitative. Q5 Do you check AI answers against your textbook? Always / Sometimes / Never – quantitative. Q6 Describe one problem you have faced with AI answers. – qualitative. Sample: about 40 Class XI students from all sections. *Secondary sources:* "generative AI" students homework filetype:pdf; "AI in education" site:.edu; "artificial intelligence" policy site:.gov.pk; evaluate each for author, date, reliability and bias. *Artefact:* a **digital report** (MS Word/Google Docs) because the principal needs the full study – method, findings, charts, conclusion, recommendations and references – in a structured document; a one-page infographic could be attached as a quick summary. Charts would be made in Google Sheets from the form responses, and every visual must match the real data.

Self-assessment: I can...

- | | |
|--|--|
| ☐ define digital literacy and explain why it matters in the Information Age. | ☐ classify data as qualitative or quantitative and explain the difference in a table. |
| ☐ name and explain the five data-collection strategies and choose the right one for a scenario. | ☐ write clear, unbiased survey questions and fix leading, double-barrelled or vague ones. |
| ☐ compare primary and secondary data by source, examples, purpose and control. | ☐ list the eight steps for designing a data-collection approach. |
| ☐ follow ethical rules: consent, anonymity, privacy and educational use. | ☐ use spreadsheet formulas such as =SUM() , =AVERAGE() and =COUNTIF() and calculate results correctly. |
| ☐ use advanced search operators and evaluate whether a source is reliable. | ☐ choose a suitable tool (spreadsheet, presentation, infographic, report, dashboard) to present findings. |
| ☐ describe the 12 steps of the digital inquiry project. | ☐ plan a digital artefact using the ten-part standard structure. |

FINAL REVISION

One-page recaps and a mock paper

Use these pages in the last week before the exam. Read one recap, close the booklet, and say it aloud. Then sit the mock paper in 2 hours 30 minutes with no notes.

امتحان سے ایک ہفتہ پہلے یہ خلاصے زبانی دہرائیں، پھر ماک پیپر بغیر نوٹس کے حل کریں۔

Chapter 1 recap — Computer Systems

- Discrete = counted (stairs). Continuous = measured (ramp). Digital systems use bits 0 and 1; they are reliable, accurate, easy to design and programmable.
- Analog signal = smooth sine wave. Digital signal = square wave HIGH/LOW. Digital data resists noise.
- AND · output 1 only if all inputs 1. OR + output 1 if any input 1. NOT ' inverts. NAND = NOT-AND, NOR = NOT-OR (universal). XOR 1 when inputs differ. XNOR 1 when inputs same.
- Truth table rows = 2^n . SOP = sum of minterms where output is 1. POS = product of maxterms where output is 0.
- K-map: group 1, 2, 4, 8 adjacent 1s (power of 2, wrap-around allowed). Each group drops the variables that change.
- SDLC: requirement → design → implementation → testing → deployment → maintenance. Waterfall = sequential; Agile = short sprints and feedback.
- OSI 7: Physical, Data Link, Network, Transport, Session, Presentation, Application. Mnemonic: Please Do Not Throw Sausage Pizza Away. TCP/IP 4: Network Access, Internet, Transport, Application.

Chapter 2 recap — Computational Thinking & Algorithm

- CT pillars: decomposition, pattern recognition, abstraction, algorithms.
- Algorithm = finite, ordered, unambiguous steps. Pseudocode = structured English, not a programming language.
- Bubble Sort: compare neighbours, swap if out of order; largest "bubbles" to the end each pass. $O(n^2)$.
- Selection Sort: find the minimum in the unsorted part and swap it to the front. $O(n^2)$.
- Linear Search: check one by one. Works on unsorted data. $O(n)$.
- Binary Search: list MUST be sorted. Repeatedly test the middle. $O(\log n)$. Trace with low / mid / high.
- Judge an algorithm on correctness, efficiency (time and space) and clarity. Prefer binary search on large sorted lists; linear search on small or unsorted lists.

Chapter 3 recap — Programming Fundamentals

- Python: high-level, interpreted. Variables are names; types int, float, str, bool, list, dict. `input()` returns str.
- Arithmetic `+` `-` `*` `/` `//` `%` `**`. Relational. Logical `and` `or` `not`. Bitwise `&` `|` `^` `~` `<<` `>>`.
- Selection: `if` / `if-else` / `if-elif-else` / nested. Test from the top; first True wins.
- `for` = known count + range. `while` = unknown count; always update the condition. `break` leaves; `continue` skips one.

- Libraries: import math / random / datetime. Third-party via pip: NumPy, Pandas, Matplotlib.
- Syntax = will not run. Runtime = crash + traceback. Logical = runs, wrong answer.

Chapter 4 recap — Data and Analysis

- Data = raw facts. Information = processed data used for decisions. Database = organised related data. DBMS = software that manages it (reduces redundancy, improves security, sharing, integrity).
- Table / relation; row / record / tuple; column / field / attribute. Primary key uniquely identifies a row. Foreign key refers to another table's primary key.
- ER model: entity (rectangle), attribute (oval), relationship (diamond), cardinality 1:1, 1:M, M:N.
- Referential integrity: you cannot orphan a child row. Cascade Update / Cascade Delete copy or remove matching child rows.
- Access objects: tables, forms, queries, reports. Queries filter with criteria, wildcards (*, ?), AND/OR. Totals: Count, Sum, Avg, Group By. Charts visualise the result.

Chapter 5 recap — Application and Impacts of Computing

- IoT = connected devices. Six parts: sensor, actuator, connectivity, processing, cloud, UI. AI = machines that learn and decide. Analytics = collect → store → clean → analyse → visualise.
- Together: IoT (eyes) → Analytics (sort) → AI (brain) → actuator (action).
- Sources: primary (firsthand), secondary (explained by others), tertiary (summaries). No author / date / references → do not trust.
- Assistive tech: mobility, sensory, communication, cognitive. Goal: equal access, independence, inclusion.

Chapter 6 recap — Digital Literacy

- Digital literacy = find, evaluate, create and share information with digital tools, safely.
- Qualitative = words/descriptions. Quantitative = numbers. Strategies: interview, survey, prototype, observation, simulation.
- Primary data you collect; secondary data already exists. Good survey questions are clear, single-idea, unbiased.
- Present with spreadsheets (formulas such as =AVERAGE()), slides, infographics and reports. Cite sources. Follow the 12-step inquiry project from question to artefact.

Mock paper — Computer Science XI

Time: 2 hours 30 minutes | Total: 75 marks | Answer all questions. Calculators allowed. Write Python with correct indentation.

Section A — Multiple choice (15 × 1 = 15)

1. Which quantity is discrete?
 - a. Room temperature
 - b. Number of desks
 - c. Car speed
 - d. Time of day
2. NAND is called universal because:
 - a. it has two inputs
 - b. any gate can be built from it
 - c. its output is always 1
 - d. it is faster than AND
3. The Boolean expression for XOR is:
 - a. $A \cdot B$
 - b. $A + B$
 - c. $A \oplus B$
 - d. $(A \cdot B)'$
4. OSI layer that chooses the best path (IP addresses) is:
 - a. Transport
 - b. Network
 - c. Data Link
 - d. Session
5. Which SDLC model delivers working software in short sprints?
 - a. Waterfall
 - b. Agile
 - c. Spiral only
 - d. Big Bang
6. Binary Search requires the list to be:
 - a. unsorted
 - b. sorted
 - c. all unique and unsorted
 - d. of even length
7. Worst-case time of Bubble Sort is:
 - a. $O(n)$
 - b. $O(\log n)$
 - c. $O(n^2)$
 - d. $O(1)$
8. `input()` in Python returns a:
 - a. int
 - b. str
 - c. float
 - d. bool
9. `17 // 5` equals:
 - a. 3.4
 - b. 3
 - c. 2
 - d. 3.0
10. A program that runs but prints the wrong total has a:
 - a. syntax error
 - b. runtime error
 - c. logical error
 - d. hardware fault

11. A field that uniquely identifies each record is a:
 - a. foreign key
 - b. primary key
 - c. composite description
 - d. query
12. An actuator in IoT:
 - a. collects temperature
 - b. opens a water valve
 - c. stores cloud files
 - d. draws a chart
13. Wikipedia is a:
 - a. primary source
 - b. secondary source
 - c. tertiary source
 - d. sensor
14. Marks out of 100 are:
 - a. qualitative data
 - b. quantitative data
 - c. tertiary data
 - d. metadata only
15. Which Access object is best for printing a formatted class list?
 - a. Table
 - b. Form
 - c. Query
 - d. Report

Section B — Short questions (10 × 3 = 30)

1. Differentiate analog and digital signals with one example of each.
2. Draw the symbol and truth table of a 2-input NOR gate.
3. List the four pillars of computational thinking in one sentence each.
4. Why can you not use Binary Search on an unsorted attendance list?
5. Write a Python `if-elif-else` that prints Pass if marks ≥ 40 , else Fail.
6. Define primary key and foreign key.
7. Name the six components of IoT.
8. Classify: (a) a photograph you took, (b) a textbook, (c) an encyclopedia.
9. Give two good and two bad survey questions about mobile-phone use in college.
10. State three advantages of a DBMS over paper files.

Section C — Long questions (3 × 10 = 30). Attempt any three.

1. Simplify $F(A,B,C) = \Sigma(1,3,5,7)$ using a 3-variable K-map. Show the grouping and the final expression. Draw the logic diagram.
2. Dry-run Bubble Sort on the list [5, 1, 4, 2]. Show every pass. Then search for 4 using Binary Search on the sorted result, with a low/mid/high table.
3. Write a complete Python program that reads n subject marks, prints the average, and prints "A" if average ≥ 80 , "B" if ≥ 70 , "C" if ≥ 60 , else "D". Include a sample run of four marks: 80, 70, 90, 60.
4. Design an ER diagram and relational schema for a school library (Student, Book, Issue). Mark keys. Write one Access/SQL query that lists all books issued to roll number 101.
5. Explain how IoT, Data Analytics and AI work together on a Sindh farm. Then discuss two positive and two negative impacts of computing on education in Pakistan.

Mock paper — answers

Section A

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. b | 7. c | 10. c | 13. c |
| 2. b | 5. b | 8. b | 11. b | 14. b |
| 3. c | 6. b | 9. b | 12. b | 15. d |

Section B

1. Analog: continuous, sine wave, e.g. human voice. Digital: two levels 0/1, square wave, e.g. computer data. Digital resists noise.
2. NOR symbol: OR-gate shape with a bubble on the output. $Y = (A+B)'$. Truth: 00→1, 01→0, 10→0, 11→0.
3. Decomposition = split a problem. Pattern recognition = reuse similarities. Abstraction = ignore extra detail. Algorithms = write ordered steps.
4. Binary Search repeatedly halves a sorted range. If the list is unsorted, the middle value does not tell you which half to keep, so the method is invalid.
5. `m = int(input("Marks: "))` then `if m ≥ 40: print("Pass")` else `print("Fail")`.
6. Primary key: unique identifier of a row (e.g. StudentID). Foreign key: a field that refers to the primary key of another table (e.g. StudentID in Issuance).
7. Sensors, actuators, connectivity, data processing, cloud computing, user interface.
8. (a) primary (b) secondary (c) tertiary.
9. Good: "How many hours per day do you use a phone for study?" / "Which apps do you use for class notes?" Bad: "Don't you agree phones are destroying students?" (leading) / "Do you like phones and also use laptops?" (double-barrelled).
10. Less redundancy, easier search and update, better security and multi-user access, integrity constraints. (Any three.)

Section C — outline

1. K-map 1s at m1, m3, m5, m7 form one group of 4 covering B=1 (or C=1 depending on labelling — with standard ABC order the 1s are all cells where C=1). $F = C$. Diagram: a single wire C to output (or a buffer). Show the map with the group circled.
2. Pass 1: [1,4,2,5]; Pass 2: [1,2,4,5]; Pass 3: already sorted. Sorted [1,2,4,5]. Search 4: low=0 high=3 mid=1 (2<4) → low=2; mid=2 (4=4) found at index 2.
3. Read n, loop n times adding `float(input())`, `avg = total/n`, then if-elif for A/B/C/D. Sample: total 300, avg 75.0, grade B.
4. Entities Student(StudentID PK, Name, Class), Book(BookID PK, Title, Author), Issue(IssueID PK, StudentID FK, BookID FK, Date). Query: `SELECT Book.Title FROM Issue INNER JOIN Book ON Issue.BookID = Book.BookID WHERE StudentID = 101;`
5. Soil sensor (IoT) → cloud spreadsheet of moisture (Analytics) → model decides "too dry" (AI) → pump (actuator). Education +: LMS, remote classes. Education -: screen addiction, urban/rural device gap.